

Environmental Studies — Study Notes with PYQ

Course Code: B21ES01AC | **Semester:** I | **Programme:** All UG Programmes (AECC) | **University:** Sreenarayanaguru Open University (SGOU)

Subject: Environmental Studies | **Coverage:** Blocks 1–4, 24 Units (Theory Only)

Source: Self Learning Material (SLM) — SGOU B21ES01AC

About These Notes

Prepared by: Abdul Vahab A A

Website: <https://abdulvahabaa.in>

These notes were created for my personal study and revision. I am sharing them here because someone else might find them useful too.

These notes follow the SGOU SLM B21ES01AC structure. Each topic has **Theory, Important Points, For Exam, and Previously Asked Questions** (PYQ number + year + marks). Related subsections are merged into tables where possible to keep revision fast. Where the SLM uses figures, a text description is included (refer your textbook for diagrams).

Wishing you all success in your semester exams and environmental studies journey. Stay disciplined, revise smartly, and keep confidence in your effort, in sha Allah.

Dua mein Yaad Rakhna.

Index

Block 1: Environment and Natural Resources — Environmental segments, natural resources, forests, water, land, energy

Block 2: Ecosystems, Biodiversity and Conservation — Ecosystems, biodiversity levels, India mega diversity, threats, conservation

Block 3: Social Issues and Sustainable Development — Human health, epidemiology, awareness, population, sustainable development

Block 4: Environmental Ethics and Contemporary Issues — Ethics, waste management, climate change, carbon, disasters, environmental acts

(Detailed subsection numbers match SGOU SLM — see each unit heading in the notes.)

Block 1: Environment and Natural Resources

Unit 1: Environmental Segments1.1.1 Environmental Segments

Theory

Environmental studies is a multidisciplinary field examining biotic and abiotic surroundings that influence survival, development and evolution of organisms. Earth is the only planet with life-sustaining land, water and air. Four environmental segments — atmosphere, hydrosphere, lithosphere and biosphere — interact dynamically; change in one segment affects all others. Mahatma Gandhi observed: "There is enough for man's needs but not for his greed" — human greed has brought untold destruction of nature, resulting in repeated disasters like tsunami, Ockhi, floods and landslides.

Important Points

- Four segments: **Atmosphere, Hydrosphere, Lithosphere, Biosphere**
- Environment scale ranges from microscopic to global (marine, atmospheric, terrestrial)
- Integrates physical sciences, economics, humanities, social sciences, ethics, law, pollution control
- Man is part of nature, not above it; both are interrelated and interconnected
- Principles from geography, anthropology, politics, sociology guide environmental studies

For Exam

Environmental studies is a multidisciplinary academic field dealing with the environment and its functions. The environment includes biotic and abiotic surroundings affecting organism survival and evolution. Four environmental segments — atmosphere, hydrosphere, lithosphere and biosphere — are interlinked and control life on Earth. Any change in one segment affects the

nature and functioning of others. Human greed, not need, drives nature's destruction.

Previously Asked Questions

- **Q64** (2 marks, Apr 2025) — What are the environmental segments, controlling life and environmental processes?

1.1.1.1–1.1.1.4 Four Environmental Segments

Theory

The four environmental segments form the physical basis of all life. Atmosphere is a 5000-million-ton gas envelope; hydrosphere covers 71% of Earth as liquid, vapour and ice; lithosphere is the rigid outer Earth of crust, soil and upper mantle; biosphere is the realm of living organisms divided into ecosystems with producers, consumers and decomposers.

Important Points

- **Atmosphere:** Layer of gases retained by Earth's gravity — N₂ 78%, O₂ 21%, CO₂ 0.33%, Ar 0.93%; also He, CH₄, O₃, Ne, Xe; water vapour varies seasonally; above 50 km only O₂, O₃, He, H
- **Hydrosphere:** Total water on Earth — surface, underground, air — 71% Earth covered; oceans, seas, rivers, lakes, glaciers, groundwater; frozen part = **cryosphere**; Earth called **blue planet**
- **Lithosphere:** Solid outer Earth — crust + brittle upper mantle — Rocks, soil with nutrients; most rigid, coolest Earth layer; rocks elastic but not viscous; Earth structure: Crust, Mantle, Outer Core, Inner Core
- **Biosphere:** Living organisms + interactions with other segments — Divided into **ecosystems**; **Producers** (green plants, CO₂ → O₂), **Consumers** (herbivores → carnivores, food chain), **Decomposers** (bacteria/fungi recycle nutrients to soil)
- Atmosphere evolution: 1st atmosphere (He, H) → 2nd atmosphere (CO₂, H₂O, NH₃, ~100× current gas) → rain/oceans dissolved CO₂ → life raised O₂, ozone layer formed
- Atmosphere functions: unequal heating, heat zones, temperature variation, pressure changes, winds, clouds, rainfall, snowfall

- Hydrological cycle: ocean evaporation exceeds sea precipitation by 10%; excess precipitates on land balancing the cycle

Atmosphere Evolution (Detailed)

- **1. Original atmosphere:** Helium and hydrogen from Earth's interior heat and solar radiation
- **2. Second atmosphere:** Volcanic eruptions released steam, CO₂, ammonia; ~100× more gas than today; virtually no oxygen
- **3. Water vapour condensed** → rain formed oceans → first aquatic environment favoured origin of life
- **4. Life evolved** → oxygen levels rose significantly → CO₂ levels lowered
- **5. Ozone layer formed** (O₃ allotrope of O₂) → protected life from UV radiation
- **6. Present atmosphere:** Oxygen-nitrogen composition supporting diverse life forms

For Exam

The four environmental segments are atmosphere (gases — N₂ 78%, O₂ 21%), hydrosphere (all water forms including cryosphere — 71% coverage), lithosphere (solid crust and upper mantle with soil nutrients), and biosphere (living organisms in ecosystems with producers, consumers and decomposers). They are mutually interdependent. The hydrosphere drives the hydrological cycle exchanging water between sea, atmosphere, land and organisms.

Previously Asked Questions

- **Q1** (1 mark, Aug 2023) — Name the gaseous layer that surround the earth.
- **Q2** (1 mark, Aug 2023) — Define biosphere.

1.1.4 Hydrological Cycle

Theory

The hydrological cycle is a continuous exchange of water between sea, atmosphere, land and living organisms through evaporation, cloud formation and precipitation. Land and water surfaces lose water by evaporation powered

by solar energy. Ocean evaporation exceeds rainfall into seas by 10%; this excess precipitates on land, maintaining global water balance.

Important Points

- **1. Evaporation** — Water → vapour from oceans, lakes, rivers, soil. Solar energy drives evaporation; oceans contribute most atmospheric moisture
- **2. Transpiration** — Plants release water vapour. Water absorbed by roots returned to atmosphere through leaf pores
- **3. Cloud formation** — Water vapour condenses. Cooling causes condensation into clouds and fog in atmosphere
- **4. Precipitation** — Rain, snow, hail fall to Earth. Excess 10% ocean evaporation precipitates over land balancing cycle
- **5. Surface runoff** — Water flows over land. Collects in streams, rivers, lakes, catchment areas, reservoirs
- **6. Infiltration** — Rain seeps into soil. Water percolates downward forming groundwater in aquifers
- **7. Capillary action** — Groundwater rises. Maintains continuous water supply to surface soil layer
- **8. Plant absorption** — Roots take up soil water. Used in photosynthesis; returned via transpiration
- **9. Animal consumption** — Animals drink water. Returned to atmosphere through excretion and evaporation
- **10. Return to ocean** — Rivers and runoff. Surface and groundwater eventually discharge back to seas
- Cycle is balanced and continuous between Earth's surface and atmosphere
- Cryosphere (glaciers, ice caps) stores frozen hydrosphere; melting feeds rivers
- Disturbing any step (deforestation ↑ transpiration loss, urbanization ↓ infiltration) disrupts the cycle
- **PYQ (Q30 — another name for water cycle): Hydrologic cycle / Hydrological cycle** (SLM Objective Answer)
- **PYQ Q88 For Exam: The hydrological (hydrologic) cycle continuously exchanges water between oceans, atmosphere, land and organisms**

through **evaporation** (solar-driven; oceans main source), **transpiration** (plants release vapour), **condensation** (cloud formation), **precipitation** (rain/snow), **surface runoff**, **infiltration** and **groundwater movement**. Ocean evaporation exceeds precipitation by 10%; excess falls on land. Deforestation and urbanisation disrupt the cycle.

For Exam

The hydrological cycle exchanges water continuously between oceans, atmosphere, land and organisms through evaporation, transpiration, condensation, precipitation, runoff, infiltration and groundwater movement. Ocean evaporation exceeds sea precipitation by 10%, with the excess falling on land. Rain infiltrates soil as groundwater, rises by capillary action, is absorbed by plants and animals, and returns to the atmosphere — completing the cycle.

Previously Asked Questions

- **Q30** (1 mark, Apr 2025) — What is another name for water cycle?
- **Q88** (5 marks, Aug 2023) — Briefly explain hydrological cycle.

Diagram (refer SLM): Fig. 1.1.1 Layers of atmosphere; Fig. 1.1.2 Global distribution of water; Fig. 1.1.3 Hydrological cycle; Fig. 1.1.4 Composition of Earth; Fig. 1.1.5 Interactions of Biosphere.

1.1.2 Atmosphere Layers

Theory

Atmosphere is stratified into layers differing in composition, temperature and radiation effects. SLM identifies three major layers (Troposphere, Stratosphere, Ionosphere) plus Mesosphere, Thermosphere and Exosphere. Troposphere holds 70% atmospheric mass and is the zone of weather; stratosphere contains the ozone layer; thermosphere/ionosphere has charged particles from photoionisation.

Important Points

- **Troposphere:** 8–18 km above surface — 70% atmospheric mass; 1% water vapour, more CO₂; dusty; temp falls 1°C/km (15°C to -56°C); zone of turbulence, climate, weather; ends at **tropopause**; most pollutants stay here; CFCs can cross upward

- **Stratosphere:** Up to 80 km — 1000× less water vapour, 1000× more ozone than troposphere; cloud-free; jet planes fly here; temp rises with altitude (−52°C to −2°C); **ozone layer** protects from UV; O₃ formed by UV/lightning + O₂; CFCs cause ozone depletion → sunburn, skin/eye cancers, cataracts; ends at **stratopause**
- **Mesosphere:** Up to 80 km above stratosphere — Temp decreases uniformly; may reach −100°C; very low pressure
- **Thermosphere/Ionosphere:** Up to 700 km — Very low density; temp rises rapidly (1200°C at 350 km); photoionisation → charged particles; also called **ionosphere**
- **Exosphere:** Up to 10,000 km — Only H and He atoms; very high temperature from solar radiation; **outermost layer**
- **PYQ Q80 For Exam:** Earth's atmosphere has five main layers: **Troposphere** (8–18 km, 70% mass, weather zone, temp decreases 1°C/km); **Stratosphere** (up to 80 km, ozone layer absorbs UV, CFCs cause depletion); **Mesosphere** (up to 80 km, −100°C); **Thermosphere/Ionosphere** (up to 700 km, ionised particles); **Exosphere** (up to 10,000 km, outermost, H and He only).

For Exam

The troposphere (8–18 km) contains 70% of atmospheric mass and is the weather zone where most pollutants remain. The stratosphere (up to 80 km) has the ozone layer protecting life from UV radiation; ozone depletion by CFCs causes skin cancers and cataracts. The thermosphere/ionosphere (up to 700 km) has ionised particles. The exosphere (up to 10,000 km) is the outermost layer with only hydrogen and helium.

Previously Asked Questions

- **Q27** (1 mark, May 2024) — Which atmospheric layer has more concentration of Ozone?
- **Q55** (2 marks, May 2024) — What is troposphere?
- **Q80** (5 marks, Aug 2023) — Explain the different layers of the atmosphere.

1.1.3 Weather and Climate

Theory

Weather consists of short-term atmospheric changes (minutes to months) over a limited area — sunshine, rain, winds, hail, floods, heat waves. Climate is the average weather over a long period (usually 30 years) for a larger region. Climate is what you expect; weather is what you get.

Important Points

Weather vs Climate

Aspect	Weather	Climate
Duration	Short (minutes to months)	Long period average (typically 30 years)
Area	Small / local	Large / continent-scale
Change	Changes frequently	Relatively permanent
Elements	One element may dominate	Collective effect of all elements
Memory tip	What you get	What you expect

- Climate study examines averages of precipitation, temperature, humidity, sunshine, wind, fog, frost, hail over long periods
- Climate changes affect people worldwide; greenhouse effect warms Earth's surface by trapping solar radiation
- **PYQ Q89 For Exam:** **Weather** is short-term atmospheric conditions (minutes to months) over a small/local area — changes frequently. **Climate** is the average weather over a long period (typically 30 years) over a large region — relatively permanent. Memory tip: climate = what you expect; weather = what you get. Key differences: duration, areal extent, and stability.

For Exam

Weather is the study of short-duration atmospheric conditions over a limited area and changes frequently. Climate is the average weather observed over a long period (usually 30 years) over a larger area and is relatively permanent. An easy distinction: climate is what you expect (hot summer), weather is what you get (hot day with thunderstorms).

Previously Asked Questions

- **Q71** (2 marks, Apr 2025, Aug 2023, May 2024) ★🔥 — Differentiate Weather and Climate.
- **Q89** (5 marks, Aug 2023, May 2024) ★🔥 — Differentiate weather and climate. / Write the differences between weather and cli...

Unit Recap (from SLM)

- Environmental studies is multidisciplinary; atmosphere, hydrosphere, lithosphere, biosphere are interlinked
- Atmosphere: N₂ 78%, O₂ 21%; troposphere = weather zone; stratosphere = ozone layer; exosphere = outermost
- Hydrosphere: 71% water; cryosphere = frozen part; hydrological cycle balances land-ocean water exchange
- Lithosphere: solid crust + upper mantle; soil formed by weathering
- Biosphere: producers, consumers, decomposers; smallest unit = ecosystem
- Weather = short-term/local; Climate = long-term average (30 years)/large area
- Earth called blue planet; composition: Crust, Mantle, Outer Core, Inner Core

Unit 2: Natural Resources

1.2.1 Concepts and Types of Natural Resources

Theory

Natural resources exist independent of human actions and include air, sunlight, water, soil, stone, plants, animals and fossil fuels. The five most important are air, water (71% Earth covered, only 2% fresh), soil, iron and forests. Based on availability, resources are classified as renewable (infinite, reusable — forest, wind, water) or non-renewable (limited, exhaustible — fossil fuels, minerals).

Important Points

Renewable vs Non-renewable Resources

Type	Availability	Examples
Renewable	Infinite quantity; usable repeatedly	Forest, wind, water, solar, tidal energy
Non-renewable	Limited abundance; may run out	Petroleum, coal, minerals, fossil fuels

- Five key natural resources: Air, Water, Soil, Iron, Forests
- Only 2% of Earth's water is fresh water

- Forests provide clean air and preserve global ecology
- Petroleum is non-renewable and gets exhausted with use

For Exam

Natural resources are materials existing independent of human actions, including air, water, soil, plants, animals and fossil fuels. Renewable resources (forest, wind, water) are available repeatedly; non-renewable resources (fossil fuels, minerals) are limited and exhaustible. Human necessities, comforts and luxuries all come from nature, but reckless overuse threatens both nature and mankind.

Previously Asked Questions

- **Q3** (1 mark, Aug 2023) — Based on the availability what are the two types of natural resources.
- **Q65** (2 marks, Apr 2025) — Give an account of renewable resources with suitable examples.

1.2.2–1.2.4 Importance, Threats and Conservation

Theory

Natural resources are essential for human survival and development — air, land, solar energy, oil/gas, minerals, forests, industrialization and dams. Major threats are overpopulation, over-exploitation, climate change and environmental pollution. Conservation means protection and skilful management for future generations through reduce-reuse-recycle, afforestation, alternative energy and strict laws.

Important Points

- **Importance:** air to breathe; land for cultivation; solar energy; oil/gas for fuel; minerals (coal, iron, gold, diamonds); forests for timber/fuel/shelter; dams for flood control, irrigation, fishing, electricity
- **Threats:** (1) Overpopulation (2) Over-exploitation (3) Climate change (4) Environmental pollution
- **Conservation methods:** 3R (Reduce, Reuse, Recycle); community cleaning; rainwater harvesting; pond/lake maintenance; dams/reservoirs; stop deforestation; efficient resource use; public awareness; alternative energy; tree plantation; pollution control (electric vehicles); strict preservation laws

Natural Resource Conservation (Detailed)

- **1. Reduce, Reuse, Recycle (3R)** — minimise waste and maximise resource reuse
- **2. Volunteer community cleaning** — local participation in environment protection
- **3. Rainwater harvesting** — collect and store rain for future use; recharge groundwater
- **4. Pond and lake maintenance** — regular cleaning and desilting of water bodies
- **5. Dams and water reservoirs** — store water for irrigation, power and flood control
- **6. Stop deforestation** — prevent reckless tree cutting and forest fires
- **7. Efficient resource use** — avoid wastage of water, energy and raw materials
- **8. Public awareness** — educate communities on importance of resource conservation
- **9. Alternative energy** — shift from fossil fuels to solar, wind, biogas
- **10. Tree plantation** — afforestation and reforestation programmes
- **11. Pollution control** — electric vehicles, cleaner industrial processes
- **12. Strict laws** — implement and enforce natural resource preservation legislation
- **PYQ Q91 For Exam:** Natural resource **conservation methods** include: **(1) 3R** — Reduce, Reuse, Recycle; **(2) rainwater harvesting**; **(3) stop deforestation / afforestation**; **(4) alternative energy** (solar, wind, biogas); **(5) public awareness**; **(6) efficient resource use**; **(7) pollution control**; **(8) dams and reservoirs**; **(9) strict preservation laws**; **(10) community participation**.
- **PYQ Q102 For Exam:** Major **threats to natural resources**: **(1) overpopulation** — demand exceeds supply; **(2) over-exploitation** — reckless use of forests, minerals, water; **(3) climate change** — alters availability; **(4) environmental pollution** — contaminates air, water, soil. These cause depletion, biodiversity loss and scarcity for future

generations. Conservation through 3R, afforestation and sustainable use is essential.

For Exam

Natural resources are vital for survival — providing air, food, fuel, minerals and industrial raw materials. Major threats include overpopulation, over-exploitation, climate change and pollution. Conservation involves the 3R principle, rainwater harvesting, afforestation, alternative energy development, public awareness and strict implementation of preservation laws to protect resources for future generations.

Previously Asked Questions

- **Q91** (5 marks, May 2024) — Explain different methods for the conservation of natural resources.
- **Q102** (5 marks, Apr 2025) — Comment on the threats faced by natural resources.

Diagram (refer SLM): Fig. 1.2.1 Renewable and Non-Renewable Resources; Fig. 1.2.2 Different types of Natural Resources; Fig. 1.2.3 Threats to natural resources.

Unit Recap (from SLM)

- Natural resources exist independent of human actions (air, water, soil, plants, animals, fossil fuels)
- Renewable: forest, wind, water; Non-renewable: fossil fuels, minerals
- Importance: survival, industry, dams, timber, fuel, minerals
- Threats: overpopulation, over-exploitation, climate change, pollution
- Conservation: 3R, rainwater harvesting, afforestation, alternative energy, strict laws

Unit 3: Forest Resources1.3.1 Types and Classification of Forests in India

Theory

Forests are important renewable resources. India's forests were first classified in 1936 by Sir H.G. Champion; Champion and Seth (1968) classified them into 16 types under 6 major groups based on climate, physiognomy, species

composition, phenology, topography, soil, altitude and biotic factors. India has ~24.16% forest cover with 16 forest types; forest cover reduced from ~30% (early 20th century) to 21.34% (2015).

Important Points

Six major groups — 16 forest types (Champion & Seth, 1968)

Group I — Moist Tropical Forests

- **1. Tropical Wet Evergreen** — Rainfall 2000–3000+ mm; Western Ghats, NE India, Andaman-Nicobar; dense 30–45 m canopy, high species diversity
- **2. Tropical Semi-evergreen** — Rainfall 1500–2500 mm; transition zones adjoining evergreen; 13.79% of Indian forests; multi-strata 24–36 m
- **3. Tropical Moist Deciduous** — Rainfall 1000–2000 mm; TN, Arunachal, Assam, Meghalaya, WB, Odisha; 19.73% of forests; shed leaves in winter
- **4. Littoral and Swamp** — Coastal/mesic; Sundarban (largest mangrove, 40% WB); Rhizophora, Avicennia; Casuarina on beaches

Group II — Dry Tropical Forests

- **5. Tropical Dry Deciduous** — Rainfall 800–1200 mm; largest type (38.2%); Kanyakumari to Himalaya foothills; dry teak (south), dry sal (north)
- **6. Tropical Thorn** — Rainfall 200–800 mm; Rajasthan, Punjab, Gujarat, MP, UP; Acacia, Prosopis, Ziziphus; open short thorny trees
- **7. Tropical Dry Evergreen** — Coastal low rain; Karnataka coast, AP east coast; Manilkara hexandra, Memecylon edule; 9–12 m height

Group III — Montane Subtropical Forests

- **8. Subtropical Broad-leaved Hill** — Altitude 1000–2000 m; Nilgiris, Palani, Tirunelveli; Quercus, Castanopsis (NE hills); Calophyllum (south)
- **9. Subtropical Pine** — Altitude 1000–1800 m; central/western Himalaya; Pinus roxburghii (chir pine); Haryana, HP, J&K, Punjab, Uttarakhand
- **10. Subtropical Dry Evergreen** — Foothills; Bhabar, Shiwalik; Olea cuspidata, Acacia modesta in J&K scrub forests

Group IV — Montane Temperate Forests

- **11. Montane Wet Temperate** — Altitude 1800–3000 m, rainfall >2000 mm; Eastern Himalaya; moss-covered short trees

- **12. Himalayan Moist Temperate** — Altitude 1500–3300 m; Kashmir, HP, Uttarakhand, Sikkim; oak, Cedrus deodara, Abies, Picea
- **13. Himalayan Dry Temperate** — Altitude 1700–3000 m, rainfall <1000 mm; Inner Himalaya; Pinus gerardiana, Juniperus; Ladakh, Chamba, Garhwal

Group V — Sub-Alpine Forests

- **14. Sub-Alpine Forests** — Above 3000 m; Rhododendron, Betula utilis; J&K, HP, Uttarakhand, Sikkim, Arunachal, WB

Group VI — Alpine Forests

- **15. Moist-Alpine Scrub** — Up to 5500 m; Rhododendron + birch scrub; Kumaun, eastern Himalaya 3350–4600 m
- **16. Dry-Alpine Scrub** — Rainfall <370 mm; Juniperus, Lonicera, Potentilla; J&K, HP, Uttarakhand, Arunachal 3000–4600 m

Other key facts

- Champion classified forests first in 1936; revised survey 1968 by Champion & Seth
- 2015 FSI: very dense 2.61%, moderately dense 9.59%, open 9.14%, scrub 1.26%
- Mizoram highest % cover (88.93%); MP largest total area (77,462 km²); Lakshadweep 84.56%
- New classification (CFRE 2013): 10 major groups, 48 sub-groups; blue pine shifted to 2700 m elevation
- Average 26,000 sq. km forests converted to agricultural land annually in India
- **PYQ Q120 For Exam:** Indian forests (Champion & Seth, 1968) are classified into **6 major groups, 16 types: (I) Moist Tropical** — wet evergreen (2000–3000+ mm), moist/semi-evergreen; **(II) Dry Tropical** — dry deciduous (largest, 38.2%), thorn, littoral/mangrove (Sundarban); **(III) Montane Subtropical**; **(IV) Montane Temperate**; **(V) Sub-Alpine**; **(VI) Alpine scrub**. India has ~24.16% forest cover; declined from ~30% to 21.34% by 2015.

For Exam

India's forests (Champion & Seth, 1968) are classified into 6 major groups and 16 types ranging from tropical moist to alpine scrub. Tropical dry deciduous

forests are the largest (38.2%). India has ~24.16% forest cover. Wet evergreen forests occur in high-rainfall Western Ghats/NE regions; Sundarban is the world's largest mangrove. Forest cover declined from 30% to 21.34% by 2015.

Previously Asked Questions

- **Q16** (1 mark, May 2024) — Which type of forest is dense and has a rainfall more than 3000 mm per year?
- **Q120** (10 marks, Apr 2025, May 2024) ★ — Explain the types and classification of forest in India.

Diagram (refer SLM): Fig. 1.3.1 Moist Tropical Forest; Fig. 1.3.2 Dry Tropical Forest; Fig. 1.3.3 Montane Subtropical; Fig. 1.3.4 Montane Temperate; Fig. 1.3.5 Alpine Forests.

1.3.2–1.3.4 Threats — Mining, Dams and Environmental Impact

Theory

Forest ecosystems face over-exploitation, timber extraction, deforestation, mining and large dams. Over 72% of IUCN Red List species are threatened by over-exploitation; 62% by agriculture. Mining degrades ~80,000 hectares in India; dams (1550+ large dams, max in Maharashtra) submerge forests, displace tribals and alter river ecosystems. Nehru called dams "Temples of modern India" but they cause irreparable environmental damage.

Important Points

- **Deforestation:** Clearance or permanent destruction of forest cover by human activities — a serious form of environmental degradation
- **Deforestation causes:** reckless tree cutting, forest fires for plantation, air pollution
- **Mining effects:** land degradation, top soil loss, drying of springs/streams, 33% forest decline in mining areas; Goa lost 50,000+ ha; Jharia/Raniganj/Singrauli deforestation
- **PYQ Q46 For Exam:** **Deforestation** is the clearance/destruction of forests by human activities — a serious form of environmental degradation. Main causes: **(1) reckless and ruthless cutting of trees, (2) forest fires set for plantation, (3) air pollution.** Forests store carbon, support biodiversity and

stabilise climate; deforestation destroys these functions, increases soil erosion, landslides and global warming.

- **Dams — 6 major impacts:** (1) Downstream morphology/water quality change (2) Forest/biodiversity loss (~40,000 km² flooded worldwide) (3) Salinity and pollution (4) Reservoir-induced earthquakes (Koyna 1967) (5) Design failures (Machu dam Gujarat 1979) (6) Resettlement of environmental refugees (mostly tribals)
- India: 1550+ large dams; Maharashtra >600, Gujarat >250, MP 130

Mining Effects (Detailed)

- Shallow deposits: surface mining; deep deposits: sub-surface mining — both degrade land, lose topsoil
- ~80,000 hectares under mining stress in India
- Dries perennial springs and streams in mountainous areas
- Mussoorie and Dehradun valleys: large-scale deforestation from mining
- Forest area declined 33% in mining zones; increased landslide risk
- Goa: >50,000 ha forest destroyed since 1961; Jharia, Raniganj, Singrauli (Jharkhand) extensive deforestation
- Khirakot, Kosi valley, Almora: 14 ha lost to magnetite/soapstone mining
- Kerala, Tamil Nadu, Karnataka: radioactive mineral mining threatens forests
- Western Ghats: copper, chromite, bauxite, magnetite mining projects threaten rich forests

Large Dam Impacts (Detailed)

- **1.** Downstream rivers lose sediment → coastal erosion; fisheries decline if runoff reduced >25%
- **2.** ~40,000 km² forest/wetland flooded worldwide; catchment deforestation → siltation
- **3.** Massive evaporation → salinity increase; delta areas face sea water ingress
- **4.** Koyna dam (Maharashtra 1967): reservoir-induced earthquake; Tehri dam controversy in Garhwal

- **5.** Machu dam burst (Gujarat 1979): poor construction caused widespread flooding
- **6.** Tribal displacement without participation or benefit-sharing; incomplete land records

For Exam

Threats to forests include over-exploitation, timber extraction, deforestation, mining and dams. Mining degrades land and forests (80,000 ha stressed in India). Large dams submerge forests, displace tribals as environmental refugees, alter downstream ecosystems, increase salinity, and can trigger earthquakes. Deforestation by reckless cutting, forest fires and pollution destroys carbon-stabilizing forest ecosystems globally.

Previously Asked Questions

- **Q46** (2 marks, Aug 2023) — Explain deforestation.

1.3.5–1.3.7 Conservation, Social Forestry and Agroforestry

Theory

Conservation preserves natural assets for future generations maintaining species, ecosystem and nutrient cycling diversity. Strategies include afforestation, reforestation, social forestry and agroforestry. Afforestation creates new forests on historically non-forest land; reforestation restores recently cleared forest areas. Social forestry (NCA 1976, NFP 1988) plants trees outside conventional forests for local community needs. Agroforestry integrates trees, crops and livestock on the same land.

Important Points

- **Afforestation:** New forest on land with no prior forest — One sapling per tree; increases forest area
- **Reforestation:** Re-establishing forest on cleared forest land — Two saplings per felled tree; costlier
- **Social Forestry:** Forestry outside conventional forests for local fuel, fodder, timber — Government programme; roadsides, wastelands, community lands
- **Agroforestry:** Trees + crops/livestock on same plot — Farmer's own land; shelter belts; nutrient cycling

- Conservation: increase plantation area, alternative materials, reserved timber forests, remote sensing monitoring, fire fighting, enforce anti-poaching laws, Joint Forest Management (WB, Haryana 1970s)
- Social forestry benefits: fuel, fodder, timber, income, community cohesion, waste land reclamation, wind/dust protection, desertification check
- Agroforestry benefits: soil fertility, erosion control, moisture retention, nutrient cycling, N-fixing trees, income diversification

Social Forestry vs Agroforestry

Aspect	Social Forestry	Agroforestry
Definition	Tree planting outside conventional forests for community benefit	Integration of trees, crops and/or livestock on the same land
Land	Vacant land, roadsides, railway tracks, panchayat lands, religious places	Farmer's own field/own land
Objective	Fuel wood, fodder, timber, wind/water erosion control	Maximise biomass per unit area; food + fodder + fuel wood
Scale	Massive government programme on community/wasteland	Localized unit of land management
Philosophy	"Forestry of the people, by the people, for the people"	Rural self-sufficiency through complementary crop-tree production
Recognition	NCA 1976, National Forest Policy 1988	—
Examples	Wind breaks on dry farms, trees along canals/streams	Cocoa, coffee, rubber, oil palm with shade trees; shelter belts

For Exam

Forest conservation includes afforestation (new forests on unused land), reforestation (restoring cleared forests), social forestry (community tree planting on non-forest land for local needs) and agroforestry (integrating trees with crops/livestock). Other strategies include alternative timber substitutes, remote sensing for deforestation hotspots, fire control, strict laws and Joint Forest Management involving local communities.

Previously Asked Questions

- **Q56** (2 marks, May 2024) — What is meant by agroforestry?

Diagram (refer SLM): Fig. 1.3.6 Kerala Man M.R. Hari Grows 400-Tree Forest in 3 Cents of Land.

Unit Recap (from SLM)

- Forests: renewable resource; Champion & Seth 1968 = 6 groups, 16 types; ~24.16% cover
- Threats: over-exploitation, timber extraction, deforestation, mining, dams
- Afforestation = new forest; Reforestation = restore cleared forest
- Social forestry: outside conventional forests for local community needs
- Agroforestry: trees + crops/livestock integration on same land
- India: 1550+ large dams; Sundarban = largest mangrove; forest cover declined to 21.34%

Unit 4: Water Resources

1.4.1–1.4.2 Global Distribution and Water Resource Types

Theory

Earth's ~332.5 million cubic miles of water is extremely unevenly distributed — over 97% saline (oceans), only 3% freshwater. Of freshwater on land, 79% locked in ice/glaciers, 20% groundwater, only 1% easily accessible surface freshwater. Water exists as saltwater, surface water, groundwater, frozen water (cryosphere) and atmospheric vapour.

Important Points

- **Oceans and Seas** — 97.4%: Saline; Na^+ , Cl^- dominant ions
- **Ice Caps and Glaciers** — 1.98%: Largest freshwater store; 90% in Antarctica
- **Groundwater and Soil water** — 0.599%: Aquifers; Great Artesian Basin Australia
- **Lakes** — 0.007%: 52% of accessible surface freshwater
- **Atmosphere** — 0.001%: Water vapour, clouds

- **Rivers** — 0.0001%: Smallest but vital; use judiciously
- **Surface water:** rivers, lakes, ponds, wetlands; replenished by precipitation
- **Groundwater:** subsurface pore space/aquifers; natural input = seepage; output = springs, ocean seepage
- **Estuary:** brackish water where fresh meets salt water; rich in organisms
- **Frozen water:** glaciers; if all melted sea rises ~6 m (20 ft); Himalayas = "Roof of the World"
- Desalinization removes salts but is costly and energy-intensive

For Exam

Over 97% of Earth's water is saline in oceans; only 3% is freshwater, mostly locked in ice caps (79% of land freshwater). Rivers constitute only 0.0001% but are critical for human use. Water resource types include surface water (rivers, lakes), groundwater (aquifers), estuaries (brackish), frozen water (glaciers/cryosphere) and atmospheric vapour. Sustainable management requires judicious river water use.

Previously Asked Questions

- **Q4** (1 mark, Aug 2023) — Fresh water that located in the subsurface pore space of soil and rocks is known as.
- **Q18** (1 mark, May 2024) — What percentage of water is distributed in ocean and seas?

Diagram (refer SLM): Fig. 1.4.1 Global distribution of water; Fig. 1.4.2 Water moves between reservoirs.

1.4.3 Water Availability, Uses and Scarcity

Theory

Global water use: agriculture ~70% (irrigation), industry ~22%, domestic ~8%. Water scarcity is lack of freshwater to meet demand — physical scarcity (not enough water, e.g. arid regions) or economic scarcity (lack of infrastructure/technology, e.g. Sub-Saharan Africa). India ranks 133rd in per capita water availability despite major rivers.

Important Points

- **Uses:** drinking/domestic (8%), agriculture/irrigation (70%), industry (22%), environment, recreation
- **Potable water:** safe for drinking without immediate/long-term harm

Physical vs Economic Water Scarcity

Type	Meaning	Example Region
Physical scarcity	Insufficient water for ecosystems and people	Central/West Asia, North Africa
Economic scarcity	Lack of investment/technology to access water	Sub-Saharan Africa

- Scarcity intensifies with development; climate change, deforestation, pollution worsen shortages
- Tamil Nadu: only state with compulsory rooftop rainwater harvesting for all houses

Water Uses

Use	Share	Examples
Agriculture/Irrigation	~70%	Crop irrigation; hydroelectric dams
Industry	~22%	Manufacturing, refineries, cooling
Domestic	~8%	Drinking, bathing, cooking, cleaning
Environment	Small but growing	Wetland watering, fish ladders
Recreation	Non-consumptive	Angling, swimming, water skiing

- **India rank:** 133rd per capita water availability despite Ganga, Yamuna, Godavari, Narmada rivers
- **PYQ Q67 For Exam:** **Water scarcity** is lack of freshwater to meet demand — **physical scarcity** (insufficient water in arid regions) or **economic scarcity** (lack of infrastructure/technology to access water). **Main causes:** population growth, industrialisation, urbanisation, agricultural overuse (~70% water use), climate change, deforestation and pollution. India ranks **133rd** in per capita water availability despite major rivers.

For Exam

Water is used mainly for agriculture (70%), industry (22%) and domestic purposes (8%). Water scarcity is either physical (insufficient water in arid regions) or economic (lack of infrastructure to access water). India's water

resources are declining due to population growth, industrialization, urbanization and agricultural modernization, making rational water use essential.

Previously Asked Questions

- **Q67** (2 marks, Apr 2025) — What is water scarcity and what are its main causes?

1.4.4–1.4.7 Threats, Water Logging, Floods and Droughts

Theory

Water threats are quality or quantity issues from man-made or natural sources. Groundwater threats include landfills, septic systems, mining, agriculture (nitrates/pesticides), urbanization (impervious surfaces, saltwater intrusion). Surface water faces point source (single identifiable) and non-point source (multiple diffuse) pollution. Water logging saturates soil restricting air; floods and droughts are major natural hazards.

Important Points

- **Groundwater threats (4 groups):** waste disposal, resource extraction, agriculture, urbanization
- **Surface pollution sources:** domestic sewage, industrial waste, agricultural sediments/fertilizers/pesticides
- **Pollution effects:** cholera, jaundice, typhoid; kills aquatic life; artificial eutrophication (excess P, N → low O₂)
- **Treatment:** primary (mechanical solids removal) → secondary (biological bacteria) → tertiary (advanced)
- **Water logging types:** natural/human-induced; temporary/permanent; rainfed/irrigated; agricultural/urban
- **Flood types:** flash, single event (2–3 days rain), multiple event (Indo-Gangetic), seasonal, coastal, estuarine, dam failure, snow/glacier melt
- **Drought types:** meteorological (low precipitation), agricultural (crop failure), hydrological (low reserves in aquifers/lakes)

Groundwater Threats (Detailed)

- **Waste disposal:** Landfills leach toxins; septic systems release bacteria, viruses, nitrates, solvents

- **Mining:** Intersects aquifers, lowers water tables, increases iron, manganese, sulphate in wells
- **Agriculture:** Over-applied manure/fertilizer → nitrates ("blue baby" disease in infants); pesticides
- **Urbanization:** Impervious surfaces (roofs, roads) block recharge; over-pumping draws saltwater into coastal aquifers
- **Natural:** Dissolved solids, calcium carbonate (hardness), corrosive acidity from rock leaching

Surface Water Threats (Detailed)

- **Point source:** Single identifiable — factory discharge, sewage plant outlet (easier to regulate)
- **Non-point source:** Diffuse — agricultural fertilizers, herbicides, sedimentation from farmland (harder to control)
- **Domestic sewage:** Homes and businesses pollute rivers and lakes
- **Industrial waste:** Chemical pollutants from refineries, manufacturing into freshwater
- **Agricultural pollution:** Sediments, fertilizers, pesticides, animal wastes → eutrophication

Causes of Water Pollution (SLM — PYQ Q79)

- **Domestic sewage** — homes and businesses discharge into rivers/lakes
- **Industrial waste** — chemicals from refineries and manufacturing (point source)
- **Agricultural runoff** — fertilizers, pesticides, sediments (non-point source)
- **Groundwater contamination** — landfills, septic systems, mining, nitrates, urban saltwater intrusion
- **Effects:** cholera, jaundice, typhoid; kills aquatic life; eutrophication (excess N, P → low O₂)
- **PYQ Q79 For Exam:** Water pollution is caused by **domestic sewage** and **industrial waste** (point sources — single identifiable outlets), **agricultural runoff** of fertilizers, pesticides and sediments (non-point/diffuse sources), **landfill leachates** and **septic systems** contaminating groundwater, **mining** intersecting aquifers, and **urbanization** (impervious surfaces blocking

recharge, saltwater intrusion). Effects include waterborne diseases (cholera, typhoid, jaundice), death of aquatic life, and artificial eutrophication from excess nitrogen and phosphorus.

Major Threats to Water Resources (SLM — PYQ Q103)

- **Quantity threats:** over-extraction, droughts, climate change reducing supply, population growth
- **Quality threats:** groundwater and surface water pollution (see above)
- **Natural hazards:** floods, water logging saturating soil
- **Human causes:** industrialization, urbanization, agricultural modernization, poor wastewater treatment
- **PYQ Q103 For Exam:** Major threats to water resources include **groundwater contamination** (landfills, septic systems, mining, agricultural nitrates/pesticides, urbanization reducing recharge and causing saltwater intrusion), **surface water pollution** (domestic sewage, industrial discharge, agricultural non-point runoff), **water scarcity** from overuse and climate change, **water logging** (soil saturation), **floods** and **droughts** affecting both quantity and quality of freshwater.

For Exam

Groundwater is threatened by landfills, septic systems, mining, agricultural nitrates and urbanization reducing recharge. Surface water pollution comes from point sources (factory discharge) and non-point sources (agricultural runoff). Floods result from heavy rainfall, storm surges and dam failures; droughts from prolonged below-normal precipitation affecting soil moisture, crops and groundwater reserves.

Previously Asked Questions

- **Q79** (5 marks, Aug 2023) — Write a short note on causes of water pollution.
- **Q103** (5 marks, Apr 2025, Aug 2023) ★ — Describe the major threats to water resources.

Diagram (refer SLM): Fig. 1.4.3 Maharashtra flooding; Fig. 1.4.4 West Bengal flooding; Fig. 1.4.5 Major Droughts; Fig. 1.4.6 Refugee camp Kenya.

1.4.8–1.4.9 Conservation and Coastal Regulation Zone

Theory

Water conservation manages water as a sustainable resource through traditional harvesting, watershed management, groundwater recharge and wastewater reclamation. **Coastal Regulation Zone (CRZ)** under Environment Protection Act 1986 (first notification February 1991, MoEF) regulates activities within 500 m of High Tide Line (HTL) and 100 m along tidal creeks/estuaries.

Important Points

Conservation methods (detailed):

- **1. Traditional water harvesting** — Johad (Rajasthan), Talab, Saza Kuva, Paar systems; suited to local conditions
- **2. Watershed management** — managing land area shedding water into a region; sustainable distribution of water, soil, plant and animal resources
- **3. Pani Panchayat** — voluntary farmer groups regulating river-to-field water supply for optimum crop use
- **4. Percolation ponds** — check dams tapping rainwater; reduce water shortage by recharging aquifers
- **5. Spring water trapping** — new solution to conserve and harvest natural spring flows
- **6. Rooftop rainwater harvesting** — rain from building roof collected into well/tank below; **Tamil Nadu made compulsory for all houses**
- **7. Wastewater reclamation** — treating wastewater to acceptable quality; direct reuse (linked processing) or indirect (discharge to stream/aquifer then mixed)
- **8. Groundwater recharge** — artificial injection of water through wells into deep aquifers where surface application is ineffective
- **9. Check dams and farm ponds** — slow runoff, increase percolation in rural areas
- **10. Desalinization** — removing salts from seawater; costly but essential for arid coastal regions
- **CRZ categories (2019):** CRZ-I (ecologically sensitive — mangroves, corals, dunes; IA/IB split), CRZ-II (urban developed, >50% built-up), CRZ-III (rural; IIIA/IIIB with NDZ 50–200 m), CRZ-IV (aquatic; IVA seaward 12 nm, IVB tidal water bodies)

- CRZ 2019: MoEFCC clearance only for CRZ-I and IV; State clearance for CRZ-II/III; 20 m NDZ for islands
- India major freshwater source: groundwater; rooftop harvesting recharges groundwater

Flood Types (Detailed)

- **Flash floods:** Cloud burst, violent convectional storms; little warning — Short duration, great volume; biggest human life loss
- **Single event:** 2–3 days widespread heavy rain — Cyclonic disturbances, storms, depressions
- **Multiple event:** Successive weather disturbances — Indo-Gangetic plains
- **Seasonal:** Monsoon patterns — South India: winter monsoon floods; others: summer monsoon
- **Coastal:** High tides, storm surges, hurricanes — Beach/soil erosion; tsunamis as coastal killer waves
- **Estuarine:** Tidal flooding in river mouths — Brackish water zone flooding
- **Dam failure:** Poor construction or earthquake damage — Machu dam Gujarat 1979
- **Snow/glacier melt:** Sudden warming melts ice — Mountainous regions; August 2021 Maharashtra, West Bengal floods

Drought Types

Type	Cause	Key Feature
Meteorological	Prolonged below-average precipitation	Usually precedes other drought types
Agricultural	Crop failure from irrigation/soil issues	Can occur even without rainfall change
Hydrological	Water reserves below threshold	Shows up slowly in aquifers, lakes, reservoirs

- Droughts: second-most costly weather events after hurricanes
- Annual dry seasons in tropics increase drought and bush fire chances
- Heat waves worsen drought by accelerating evaporation

CRZ Details (2019 Notification)

- Coastal land up to 500 m from HTL + 100 m along tidal creeks, estuaries, backwaters, rivers
- CRZ-I A: mangroves, corals, sand dunes, salt marshes, national parks, sea grass beds (11 sensitive areas)
- CRZ-I B: intertidal zone between Low Tide Line and High Tide Line
- CRZ-II: developed urban coastal areas; floor space index norms de-frozen in 2019
- CRZ-III A: population density $>2161/\text{km}^2$ (2011 census) — NDZ 50 m; CRZ-III B: $<2161/\text{km}^2$ — NDZ 200 m
- CRZ-IV A: water area seaward up to 12 nautical miles; CRZ-IV B: tidal water bodies up to 5 ppt salinity
- Promotes sustainable coastal tourism; pollution abatement facilities in CRZ-IB

For Exam

Water conservation includes traditional harvesting (Johad, Talab), watershed management, rooftop rainwater harvesting, percolation ponds and groundwater recharge. CRZ notification (1991, amended 2003/2011/2019) regulates coastal activities within 500 m of HTL in four categories (CRZ-I to IV), with CRZ-I as ecologically sensitive and CRZ-III having No Development Zones of 50–200 m.

Previously Asked Questions

- **Q5** (1 mark, Aug 2023, May 2024) ★ — Expand CRZ.

Unit Recap (from SLM)

- 97%+ water saline; rivers only 0.0001% — use judiciously; cryosphere = frozen hydrosphere
- Uses: agriculture 70%, industry 22%, domestic 8%; physical vs economic scarcity
- Threats: groundwater (landfills, mining, nitrates) and surface (point/non-point pollution)
- Floods: flash, seasonal, coastal; Droughts: meteorological, agricultural, hydrological

- Conservation: rainwater harvesting, watershed management, groundwater recharge
- CRZ 1991: 500 m from HTL; categories I–IV; TN compulsory rooftop harvesting

Unit 5: Land and Mineral Resources

1.5.1–1.5.2 Land Resources and Degradation

Theory

Land resources include hills, valleys, plains, river basins and wetlands essential for agriculture and terrestrial ecosystems. Land degradation is undesirable change in biophysical environment value from human-induced processes (excluding natural hazards). IPCC: ~25% ice-free land degraded; UN: ~30% land degraded globally, 12 million ha/year lost (size of Greece). SDG 15 targets land degradation-neutral world by 2030.

Important Points

- Land = essential for agriculture, survival, prosperity, ecosystem maintenance
- **Causes:** deforestation, clear-cutting, nutrient depletion, overgrazing, over-irrigation, urban sprawl, quarrying, vehicle off-roading, enlarged field sizes
- Up to 4% world's agricultural land seriously degraded
- Soil erosion rate 10–100× higher than soil formation rate (conventional tillage)
- Humankind must act as stewards, not exploiters, for unborn generations

For Exam

Land degradation is undesirable human-induced decline in land's biophysical value affecting agricultural productivity, environment and food security. Causes include deforestation, overgrazing, over-irrigation, urbanization and quarrying. About 30% of global land is degraded, losing 12 million hectares annually. SDG 15 aims for land degradation neutrality by 2030.

1.5.3–1.5.5 Landslides, Soil Erosion and Desertification

Theory

Human-induced landslides (HIL) result from topography modification, water circulation changes, land use changes and infrastructure ageing. Soil erosion loses >36 billion tons fertile soil/year from agriculture. Desertification is reduction/destruction of land's biological potential creating desert-like conditions — affects ~25% of world's land; caused by human activities, not natural expansion of deserts.

Important Points

- **Landslide triggers:** deforestation, cultivation, construction, vibrations, blasting, mining, earthwork, logging, urbanization, LU/LC changes
- **Erosion types:** sheet (invisible, light soil), wind (visible), rill (heavy rain, hillside), gully (large channels), ephemeral (natural depressions)
- **Desertification causes:** over-cultivation, over-grazing, over-irrigation (salination), deforestation, mining rejects, population growth/land encroachment
- **Desertification consequences:** productivity loss, poverty, water resource drying, biotic/abiotic damage — largely irreversible
- Arthur Young: "God sleeps in minerals, awakens in plants, walks in animals, and thinks in man"

Soil Erosion (Detailed)

- 36 billion tons fertile soil lost annually from world agriculture
- Sediment from sheet/rill erosion enters streams and waterways
- Controlling sediment integral to soil management and water quality
- Conventional tillage: erosion rate 10–100× higher than soil formation rate
- Light-coloured soils indicate years of invisible sheet erosion damage

Desertification (Detailed)

- Affects ~25% of world's land area; fertile land near existing deserts most vulnerable
- **Over-cultivation:** cash crops displace traditional agriculture to marginal lands
- **Over-grazing:** cattle remove vegetation; wind blows away loose fertile soil particles

- **Over-irrigation:** waterlogging → salination → soil unfit for cultivation
- **Deforestation:** exposes topsoil to erosion by wind and water
- **Mining rejects:** disposal on fertile land causes total soil erosion
- **Population growth:** land encroachment increases pressure on fragile ecosystems
- Consequences largely irreversible; especially severe in developing countries

Soil Erosion Control Methods (SLM — PYQ Q66)

- **Afforestation/Reforestation** — tree cover binds soil; forests provide anti-erosion benefits (SLM 1.3.4–1.3.6)
- **Wind breaks / shelter belts** — social forestry on dry farmlands (SLM 1.3.6)
- **Vegetation cover maintenance** — prevents loose fertile soil particles blowing/washing away
- **Agroforestry** — trees + crops improve soil fertility and erosion control
- **Controlling sediment runoff** — integral to soil management (SLM 1.5.4)
- **PYQ Q66 For Exam:** Methods to control soil erosion include **afforestation and reforestation** (tree cover binds soil — SLM: forests provide anti-erosion and soil protection), **wind breaks/shelter belts** on farmlands (social forestry), **maintaining vegetation cover**, **agroforestry** (trees with crops), and **controlling sediment runoff**. Erosion types (sheet, wind, rill, gully) are reduced by vegetative cover and proper land management.

For Exam

Man-induced landslides result from deforestation, mining, construction and land use changes. Soil erosion (>36 billion tons/year lost) includes sheet, wind, rill and gully types. Desertification degrades ~25% of Earth's land through over-cultivation, over-grazing, over-irrigation, deforestation and mining — reducing productivity irreversibly and causing poverty in developing nations.

Previously Asked Questions

- **Q47** (2 marks, Aug 2023) — Why do landslides happen?
- **Q53** (2 marks, Aug 2023) — What are the causes of desertification?
- **Q54** (2 marks, Aug 2023) — What is soil erosion?

- **Q66** (2 marks, Apr 2025, Aug 2023) ★ — What are the methods to control soil erosion?

Diagram (refer SLM): Fig. 1.5.1 August 2020 Landslide in Idukki, Kerala.

1.5.6 Mineral Resources

Theory

Minerals have been used since ancient times (water, salt, stone tools). Use depends on deposits and properties — gold/silver/platinum (jewellery), copper (coins, pipes, wires), silicon from quartz (computers), aluminium (aircraft, shipping). Exploitation accelerated in the 20th century; ~80% world energy from fossil fuel extraction. Mining causes degradation, water/air pollution, deforestation and forced migration.

Important Points

- **Environmental effects of mining:** (1) Land degradation (2) Surface/groundwater pollution (3) Vegetation leaching (4) Occupational health hazards (5) Air pollution (6) Deforestation (7) Rehabilitation of displaced (8) High-grade mineral depletion (9) Forced migration (10) Topsoil loss (11) Soil erosion (12) Ozone depletion (13) Environmental pollution (14) Natural hazards
- Geological Survey of India (GSI) maps and explores minerals
- **Most abundant Earth crust element: Oxygen;** silicon from quartz for computer industry
- Mining = process of extracting minerals from buried rocks

Major Minerals and Uses

- **Gold, Silver, Platinum:** Ores in crust — Jewellery, electronics, investment
- **Copper:** Sulphide ores — Coins, electric wires, pipes, alloys
- **Iron:** Hematite, magnetite — Steel, weapons, buildings, machinery
- **Silicon:** Quartz — Computer chips, glass, solar cells
- **Aluminium:** Bauxite — Aircraft, shipping, cars, packaging
- **Coal:** Sedimentary basins — Thermal power, coke for steel, chemicals

- **Petroleum:** Sedimentary rocks 500+ m deep — Transport fuel (40% commercial energy), plastics
- **Natural Gas:** Associated with oil deposits — Heating, cooking, electricity (21% commercial energy)
- **Uranium:** Rare-earth minerals, granitic rocks (2 ppm) — Nuclear power (17% world electricity)

For Exam

Mineral resources (gold, copper, iron, silicon) are essential for industries and development. Mindless exploitation causes land degradation, water and air pollution, deforestation, health hazards and forced migration of affected populations. About 80% of world energy consumption depends on fossil fuel extraction, with severe environmental consequences at every stage from mining to disposal.

Previously Asked Questions

- **Q34** (1 mark, Apr 2025) — What is the most abundant element on the earth's crust?

Unit Recap (from SLM)

- Land resources: hills, plains, wetlands; degradation from human-induced processes
- ~30% global land degraded; 12 million ha/year lost; SDG 15 target 2030
- Landslides: deforestation, mining, construction; erosion types: sheet, wind, rill, gully
- Desertification: over-cultivation, grazing, irrigation, deforestation — affects ~25% land
- Minerals: gold, copper, silicon (quartz); mining causes pollution, deforestation, displacement
- GSI responsible for mineral mapping; soil erosion >36 billion tons/year

Unit 6: Energy Resources1.6.1–1.6.2 Types and Classification of Energy

Theory

Energy is the capacity to perform work; total usable energy = energy supply. Chief sources: fossil fuels, flowing water, biomass, atomic minerals, plus solar, wind, tidal, chemical and geothermal. Sources classified as renewable (hydro, solar, wind — available repeatedly) or non-renewable (coal, oil, gas, nuclear, geothermal — permanently consumed). ~85% world commercial energy from coal, oil, natural gas.

Important Points

- **Availability:** Infinite; naturally replenished — Finite; permanently consumed
- **Examples:** Solar, wind, hydro, tidal, biogas — Coal, petroleum, natural gas, nuclear, geothermal
- **Environmental impact:** Low pollution; clean energy — High pollution; SO₂, NO_x, CO₂, ash, radioactive waste
- **Cost:** High initial setup; low running cost — Lower setup historically; rising as reserves deplete
- **Depletion:** Never exhausted at human scale — Exhaustible; coal, oil, gas will run out
- **Regeneration time:** Immediate to seasonal (wind 5 months) — Millions of years (fossil fuels from prehistoric organisms)
- **Share of commercial energy:** Growing but <15% globally — ~85% (coal + oil + gas)
- **Sustainability:** Sustainable long-term — Unsustainable at current consumption rates
- **Geographic dependence:** Site-specific (solar needs sunlight, wind needs wind) — Widespread where deposits exist
- **By-products:** None/minimal (biogas → manure) — Fly ash, oil spills, mine waste, nuclear waste
- **India examples:** Solar parks (Rajasthan, Gujarat), wind (TN, Maharashtra), biogas (rural) — Coal (Jharia, Singrauli), oil (Bombay High), uranium (Jaduguda)
- **Fossil Fuels:** Coal, Oil (Petroleum), Natural Gas — 85% commercial energy; coal types: Anthracite > Bituminous > Sub-bituminous > Lignite; peat precursor; oil = 40% commercial energy; gas = 21%, cleanest fossil fuel (85% methane)

- **Solar & Wind:** Photovoltaic, solar thermal, windmills — Solar: unlimited, pollution-free; expensive cells; wind: 5-month generation, cheap power, heavy installation cost
- **Biogas & Tidal:** Organic waste methane, tidal dams — Biogas from sewage/manure; community/family plants; tidal power expensive, seasonal
- **Nuclear:** Uranium-235, Plutonium-239 fission — 17% world electricity; 1 kg U = 25,000 kg coal energy; waste disposal problem
- **Geothermal:** Earth's interior heat — Iceland, Italy, Japan, Philippines, NZ, USA; pipes + turbines
- **Hydrogen:** Future fuel/emissions-free carrier — Most abundant element; combusts to water vapour; hydrogen economy needs renewable electricity
- Coal: thermal power plants; low sulphur = less pollution; ash content affects efficiency; 1 MW plant needs ~1 acre for ash dump
- Clean energy: renewables + energy efficiency; biomass/large hydro have wildlife tradeoffs
- Less than 5% electricity from solar, wind, hydro currently; shift to renewables needed for hydrogen economy

Coal Types

Coal Type	Carbon Content	Key Feature	Main Use
Anthracite	Highest	Hardest, rarest; most heat	Premium fuel, steel industry
Bituminous	High	Most common industrial coal	Thermal power, coke production
Sub-bituminous	Medium	Less carbon than bituminous	Power generation
Lignite	Lowest	Brown coal; high moisture	Local power plants
Peat	Precursor	Partially decayed swamp matter	Precursor to lignite

Solar Energy (Detailed)

- Two conversion methods: (1) Photovoltaic (solar cells — silicon crystals) (2) Solar thermal (copper pipe on blackened base under glass)

- Solar energy falling every 29 seconds = one day's human energy requirement
- Advantages: unlimited, no air/water/thermal/noise pollution, no disasters, conserves resources, technology available
- Disadvantages: expensive cells, sparse distribution, needs collection/concentration devices, cannot fully replace conventional fuels
- Applications: water pumping, refrigeration, solar ponds, building temperature control

Wind Energy (Detailed)

- Windmill blades → turbine → power generator
- Generation period: ~5 months (seasonal wind availability)
- 200 kW/250 kW generator produces 6–7 lakh units/year
- Merits: immediate power after commissioning, cost-free generation, low recurring cost, pollution-free, income source
- Demerits: heavy installation cost, expensive maintenance, seasonal availability

Biogas & Nuclear (Detailed)

- Biogas: methane from anaerobic decomposition of sewage, garbage, manure, crop residues in sealed containers
- Plant types: community, industrial, night soil, family size, improved chulhas
- Residue after biogas = high-quality organic manure
- Nuclear: 17% world electricity; U-235 and Pu-239 fission; 1 kg U = 25,000 kg coal; fuel bundles last 1–2 years
- Nuclear waste: radioactive liquids and gases severely affect environment; disposal is major challenge
- Geothermal: Iceland, Italy, Japan, Philippines, NZ, USA — pipes into wells connected to turbines
- Hydrogen: most abundant element; combusts to H₂O; PEM fuel cells (Honda FCX Clarity: 1 kg = 68 miles); needs renewable electricity for sustainable production

Biogas — What it is & Energy Production (SLM 1.6.2.3 — PYQ Q92)

- **Definition:** Biogas = methane (CH_4) gas produced/released from organic wastes — sewage, garbage, manure, crop residues
- **Process:** Anaerobic decomposition — wastes decompose in the **absence of air** inside sealed containers (similar to natural gas in origin)
- **Production setup:** Container filled with organic waste (mainly animal dung/excreta), closed at top with **gas tap** for trapping methane
- **Energy production:** Methane collected from tap → used as fuel for **cooking, lighting, and power generation** — converts waste into usable energy
- **By-product (manure):** After gas production, spent slurry = **high-quality organic manure** for croplands (dual benefit: energy + fertiliser)
- **Plant types:** Community biogas plants, industrial biogas plants, night soil biogas plants, family-size biogas plants, improved chulhas — operational worldwide
- **India:** Widely used in rural areas; Haritha Keralam promotes biogas plants for waste-to-energy
- **Advantages:** Renewable non-conventional source; uses waste; reduces pollution; low-cost rural energy; produces manure alongside fuel
- **PYQ Q92 For Exam:** Biogas is methane (CH_4) gas produced from organic wastes such as sewage, garbage, manure and crop residues through anaerobic decomposition (in the absence of air) — similar to natural gas. For energy production, organic wastes (mainly animal dung/excreta) are placed in a sealed container with a gas collection tap; bacterial decomposition releases methane which is trapped and used as fuel for cooking, lighting and power generation. After gas production, the spent material serves as high-quality organic manure for agriculture. Community, industrial, night soil and family-size biogas plants operate worldwide including rural India. Biogas is a renewable, non-conventional energy source that converts waste into useful fuel while reducing environmental pollution.
- **PYQ Q81 For Exam:** **Renewable energy** (solar, wind, hydro, tidal, biogas, geothermal) can be used repeatedly and is inexhaustible on human timescale. **Non-renewable energy** (coal, petroleum, natural gas, nuclear) exists in limited quantities and gets exhausted. Fossil fuels supply ~85%

commercial energy. Coal: anthracite > bituminous > lignite. Solar/wind are clean but costly; biogas produces CH₄ from organic waste.

For Exam

Energy sources are renewable (solar, wind, hydro — reusable) or non-renewable (fossil fuels, nuclear — exhaustible). Fossil fuels provide 85% commercial energy: coal (4 types: anthracite to lignite), petroleum (40%) and natural gas (21%, cleanest). Non-conventional sources include solar, wind, biogas, tidal, geothermal and nuclear energy. Hydrogen is proposed as a future emissions-free fuel requiring renewable energy for sustainable production.

Previously Asked Questions

- **Q31** (1 mark, Apr 2025) — Name one non-renewable source of energy?
- **Q81** (5 marks, Aug 2023) — Differentiate renewable and nonrenewable energy resources.
- **Q92** (5 marks, May 2024) — What is biogas? Comment on the production of energy from biogas.

Diagram (refer SLM): Fig. 1.6.1 Energy Sources; Fig. 1.6.2 Solar Energy; Fig. 1.6.3 Wind Energy; Fig. 1.6.4 Geothermal Energy.

1.6.2.1–1.6.2.4 Energy Sources — Details

Theory

Coal is altered plant residue under sediment pressure; burned for steam/electricity or converted to coke for iron/steel. Petroleum (petro=rock, oleum=oil) is complex hydrocarbon mixture from organic-rich sediments at 500+ m depth. Biogas is methane from anaerobic decomposition of organic wastes. Nuclear reactors use fissionable U-235/Pu-239 with controlled chain reactions.

Important Points

- **Coal pollution:** SO₂, NO_x emissions; fly ash/bottom ash solid waste; mining noise, dust, trace elements
- **Solar advantages:** unlimited supply, no pollution, no large-scale disasters, conserves resources; disadvantage: expensive, cannot fully replace conventional fuels

- **Wind merits:** quick commissioning, cost-free power, pollution-free; demerits: seasonal (5 months), heavy installation/maintenance
- **Biogas plants:** community, industrial, night soil, family size; residue = organic manure
- **Nuclear:** clean/compact but radioactive waste severely affects environment; fuel bundles last 1–2 years
- **Geothermal:** only where hot rocks near surface; simple pipe + turbine method

For Exam

Coal (anthracite highest carbon content) powers thermal plants but pollutes via SO_2/NO_x and ash waste. Petroleum provides 40% commercial energy as hydrocarbons from sedimentary deposits. Solar and wind are renewable and pollution-free but face cost/seasonal limitations. Biogas from organic waste provides methane fuel plus manure. Nuclear energy (17% world electricity) is concentrated but creates hazardous radioactive waste.

Unit Recap (from SLM)

- Energy = capacity to perform work; supply = total usable energy available
- Renewable: hydro, solar, wind; Non-renewable: coal, oil, gas, nuclear, geothermal
- Fossil fuels: 85% commercial energy; coal types: anthracite, bituminous, sub-bituminous, lignite
- Solar/wind: clean but costly; biogas: methane from organic waste; tidal: seasonal
- Nuclear: 17% electricity; U-235/Pu-239; 1 kg U = 25,000 kg coal
- Hydrogen: future emissions-free fuel; needs renewable energy infrastructure

Block 2: Ecosystems, Biodiversity and Conservation

Unit 1: Concept, Structure and Function of an Ecosystem

2.1.1 Concept, Structure and Function

Theory

Ecology (Ernst Haeckel, 1869; Greek oikos=home, logos=study) examines relations between organisms and surroundings. An ecosystem is a functional unit of nature where living organisms interact among themselves and with the physical environment. Structure arises from abiotic-biotic interactions varying over space and time. AG Tansley proposed the term ecosystem.

Important Points

- **Ecology** = study of natural environment including organism relations to each other and surroundings
- Ecosystem = basic functional unit of ecology; interrelating, interdependent, ever-changing system
- Two components: **Abiotic** (non-living) and **Biotic** (living)
- Habitat = natural residence of every organism; biodiversity = variety of living organisms

For Exam

Ecology (coined by Ernst Haeckel, 1869) studies relationships between organisms and their environment. An ecosystem is a functional unit where biotic and abiotic components interact. Ecosystem structure results from these interactions across space and time. The ecosystem is the basic functional unit of ecology, divided into abiotic and biotic components.

Previously Asked Questions

- **Q17** (1 mark, May 2024) — Name the system that comprise living organisms, their physical environment and all...
- **Q33** (1 mark, Apr 2025) — Who coined the term ecology?

2.1.1.1–2.1.1.2 Abiotic and Biotic Components

Theory

Abiotic components are physical/non-living factors essential for organism survival — inorganic substances (CO₂, water, N, Ca, P in material cycles = standing state), organic compounds (proteins, carbohydrates synthesized by biota), climatic factors (rain, light, temperature, humidity, wind) and edaphic

factors (minerals, soil, pH, topography). Biotic components are categorized by nutrition into producers, consumers and decomposers.

Important Points

- **Producers:** Photoautotrophs, chemoautotrophs — Green plants, algae, phytoplankton; photosynthesis/chemosynthesis; primary production; first food chain link
- **Consumers:** Primary, Secondary, Tertiary, Quaternary — Heterotrophs lacking chlorophyll; herbivores, carnivores, omnivores, detritivores; humans = secondary consumers (omnivores)
- **Decomposers:** Saprotrophs — Bacteria, fungi, worms, insects; break dead organic matter to inorganic nutrients for producers; complete nutrient recycling
- Standing state = amount of inorganic substances at given time in ecosystem
- Chemosynthetic bacteria (sulphur bacteria) synthesize food without sunlight
- Decomposers: bacteria, protozoa, earthworms; essential for nutrient recycling

Consumer Types (Detailed)

- **Primary consumers:** Producers (plants) directly — Zooplankton, butterflies, rabbits, giraffes, pandas, elephants, deer
- **Secondary consumers:** Primary consumers — Ants, snakes, rats, crabs, lions, blue whales (via krill), humans (omnivores)
- **Tertiary consumers:** Secondary consumers — Top carnivores feeding on smaller carnivores
- **Quaternary consumers:** Tertiary consumers — White shark, polar bear, alligator — apex of food chain
- **Herbivores:** Plants only — Cow, deer, caterpillar
- **Carnivores:** Other animals — Tiger, eagle, spider
- **Omnivores:** Both plants and animals — Bear, crow, human
- **Detritivores:** Dead organic matter — Earthworm, millipede, maggot

- **Biotic components:** Living organisms — **Producers, Consumers, Decomposers** (classified by mode of nutrition)
- **PYQ Q68 For Exam:** **Biotic components** are living parts of an ecosystem: **Producers** (autotrophs — green plants, algae, phytoplankton; photosynthesis); **Consumers** (heterotrophs — primary/herbivores, secondary/carnivores, tertiary/quaternary; e.g. deer, tiger); **Decomposers** (saprotrophs — bacteria, fungi, earthworms; recycle dead matter to inorganic nutrients). They form trophic levels in food chains.
- **PYQ Q94 For Exam:** **Abiotic components** are non-living factors: **(1) inorganic substances** — CO₂, water, nitrogen, minerals; **(2) organic compounds** — proteins, carbohydrates; **(3) climatic factors** — rainfall, temperature, humidity, light, wind; **(4) edaphic (soil) factors** — texture, pH, topography. Examples: sunlight, water, air, minerals, rocks, temperature. They regulate energy flow and nutrient cycling in ecosystems.

For Exam

Abiotic components include inorganic substances (CO₂, water, nitrogen in material cycles), organic compounds, climatic factors (rain, temperature, humidity) and edaphic factors (soil, pH). Biotic components are producers (autotrophs making food via photosynthesis), consumers (heterotrophs — primary to quaternary) and decomposers (saprotrophs recycling dead matter to inorganic nutrients for producers).

Previously Asked Questions

- **Q68** (2 marks, Apr 2025, May 2024) ★ — What do you mean by biotic components of an ecosystem? Give examples.
- **Q94** (5 marks, May 2024) — What are the different abiotic components of an ecosystem? Give examples.

2.1.2–2.1.4 Food Chain, Food Web and Ecological Pyramids

Theory

A food chain is a linear sequence transferring nutrients and energy from producer through consumers to decomposers at trophic levels. Two types: grazing food chain (plants → herbivores → carnivores) and detritus food chain (dead organic matter → decomposers → carnivores). Food webs are

interconnected food chains showing complex energy flow. Ecological pyramids (Charles Elton, 1927) graphically represent trophic relationships.

Important Points

- **Grazing food chain:** Starts with green plants; energy from photosynthesis; plants → herbivores → carnivores; most ecosystems follow this
- **Detritus food chain:** Starts with dead organic matter; decomposers → detritivores → carnivores; pyramid of numbers may be inverted
- **Food web:** Multiple interconnected chains; more complex = more sustainable; multiple energy flow paths
- **Pyramid of Numbers:** Organism count per trophic level; usually upright except detritus chains
- **Pyramid of Biomass:** Mass per trophic level; upright except oceans (zooplankton > phytoplankton biomass)
- **Pyramid of Energy:** Energy per trophic level; **always upright**; only ~10% energy transfers to next level (90% lost as heat, respiration)

Ecological Pyramids — Types

Pyramid Type	Based On	Shape	Key Point
Numbers	Organism count per trophic level	Usually upright; inverted in detritus chains	May be inverted in detritus food chains
Biomass	Total mass per trophic level	Upright on land; inverted in oceans	Zooplankton biomass > phytoplankton in oceans
Energy	Energy content per trophic level	Always upright	Only ~10% energy transfers to next level (Elton, 1927)

- **PYQ Q90 For Exam:** Ecological pyramids (Charles Elton, 1927) graphically represent trophic relationships. **Three types: (1) Pyramid of Numbers** — organism count per trophic level; usually upright except detritus chains; **(2) Pyramid of Biomass** — mass per level; upright on land but inverted in oceans; **(3) Pyramid of Energy** — energy per level; **always upright** because only ~10% energy transfers to the next level (90% lost as heat and respiration).

- Trophic level = position in food chain (producers at bottom)
- Apex predators receive only ~0.01% of starting energy
- 10% law: only ~10% energy creates biomass at next trophic level

Food Chain — Concept (SLM 2.1.2 — PYQ Q95)

- **Definition:** Linear sequence of organisms where **nutrients and energy** are transferred from one organism to another (one organism consumes another)
- **Feeding relationship:** Shows order of events in ecosystem — who eats whom
- **Flow:** Begins with **producer** (autotroph) → passes through consumers at different **trophic levels** → ends with **decomposer**
- **Trophic levels:** Producers (bottom) → Primary consumers (herbivores) → Secondary consumers (carnivores) → Tertiary/quaternary consumers
- **Unidirectional:** Energy flows in a **single linear path** — not recycled back up the chain

Grazing vs Detritus Food Chain

Type	Starts With	Energy Source	Example
Grazing food chain	Green plants (producers)	Photosynthesis	Grass → Grasshopper → Frog → Snake → Hawk
Detritus food chain	Dead organic matter	Decomposers on detritus	Dead leaves → Fungi/Bacteria → Earthworm → Bird

- **10% energy rule:** Only ~10% energy passes to next trophic level; 90% lost as heat, respiration, hunting effort

Food Chain Examples

- **Grazing (grassland):** Grass → Grasshopper → Frog → Snake → Hawk — Producer → Primary → Secondary → Tertiary → Quaternary
- **Grazing (aquatic):** Phytoplankton → Zooplankton → Small fish → Large fish → Shark — Producer → Primary → Secondary → Tertiary → Apex
- **Grazing (forest):** Tree leaves → Deer → Tiger — Producer → Primary herbivore → Apex carnivore

- **Detritus (forest floor):** Dead leaves → Fungi/Bacteria → Earthworm → Bird
— Detritus → Decomposer → Detritivore → Carnivore
- **PYQ Q95 For Exam:** A food chain is a **linear sequence** of organisms through which nutrients and energy are transferred when one organism consumes another — showing feeding relationships in an ecosystem. It begins with **producers** (green plants/autotrophs) at the first trophic level, passes through **primary consumers** (herbivores), **secondary and tertiary consumers** (carnivores), and ends with **decomposers**. Two types: **(1) Grazing food chain** — starts with green plants; energy from photosynthesis; plants → herbivores → carnivores (most common); **(2) Detritus food chain** — starts with dead organic matter; decomposers → detritivores → carnivores. Energy flows **unidirectionally** along a single path; only ~10% transfers to the next level. Examples: Grass → Grasshopper → Frog → Snake → Hawk; Phytoplankton → Zooplankton → Fish → Shark; Dead leaves → Fungi → Earthworm → Bird.

Food Web Examples

- **Pond ecosystem:** Algae eaten by zooplankton, tadpoles and small fish; small fish eaten by large fish and birds; large fish eaten by snakes; all dead matter decomposed by bacteria — multiple interconnected chains
- **Grassland ecosystem:** Grass supports mice, rabbits, insects; mice/rabbits eaten by snakes, foxes, hawks; snakes also eat frogs that eat insects; hawk also scavenges — one organism in multiple chains
- **Ocean ecosystem:** Phytoplankton → krill → whale; phytoplankton → small fish → tuna → shark; seaweed → sea urchin → otter — complex cross-feeding
- Food web complexity increases ecosystem sustainability; single chain failure in web does not collapse entire system

Ecological Pyramid — Numerical Example (10% Energy Rule)

- Producers (plants): 100% energy captured from sun
- Primary consumers (herbivores): receive ~10% of producer energy
- Secondary consumers (carnivores): receive ~1% (10% of 10%)
- Tertiary consumers: receive ~0.1%
- Apex predators: receive only ~0.01% of starting energy

- 90% energy lost at each level through respiration, hunting, heat to surroundings
- This is why productivity/energy pyramids are always upright
- Explains why top predators are few in number compared to producers
- **PYQ Q110 For Exam: Food chain = linear sequence** transferring energy when one organism eats another — single path, unidirectional (producers → consumers → decomposers). **Food web = network of interconnected chains** — multiple paths, more complex and **more sustainable**. If one species is lost from a food web, alternate paths exist; a broken food chain collapses entirely. Food webs reflect real ecosystem feeding more accurately.

For Exam

Food chains transfer energy linearly through trophic levels: grazing chains start with producers, detritus chains start with dead matter. Food webs are interconnected chains providing more sustainable ecosystems. Ecological pyramids (Elton, 1927) show trophic relationships — numbers, biomass and energy types. Pyramid of energy is always upright because energy decreases ~90% at each level due to heat loss and respiration.

Previously Asked Questions

- **Q6** (1 mark, Aug 2023) — Explain food web.
- **Q21** (1 mark, May 2024) — Name the graphical representation which shows relationship between different livin...
- **Q90** (5 marks, Aug 2023) — Write a note on types of ecological pyramid.
- **Q95** (5 marks, May 2024) — Briefly describe the concept of food chain.
- **Q110** (5 marks, Apr 2025, Aug 2023, May 2024) ★🔥 — Differentiate food chain and food web.

Diagram (refer SLM): Fig. 2.1.1 Terrestrial and Aquatic Food Chains; Fig. 2.1.2 Pyramid of Numbers; Fig. 2.1.3 Pyramid of biomass in oceans; Fig. 2.1.4 Pyramid of Energy; Fig. 2.1.5 Ecological pyramid.

Unit Recap (from SLM)

- Ecology: Haeckel 1869; ecosystem = basic functional unit; abiotic + biotic components

- Producers (autotrophs), consumers (heterotrophs), decomposers (saprotrophs)
- Food chain: linear energy transfer; food web: interconnected chains
- Grazing chain (plants first) vs detritus chain (dead matter first)
- Ecological pyramids: numbers, biomass, energy; energy pyramid always upright (~10% rule)
- Decomposers recycle nutrients completing the life cycle

Unit 2: Basic Awareness on Various Ecosystems

2.2.1 Terrestrial and Aquatic Ecosystems

Theory

An ecosystem includes all living and non-living things in a specific natural setting. Ecosystems vary from small ponds to global biosphere. Divided into terrestrial (land-based, ~140–150 million km², 25–30% Earth surface) and aquatic (water-based). Distribution influenced by temperature and precipitation. Major types: forests, grasslands, deserts, tundra, freshwater and marine.

Important Points

Terrestrial vs Aquatic Ecosystems

Ecosystem	Type	Coverage / Area	Key Feature
Forest	Terrestrial	30–35 million km ²	Highest species diversity; tropical/temperate/boreal
Grassland	Terrestrial	Prairies, savannas, steppes	Semi-arid; sparse trees; grazing animals
Desert	Terrestrial	~17% Earth surface	<25 cm rain/year; hot (Sahara) or cold (Ladakh)
Freshwater	Aquatic	~0.8% Earth surface	Lentic (standing) and lotic (flowing); high O ₂ in fast flow
Marine	Aquatic	~70% Earth surface	Saltwater; estuaries, coral reefs, mangroves
Wetland	Aquatic	Transitional land–water	“Kidneys of the Earth”; storm barriers, water cleansing

- Biome = large geographic terrestrial ecosystem (e.g. tundra)
- Global biosphere = composite of all local ecosystems
- Wetlands: transitional between terrestrial and aquatic; sensitive to human disturbance
- Ladakh = cold desert example; Sahara = hot desert

Terrestrial Ecosystems (Detailed)

Ecosystem	Climate	Vegetation	Animals
Tropical rainforest	High rainfall, constant warmth	Dense canopy, lianas, epiphytes	Orangutan, jaguar, toucan
Temperate deciduous	Four seasons, moderate rain	Oak, maple — shed leaves in autumn	Deer, fox, bear, squirrels
Boreal (taiga)	Cold, long winters	Coniferous — pine, spruce, fir	Moose, lynx, wolf
Tropical grassland (savanna)	Wet/dry seasons	Tall grasses, scattered acacia	Elephant, giraffe, zebra, lion
Temperate grassland	Semi-arid	Short grasses, few trees	Bison, prairie dog, coyote
Hot desert	<25 cm rain, extreme heat	Cactus, acacia, date palm	Camel, fox, scorpion, lizard
Cold desert	<25 cm rain, extreme cold	Sparse drought-resistant shrubs	Yak, ibex, snow leopard (Ladakh)

Grassland Ecosystem — Features (SLM 2.2.1.2 — PYQ Q93)

- **Dominance:** Most dominant type of vegetation; occurs naturally in several parts of the world
- **Location/types:** Prairies, savannas, steppes — found in tropical or temperate regions; also colder areas (e.g. Siberian steppe)
- **Climate:** Semi-aridity — pleasant atmosphere; climatic conditions similar to semi-arid regions
- **Vegetation:** Mainly **grasses** with some shrubs and trees; trees sparse or nonexistent; flowers may be interspersed with grasses
- **Ideal for:** Grazing animals — perfect region for herbivorous, insectivorous organisms

- **Animal diversity:** Elephants, giraffes, hyenas, jackrabbits, lions, rhinos, warthogs, zebras, bison, prairie dogs, coyotes
- **Two types:** (1) **Tropical grasslands (savanna)** — wet/dry seasons; tall grasses, scattered acacia; elephant, giraffe, zebra, lion, hyena (2) **Temperate grasslands (steppe/prairie)** — semi-arid; short grasses, few trees; bison, prairie dog, coyote
- **Food chain example:** Grass → Grasshopper → Frog → Snake → Hawk (producer → primary → secondary → tertiary → quaternary consumers)
- **Food web:** Grass supports mice, rabbits, insects; eaten by snakes, foxes, hawks — one organism may appear in multiple chains
- **PYQ Q93 For Exam:** Grasslands are the most dominant vegetation type, found naturally in prairies, savannas and steppes across tropical, temperate and even colder regions (e.g. Siberian steppe). They share a **semi-arid climate** with sparse or no trees and grasses as the main vegetation (some shrubs, trees and flowers interspersed). Grasslands provide an ideal environment for **grazing animals** — elephants, giraffes, lions, zebras, hyenas, bison, prairie dogs and coyotes. Two main types: **tropical grasslands (savanna)** with wet/dry seasons, tall grasses and scattered acacia; and **temperate grasslands (steppe/prairie)** with short grasses and few trees. Organisms are mainly herbivorous and insectivorous, linked through grazing food chains and food webs.

Aquatic Ecosystems (Detailed)

Type	Category	Key Feature	Examples
Freshwater	Lentic (standing)	Low flow; ponds, lakes	Lake Baikal, Dal Lake
Freshwater	Lotic (flowing)	High dissolved O ₂	Ganga, Amazon; fish, otters
Marine	Open ocean	High salinity; 70% Earth	Sharks, whales, corals
Marine	Estuary	Brackish; river meets sea	Crabs, migratory birds
Marine	Coral reef	Shallow warm waters	Highest marine biodiversity
Wetland	Transitional	Anaerobic soils; water purification	Marshes, swamps, mangroves

- **PYQ Q104 For Exam:** **Aquatic ecosystems** are water-based. **Marine** (~70% Earth): oceans, coral reefs, estuaries — high salinity. **Freshwater** (0.8%): **lentic** (ponds, lakes — standing) and **lotic** (rivers — flowing). **Wetlands** (marshes, swamps) — transitional; called **"Kidneys of the Earth"** for water purification and flood control. **Estuarine** — river meets sea, high productivity. Provide fisheries, water supply and biodiversity.

For Exam

Ecosystems are terrestrial (forest, grassland, desert, tundra — 25–30% Earth surface) or aquatic (freshwater 0.8%, marine 70%). Forests have highest species diversity; deserts cover 17% with <25 cm annual rainfall. Freshwater divides into lentic (standing) and lotic (flowing) waters. Wetlands are transitional ecosystems called "Kidneys of the Earth" providing storm barriers and water cleansing.

Previously Asked Questions

- **Q19** (1 mark, May 2024) — Which type of ecosystem is featured with low rainfall, generally less than 25 cent...
- **Q32** (1 mark, Apr 2025) — What are the major types of ecosystems?
- **Q69** (2 marks, Apr 2025) — What are the different types of terrestrial ecosystem?
- **Q93** (5 marks, May 2024) — Describe the features of grassland ecosystem.
- **Q104** (5 marks, Apr 2025) — Describe aquatic ecosystems.

Diagram (refer SLM): Fig. 2.2.1 Wetland Ecosystems.

Unit Recap (from SLM)

- Two categories: terrestrial (land) and aquatic (water)
- Terrestrial: forest, grassland, desert, tundra; distribution related to precipitation/temperature
- Forests: 30–35 million km²; grasslands: semi-arid, grazing animals
- Deserts: <25 cm rain, 17% Earth; freshwater 0.8%, marine 70%
- Freshwater: lentic (standing) and lotic (flowing); wetlands = transitional
- Wetlands called "Kidneys of the Earth"; sensitive to disturbance

Unit 3: Definition and Levels of Biodiversity

2.3.1–2.3.3 Biodiversity and Its Levels

Theory

Biodiversity (coined 1985; Walter Rosen; Bio + diversity) describes richness and variety of life — essential for ecosystem health and sustainable livelihood.

Measured by species richness (number of species) and species evenness (relative abundance). Three levels: genetic diversity (within species), species diversity (between species) and ecosystem diversity (between ecosystems).

Alpha, beta and gamma diversity measure diversity at local, between-ecosystem and regional scales.

Important Points

- **Genetic:** Genetic characteristics within a species — Homo sapiens: Chinese, Indian, African; 98.4% genes shared with chimpanzees; gene pool for crop breeding
- **Species:** Species population/total organisms ratio — Decreases from equator to poles; tropics (23.5°N–23.5°S) richest; ~1.8 million documented species
- **Ecosystem:** Variation among ecosystems in a region — Natural vs modified; India exceptionally rich; forests, grasslands, deserts, aquatic, farmland
- **Alpha diversity:** Diversity within one ecosystem — Species richness of that area
- **Beta diversity:** Diversity between ecosystems — Species change between ecosystems
- **Gamma diversity:** Overall regional diversity — Diversity across all ecosystems in region
- **Species richness:** counts species only, not abundance
- **Species evenness:** Shannon-Weiner index (H, entropy), Simpson index (D, 1949, concentration)
- Hotspots = areas rich in species diversity; India among world's 15 exceptionally rich nations

- Genetic diversity prevents inbreeding extinction; wild relatives = gene pool for crops

Biodiversity Measurement (Detailed)

- **Species richness example:** Forest A and B both have 3 species (tigers, deer, rabbits) but Forest A has more even distribution → higher evenness
- **Shannon-Weiner index (H):** measures entropy/surprise in species distribution; higher H = more diverse
- **Simpson index (D, 1949):** measures concentration when individuals classified into types
- **Whittaker (1965):** divided diversity into alpha (within ecosystem), beta (between ecosystems), gamma (regional total)
- **Species diversity formula concept:** ratio of one species population to total organisms; 0 = infinite diversity, 1 = single species
- Tropics (23.5°N–23.5°S) have maximum species richness; decreases toward poles
- Natural undisturbed tropical forests >> plantation forests in species richness
- ~1.8 million species documented; estimated total 1.5–20 billion (most undiscovered)

For Exam

Biodiversity is the variety of life at genetic (within species), species (between species) and ecosystem (between ecosystems) levels. It is measured by species richness (species count) and species evenness (abundance distribution, using Shannon-Weiner or Simpson indices). Species diversity decreases from equator to poles; tropics harbour most species. India is among 15 biodiversity-rich nations with ~1.8 million known species.

Previously Asked Questions

- **Q58** (2 marks, May 2024) — Define biodiversity.

Diagram (refer SLM): Fig. 2.3.1 Species Diversity; Fig. 2.3.2 Biodiversity at different levels.

2.3.4 Values of Biodiversity

Theory

Biodiversity provides essential environmental services — nutrient cycling, soil formation, air/water cleansing, watershed protection, flood reduction. It also has direct and indirect value to humans through consumptive, productive, social, ethical, aesthetic and option values. Only a tiny fraction of known species has been investigated for food, medicinal or industrial potential.

Important Points

- **Consumptive use:** Direct utilization by local communities — Timber for housing, fuel wood for cooking, fodder for cattle, fish by fisherfolk, medicinal plants (neem, tulsi)
- **Productive use:** Commercial/industrial exploitation — Bioprospecting for drugs (Rauvolfia for blood pressure), rubber/oil palm plantations, wild crop relatives for rice/wheat breeding
- **Social values:** Cultural and community dependence — Traditional multi-crop farming as insurance against crop failure; sacred groves (deorais) preserving gene banks; tribal forest livelihoods
- **Ethical/moral:** Right of all species to exist — Indian sacred groves in tribal areas; lion in Hinduism, elephant in Buddhism, eagle in Egyptian civilization
- **Aesthetic:** Beauty, tourism, recreation — Wildlife safaris (Kaziranga rhinos), bird watching (Bharatpur), coral reef diving; forest meditation and eco-tourism
- **Option value:** Future potential not yet realized — Wild rice varieties for drought resistance; Asian wild gaur for cattle breeding; undiscovered medicinal plants in tropical forests
- Biodiversity services: nutrient cycling, soil formation, air/water cleansing, watershed protection, flood reduction, pollination
- Only tiny fraction of ~1.8 million known species investigated for food, medicinal or industrial potential
- Biological prospecting = searching wild genetic properties for crop/livestock/drug development
- Loss of wild relatives = irreversible economic loss to future generations
- **PYQ Q118 For Exam: Values of biodiversity: (1) Consumptive** — timber, fuelwood, fish, medicine; **(2) Productive** — bioprospecting,

pharmaceuticals; **(3) Social** — traditional livelihoods, sacred groves; **(4) Ethical** — right of species to exist; **(5) Aesthetic** — ecotourism, natural beauty; **(6) Option value** — future potential (wild crop relatives). Also provides ecosystem services: nutrient cycling, watershed protection, flood control.

For Exam

Biodiversity values include consumptive (direct local use of timber, food, medicine), productive (commercial bioprospecting, pharmaceuticals, crop breeding), social (traditional community dependence), ethical (right of all species to exist, sacred groves), aesthetic (tourism, natural beauty) and option (future potential for crop/animal improvement). Biodiversity provides essential ecosystem services like nutrient cycling, watershed protection and flood control.

Previously Asked Questions

- **Q118** (10 marks, May 2024) — Describe the values of biodiversity.

Diagram (refer SLM): Fig. 2.3.3 Values of Biodiversity.

Unit Recap (from SLM)

- Biodiversity: variety of life; coined 1985; measured by richness and evenness
- Levels: genetic, species, ecosystem; alpha, beta, gamma diversity
- Species diversity decreases poleward; tropics richest; ~1.8 million known species
- Values: consumptive, productive, social, ethical, aesthetic, option
- Shannon-Weiner (H) and Simpson (D) indices measure evenness
- India among 15 exceptionally biodiversity-rich nations

Unit 4: India as a Mega Diversity Nation

2.4.1–2.4.3 India as Mega Diversity Nation and Hotspots

Theory

India is a mega diversity nation with only 2.4% of world's land area but 7–8% of global flora and fauna (>45,000 plant species, >91,000 animal species). Situated at tri-junction of Afro-tropical, Indo-Malayan and Palearctic realms. Norman Myers (1988) introduced biodiversity hotspots — regions with high endemism under threat. India has 4 of 34 global hotspots.

Important Points

- **Land area:** 2.4% of world
- **Species share:** 7–8% global flora/fauna
- **Plants / Animals:** >45,000 / >91,000 species
- **Mammals / Birds / Reptiles / Plants:** 350 (8th) / 1,200 (8th) / 453 (5th) / 45,000 (15th)
- **Endemic plants:** 18% endemic to India; 1/3 flowering plants endemic
- **Endemic amphibians / lizards:** 62% / 50% endemic
- **Ferns / Orchids / Insects:** 1,022 / 1,082 / 50,000 (13,000 butterflies/moths)
- **Crop diversity:** 30,000–50,000 rice varieties; 27 cattle, 40 sheep, 22 goat, 8 buffalo breeds
- **Genebanks:** 34,000 cereals + 22,000 pulses collected
- **Recognized centre:** Crop diversity — rice, maize, millets and hundreds of crop varieties
- **Biogeographic position:** Tri-junction of Afro-tropical, Indo-Malayan, Palearctic realms
- **Ecosystems:** Forests, grasslands, wetlands, deserts, coastal, marine across diverse climates

India's Four Biodiversity Hotspots (Detailed)

- **Himalaya:** J&K, HP, Uttarakhand, Darjeeling (WB), Sikkim, N Assam, Arunachal (+ Nepal, Bhutan, Tibet, China, Myanmar); Full Indian Himalayan region; Elevation 500 m–8000+ m; world's tallest alluvial grasslands; subtropical broadleaf → temperate conifer → alpine meadows; snow leopard, red panda, Himalayan monal
- **Indo-Burma:** NE India except Assam, Andaman Islands (+ Myanmar, Thailand, Vietnam, Laos, Cambodia, S China); 2,308,815 km² (largest global hotspot); Evergreen, semi-evergreen, mixed deciduous forests; limestone

karst ecosystems with unique reptile/mollusc endemism; hoolock gibbon, clouded leopard

- **Western Ghats & Sri Lanka:** Full Western Ghats: Gujarat to Kanyakumari (Sahyadri, 1,600 km) + Sri Lanka highlands; 1,89,611 km² (1,64,280 km² WG alone = 5% India); Gondwanaland origin; Palakkad Gap 24–30 km; lion-tailed macaque, Nilgiri tahr, Malabar civet; shola forests, 1022 fern + 1082 orchid species
- **Sundaland (SLM also writes Sundalands):** Nicobar Islands (+ Indonesia, Malaysia, Singapore, Brunei, Philippines); ~17,000 islands along equator between Asia and Australia; ~1.6 million km² (Borneo 725,000 km², Sumatra 427,300 km²); Mount Kinabalu 4,101 m (highest peak); 23 active volcanoes; alluvial plains, lakes, swamps, shallow coastal waters; adjacent to Wallacea, Indo-Burma, Philippines hotspots; orangutan, proboscis monkey
- **For Exam (Sundaland / Sundalands short note):** Sundaland is one of India's four biodiversity hotspots (SLM spelling: Sundalands). It includes the Nicobar group of Islands and extends through Indonesia, Malaysia, Singapore, Brunei and the Philippines — about 17,000 islands stretching ~5,000 km along the equator between Asia and Australia, covering ~1.6 million km² (major islands: Borneo 725,000 km², Sumatra 427,300 km²). Adjacent hotspots: Wallacea (east), Indo-Burma (west), Philippines (north). Landscape features high mountains (Mount Kinabalu 4,101 m), 23 active volcanoes, plains, lakes and coastal waters — rich equatorial biodiversity including orangutan and proboscis monkey.
- **Hotspot criteria (CI):** (1) $\geq 1,500$ endemic vascular plant species ($>0.5\%$ world total) (2) Lost $\geq 70\%$ original habitat ($\leq 30\%$ vegetation remains)
- India harbours ~8% of globally known flora; 28% of floral elements endemic to country
- 1022 fern species, 1082 orchid species, 50,000 insect species (13,000 butterflies/moths) in India
- Genebanks: 34,000 cereals + 22,000 pulses collected; 30,000–50,000 traditional rice varieties
- Geological split 70 million years ago created India's unique radiation of species
- 11 biogeographical regions; among 17 mega diversity countries (Mittermeier, 1988)

- 36 global hotspots cover 2.5% land but support 50%+ endemic plants, 43% endemic vertebrates
- **PYQ Q105 For Exam: Hotspot criteria** (CI, Myers 1988): **(1) $\geq 1,500$ endemic vascular plant species; (2) $\geq 70\%$ loss of original primary vegetation.** India has 4 hotspots: **Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland.** Hotspots cover 2.5% land but hold 50%+ endemic plants and 43% endemic vertebrates — priority for conservation.

For Exam

India (2.4% land area) hosts 7–8% of world's species — a mega diversity nation at the tri-junction of three biogeographic realms. It has 4 biodiversity hotspots: Himalaya, Indo-Burma, Western Ghats-Sri Lanka and Sundaland. Hotspots (Myers, 1988) require $\geq 1,500$ endemic plant species and $\geq 70\%$ habitat loss. India has 350 mammals, 1,200 birds, 18% endemic plants and exceptional crop/livestock diversity.

Previously Asked Questions

- **Q22** (1 mark, May 2024) — Name two biodiversity hotspots in India.
- **Q70** (2 marks, Apr 2025) — Write short note on Sundalands.
- **Q105** (5 marks, Apr 2025, May 2024) ★ — What are the criteria of hotspots?

Unit Recap (from SLM)

- India: 2.4% land, 7–8% species; >45,000 plants, >91,000 animals
- Tri-junction: Afro-tropical, Indo-Malayan, Palearctic realms
- 4 hotspots: Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland
- Hotspot criteria: $\geq 1,500$ endemic plants + $\geq 70\%$ habitat lost (Myers 1988)
- 18% plants endemic; 62% amphibians, 50% lizards endemic
- 11 biogeographical regions; among 17 mega diversity countries

Unit 5: Threats to Biodiversity

2.5.1–2.5.3 Habitat Loss, Poaching and Man-Wildlife Conflict

Theory

Biodiversity faces habitat destruction, invasive species, climate change, pollution and overexploitation. Habitat destruction is the primary threat — reducing biodiversity when organisms lose homes. Poaching is the second biggest direct threat after habitat loss. Man-wildlife conflict arises from human-wildlife negative interactions affecting both sides, intensified by population growth.

Important Points

- **Main threats:** habitat loss/degradation/fragmentation, invasive alien species, climate change, nutrient pollution, overexploitation
- **Habitat loss:** forests/grasslands turned to deserts/wastelands; tropical forests losing ~1%/year → 5–10% species reduction in 30 years
- **Poaching:** illegal hunting for trade — tiger skin/bones, elephant ivory, rhino horns, musk deer perfume, orchids, Rauvolfia, Datura
- UN 2015 resolution on illicit wildlife trafficking; SDG targets on poaching
- **Man-wildlife conflict (WWF):** negative impact on human social/economic/cultural life and wildlife conservation
- **Consequences:** habitat destruction, injury/death, crop damage, livestock depredation, property damage, population decline, trophic cascades
- Scientists estimate ~10 million species lost by 2050; ~25% world species may extinct rapidly

Man-Wildlife Conflict (Detailed)

- WWF definition: interaction causing negative impact on human social/economic/cultural life AND wildlife conservation
- As old as civilization but intensified by population growth in recent centuries
- **Causes:** habitat encroachment, crop raiding by elephants/monkeys, livestock depredation by predators, property damage
- **Consequences:** injury/death of humans and wildlife, crop damage, livestock loss, property destruction, population decline, trophic cascades
- **India examples:** elephant crop raids, tiger/leopard livestock attacks, monkey urban nuisance, wild boar crop damage

- Mangrove clearing for prawn farming → reduced marine fish breeding habitat
- Wetland drainage for agriculture → lost biodiversity and water regulation
- Scientists estimate human activities may eliminate ~10 million species by 2050
- 25% of world species could become extinct within 20–30 years from habitat loss and overexploitation

For Exam

Major biodiversity threats are habitat destruction (primary — forests, wetlands, coral reefs), poaching (second biggest — tigers, elephants, rhinos for illegal trade) and man-wildlife conflict from population growth and land use changes. Habitat loss reduces biodiversity as species cannot survive without ecosystems. Poaching for skins, ivory, horns and medicinal plants drives species toward extinction. Human activities may eliminate 25% of world species within decades.

2.5.4 Species Categories and Red Data Book

Theory

IUCN (1948) classifies species by extinction risk. Red Data Book (origin: Russian Federation, 1961–1964) records endangered/rare species with colour-coded threat levels. Key concepts include endemic species (unique geographic area), keystone species (disproportionate ecosystem impact), foundation species (dominant primary producers) and flagship species (charismatic conservation symbols).

Important Points

- **Rare:** Infrequently encountered; IUCN “at risk” but not endangered — Found in isolated locations; ~75% known species potentially rare
- **Threatened:** Vulnerable to near-future extinction — IUCN categories: Vulnerable, Endangered, Critically Endangered
- **Critically Endangered (CR):** Highest risk; 80% population decrease in 3 generations — Extremely high extinction risk
- **Endangered:** Serious extinction danger if negative factors continue — Giant panda, tiger, blue whale, Asian elephant, snow leopard

- **Vulnerable (VU):** High extinction risk unless threats improve — IUCN category
- **Extinct:** No doubt last individual died — Dodo, passenger pigeon
- **Endemic:** Found only in specific geographic area — Opposite of cosmopolitan distribution
- **Keystone:** Loss/addition causes major ecosystem changes — Tiger, lion, crocodile, elephant; bats pollinating ebony
- **Foundation:** Dominant primary producer — Kelp in kelp forests, corals in coral reefs
- **Flagship:** Charismatic species representing conservation cause — Indian tiger, African elephant, giant panda

IUCN Threat Categories

Category	Code	Risk Level
Critically Endangered	CR	80% population decline in 3 generations; highest risk
Endangered	EN	High extinction risk if threats continue
Vulnerable	VU	High risk unless conditions improve
Near Threatened	NT	Close to qualifying for threatened categories
Least Concern	LC	Widespread and abundant
Extinct	EX	No doubt last individual died

Red Data Book Colour Codes

Colour	Status
Black	Extinct
Red	Endangered
Amber	Vulnerable
White	Rare
Green	Recovering
Grey	Insufficient data

- **Extinct in Wild:** EW; Only survives in captivity; Some plant/animal species

Special Species Designations

- **Endemic:** Found only in one geographic region — Lion-tailed macaque (Western Ghats only)
- **Native/Indigenous:** Naturally occurs without human introduction — Indian peafowl in India
- **Keystone:** Disproportionate ecosystem impact — Tiger (regulates prey populations), bats (pollinate ebony)
- **Foundation:** Dominant primary producer shaping ecosystem — Kelp in kelp forests, corals in reefs
- **Flagship:** Charismatic species for conservation campaigns — Indian tiger, African elephant, giant panda, leatherback turtle
- **Rare:** Infrequently encountered; ~75% known species potentially rare — Giant otter, Sumatran rhino, vaquita, red wolf

For Exam

IUCN classifies threatened species as Vulnerable, Endangered or Critically Endangered (80% decline in 3 generations). The Red Data Book records rare/endangered species with colour codes (black=extinct, red=endangered, amber=vulnerable). Endemic species exist only in specific areas; keystone species (tiger, elephant) disproportionately affect ecosystems; flagship species (tiger, panda) promote public conservation support.

Previously Asked Questions

- **Q29** (1 mark, May 2024) — Name the category of species which are vulnerable to extinction in the near future.
- **Q35** (1 mark, Apr 2025, May 2024) ★ — What are flagship species?
- **Q57** (2 marks, May 2024) — Define flagship species.

Unit Recap (from SLM)

- Threats: habitat loss (primary), poaching (second), man-wildlife conflict, climate change, pollution
- Habitat destruction: forests, wetlands, coral reefs most threatened
- Poaching: tigers, elephants, rhinos for illegal trade; UN 2015 anti-trafficking resolution
- IUCN categories: Vulnerable, Endangered, Critically Endangered

- Red Data Book: colour-coded threat levels; originated in Russia 1961–64
- Keystone, endemic, flagship species concepts; ~25% species at rapid extinction risk

Unit 6: Biodiversity Conservation at Global, National and Local Levels

2.6.1 Conservation Concepts and Global Framework

Theory

Biodiversity is distributed across 1000+ ecoregions; Global 200 are richest/distinctive areas. Mega diversity nations have large hotspot proportions. Conservation manages natural resources retaining balance for present and future generations. India signed World Heritage Convention and CITES (controls endangered species trade). Convergent evolution produces similar features in unrelated species (sharks/dolphins); divergent evolution splits species from common ancestor (Darwin's finches).

Important Points

- **Conservation goals:** (a) prevent genetic diversity loss (b) save species from extinction (c) protect ecosystems from degradation
- **Global 200:** richest 200 ecoregions; 18 hotspots contain 50,000 endemic plants (20% global)
- **India hotspots:** NE forests, Western Ghats, Andaman-Nicobar endemic subspecies
- **CITES:** Convention on International Trade in Endangered Species
- **World Heritage sites:** Manas, Kaziranga, Bharatpur, Nandadevi, Sundarbans
- Biodiversity = global resource; developed North exploits biodiversity of biodiverse South
- Convergent evolution: similar features, different ancestors; Divergent: different features, common ancestor

For Exam

Biodiversity conservation protects genetic diversity, prevents species extinction and safeguards ecosystems for sustainable development. India participates in global frameworks (World Heritage Convention, CITES). The Global 200 identifies richest ecoregions. India's national hotspots are NE forests, Western Ghats and Andaman-Nicobar. Biodiversity is increasingly treated as a shared global resource requiring international cooperation.

Previously Asked Questions

- **Q25** (1 mark, May 2024) — Expand CITES.

2.6.2–2.6.4 In-situ Conservation — National Parks, Sanctuaries, Biosphere Reserves

Theory

In-situ (on-site) conservation protects organisms in natural habitats through protected areas — cost-effective, large-scale, allows natural evolution. Three types: National Parks (strict protection), Wildlife Sanctuaries (limited human activity) and Biosphere Reserves (multi-purpose with core/buffer/transition zones, UNESCO MAB Programme 1971).

Important Points

Protected Areas Comparison

Aspect	National Park	Wildlife Sanctuary	Biosphere Reserve
Purpose	Strict conservation of natural environment	Protect wild animals and habitat	Multi-purpose: wildlife + tribal life + domesticated species
Human activities	Prohibited — no grazing, forestry, cultivation	Limited — timber/collection if not harming conservation	Zoned — varies by core/buffer/transition
Boundaries	Well demarcated by government	Less rigidly defined	Large area with three zones
Tourism	Allowed (public recreation)	Allowed (recreation)	Allowed (research + tourism)
Commercial use	Not permitted	Not for animal breeding/commercial sale	Sustainable activities in transition zone

Aspect	National Park	Wildlife Sanctuary	Biosphere Reserve
Research	Limited	Biologists permitted for study	Core = strict; buffer = research/education
Tribal rights	Generally excluded	Some traditional rights may continue	Transition zone protects tribal lifestyle
Legal status	Highest protection under Wildlife Act	Protected but less strict than NP	UNESCO MAB Programme (1971); 18 in India
Examples	Corbett, Kanha, Kaziranga, Gir, Periyar	Bharatpur, Chilika, Periyar, Madumalai	Nilgiri, Sundarbans, Gulf of Mannar, Nanda Devi
India count	100+ national parks	Many sanctuaries nationwide	18 biosphere reserves

- **National Park:** Strict protection; no grazing, forestry, cultivation; well-demarcated boundaries; public recreation — Hailey NP (first — now Corbett); Kanha, Kaziranga, Gir, Periyar, Ranthambore; 100+ in India
- **Wildlife Sanctuary:** Wild animals protected; limited timber/collection allowed; tourism permitted; no commercial breeding — Bharatpur (Rajasthan), Chilika (Odisha), Periyar (Kerala), Madumalai (TN)

Major Wildlife Sanctuaries of India

- **Bharatpur Bird Sanctuary** — Rajasthan: Migratory birds, Siberian crane
- **Chilika Lake Bird Sanctuary** — Odisha: Largest coastal lagoon; dolphins, birds
- **Periyar Wildlife Sanctuary** — Kerala: Elephants, tigers, lake Periyar
- **Madumalai Sanctuary** — Tamil Nadu: Tigers, elephants, gaurs
- **Govind Wildlife Sanctuary** — Uttarakhand: Snow leopard, Himalayan wildlife
- **Gir NP & Wildlife Sanctuary** — Gujarat: Asiatic lions
- **Chinnar Wildlife Sanctuary** — Kerala: Grizzled giant squirrel, star tortoise

Reasons for Establishing Wildlife Sanctuaries

- 1. Protect endangered species from hunting, poaching and predation
- 2. Protect animals in natural habitat rather than relocating them

- **3.** Monitor and breed endangered species; release specimens for survival
- **4.** Permit biologist research and ecological studies
- **5.** Rehabilitate injured/abandoned animals before forest release
- **6.** Educate humans on treating animals with respect
- **Biosphere Reserve:** Multi-purpose; protects wildlife + tribal lifestyle + domesticated species; research/tourism; UNESCO MAB 1971; 18 in India — Nilgiri, Sundarbans, Gulf of Mannar, Nanda Devi, Nokrek, Khangchendzonga

Biosphere Reserve Zones

Zone	Human Activity	Purpose
Core Zone	Strictly protected; no human intervention	Undisturbed ecosystem
Buffer Zone	Research and education only	Manages vegetation, agriculture, fisheries
Transition Zone	Cropping, recreation, forestry, settlements	Community cooperation with sustainable use

Major National Parks of India

- **Corbett (Hailey — first NP)** — Uttarakhand: Tigers, elephants; India's first national park
- **Kaziranga** — Assam: One-horned rhinoceros; World Heritage Site
- **Kanha** — Madhya Pradesh: Tigers, barasingha (swamp deer)
- **Bandipur** — Karnataka: Tigers, elephants, sandalwood forests
- **Gir** — Gujarat: Asiatic lions (only wild population)
- **Periyar** — Kerala: Elephants, tigers, lake ecosystem
- **Ranthambore** — Rajasthan: Tigers, historic fort, lakes

SLM Table 2.6.1 — National Parks (selected)

- **Bandipur** — Karnataka | **Bandhavgarh** — Madhya Pradesh | **Corbett (Hailey — first NP)** — Uttarakhand
- **Gir** — Gujarat | **Kanha** — Madhya Pradesh | **Kaziranga** — Assam | **Keoladeo Ghana** — Rajasthan

- **Manas** — Assam | **Nagarhole** — Karnataka | **Periyar** — Kerala | **Ranthambore** — Rajasthan
- **Great Himalayan NP** — Himachal Pradesh | **Chinnar Wildlife Sanctuary** — Kerala (*listed in SLM Table 2.6.1*)
- **PYQ Q7 (any two national parks in Kerala — SLM Table 2.6.1): Periyar National Park + Chinnar Wildlife Sanctuary** (*both Kerala entries in SLM table; Chinnar is technically a sanctuary but listed under National Parks table in SLM*)
- **Sundarbans** — West Bengal: Royal Bengal tigers, mangroves; World Heritage
- **Great Himalayan NP** — Himachal Pradesh: Alpine ecosystems, snow leopard
- **Manas** — Assam: Golden langur, pygmy hog; World Heritage

18 Biosphere Reserves of India

- **Nilgiri** — TN, Kerala, Karnataka: 2000
- **Sundarbans** — West Bengal: 2001
- **Gulf of Mannar** — Tamil Nadu: 2001
- **Nanda Devi** — Uttarakhand: 2004
- **Simlipal** — Odisha: 2009
- **Pachmarhi** — Madhya Pradesh: 2009
- **Nokrek** — Meghalaya: 2009
- **Achanakmar-Amarkantak** — Chhattisgarh, MP: 2012
- **Great Nicobar** — Great Nicobar: 2013
- **Agasthyamalai** — Kerala, Tamil Nadu: 2016
- **Khangchendzonga** — Sikkim: 2018
- **Cold Desert** — Himachal Pradesh: 2009
- **Dehang-Debang** — Arunachal Pradesh: 2008
- **Dibru-Saikhowa** — Assam: 2007
- **Seshachalam** — Andhra Pradesh: 2010
- **Manas** — Assam: 1989

- **Kachchh — Gujarat: 2010**

In-situ vs Ex-situ Conservation

Aspect	In-situ	Ex-situ
Location	Natural habitat / protected areas	Zoos, gardens, gene banks, labs
Cost	Cost-effective and convenient	Expensive; cryopreservation costly
Scale	Large number of organisms conserved	Limited individuals; breeding programmes
Evolution	Natural evolution and adaptation	Artificial conditions; may lose wild traits
Methods	National Parks, Sanctuaries, Biosphere Reserves	Zoos, botanical gardens, seed/gene banks, cryopreservation
Advantage	Maintains complete ecosystems and inter-species relationships	Controlled breeding, reintroduction, genetic techniques
Disadvantage	Vulnerable to habitat destruction	Limited space; behavioural changes in captivity
India examples	100+ NPs, 18 Biosphere Reserves	Project Tiger, Crocodile Breeding, Cryo-banks

Biodiversity Conservation Strategies (SLM — PYQ Q121)

- **In-situ (on-site):** National Parks, Wildlife Sanctuaries, Biosphere Reserves — protect in natural habitat
- **Ex-situ (off-site):** Zoos, botanical gardens, gene/seed banks, cryopreservation, captive breeding
- **India projects:** Project Tiger (1973), Crocodile Conservation (1975), Project Elephant (1992)
- **Global frameworks:** CITES, World Heritage Convention, Global 200 ecoregions
- **PYQ Q121 For Exam:** Biodiversity conservation strategies include **in-situ conservation** — protecting species in natural habitats through **National Parks** (strict protection, no exploitation), **Wildlife Sanctuaries** (limited human activity) and **Biosphere Reserves** (core/buffer/transition zones, UNESCO 1971); and **ex-situ conservation** — protecting threatened species

outside habitats via **zoos, botanical gardens, gene banks, seed banks, cryopreservation** and captive breeding (**Project Tiger 1973, Crocodile Conservation 1975, Project Elephant 1992**). Global cooperation through **CITES**, World Heritage sites and **Global 200** priority ecoregions.

For Exam

In-situ conservation protects biodiversity in natural habitats. National Parks prohibit all exploitation (grazing, forestry, cultivation) — India has 100+ parks, first being Hailey (Corbett). Wildlife Sanctuaries protect animals allowing limited resource collection and tourism. Biosphere Reserves (18 in India, UNESCO 1971) have three zones: core (no human activity), buffer (research/education) and transition (sustainable human activities with community participation).

Previously Asked Questions

- **Q7** (1 mark, Aug 2023) — Name any two national parks in Kerala.
- **Q20** (1 mark, May 2024) — Give an example for methods of In-situ conservation of animals.
- **Q36** (1 mark, Apr 2025) — What is a biosphere reserve?
- **Q37** (1 mark, Apr 2025) — Name the state in which Chilka Lake bird sanctuary is located.
- **Q45** (2 marks, Aug 2023) — Define the term Sanctuary.
- **Q121** (10 marks, Apr 2025, May 2024) ★🔥 — Write an essay on biodiversity conservation strategies.

Diagram (refer SLM): Fig. 2.6.1 Biodiversity and conservation concept map.

2.6.5 Ex-situ Conservation and Conservation Projects

Theory

Ex-situ (off-site) conservation protects endangered species outside natural habitats — zoos, botanical gardens, gene banks, seed banks, field gene banks, cryopreservation (−196°C liquid nitrogen). Advantages: longer breeding time, reintroduction possible, genetic techniques applicable. India launched Project Tiger (1973, WWF), Crocodile Breeding (1975) and Project Elephant (1992).

Important Points

- **Botanical gardens/Zoos:** Conventional ex-situ; housing, breeding, reintroduction; educational value
- **Gene banks:** Store crop genetic resources; monitor viability; regeneration; IPGRI/FAO protocols
- **Seed banks:** Most cost-effective long-term storage; minimizes genetic erosion
- **Field gene banks:** Living collections for perennials/recalcitrant species; educational and research access
- **Cryopreservation:** -80°C (CO_2) or -196°C (N_2); prevents ice crystal damage; stabilizes cells
- **Tissue culture:** In-vitro conservation biotechnology method
- **Project Tiger (1973):** 9 reserves ($16,339 \text{ km}^2$) → 27 reserves ($37,761 \text{ km}^2$); tigers 268 → 1500; protects habitat not just species
- **Crocodile Conservation (1975):** 30+ breeding centres; restocked 20 water bodies; most successful ex-situ project
- **Project Elephant (1992):** 12 states; viable elephant populations in natural habitats
- Ex-situ advantages: controlled breeding, reintroduction, genetic preservation
- Best preservation: in-situ network of National Parks + Sanctuaries covering all ecosystems

Conservation Projects (Detailed)

- **Project Tiger:** 1973; Govt of India + WWF; 9 → 27 tiger reserves; $16,339 \rightarrow 37,761 \text{ km}^2$; Tigers: 268 (1972) → ~1500 (1997); protects entire habitat
- **Crocodile Breeding:** 1975; Govt of India; 30+ captive breeding centres, zoos; All 3 species bred; thousands restocked in 20 water bodies
- **Project Elephant:** 1992; Govt of India; 12 states (north, NE, south India); Long-term viable elephant populations; habitat still shrinking

Ex-situ Methods (Detailed)

- **Gene banks:** store, maintain, reproduce crop varieties and wild relatives; periodic regeneration when viability declines

- **Seed banks:** most cost-effective; IPGRI/FAO protocols; large populations, minimal genetic erosion
- **Field gene banks:** for perennials and recalcitrant species; susceptible to disease and disasters
- **Cryopreservation:** -196°C liquid nitrogen; cells stabilized; prevents chemical reactions; widely used for in-vitro material
- **Botanical gardens/zoos:** housing, breeding, reintroduction, education
- **Tissue culture:** biotechnological in-vitro conservation method
- Convergent evolution: sharks and dolphins (similar form, different ancestors)
- Divergent evolution: Darwin's finches (common ancestor, different beak adaptations)

India's Fragile Ecosystems Requiring Protection

- **Mountain ecosystems:** Extremely fragile; forest cover loss → severe soil erosion, river course changes
- **Island ecosystems:** Andaman-Nicobar unique species; any human activity → rapid extinction
- **Evergreen forests:** Species-rich but highly vulnerable to logging and conversion
- **Coral reefs:** Sensitive to pollution, temperature rise, ocean acidification
- **River/wetland ecosystems:** Seriously affected by pollution and surrounding land use changes
- **Elephant habitats:** Require large diverse areas — open grasslands (post-monsoon) + forest foliage (dry season)
- Protected area for elephants must include multiple habitat types for complete species complement
- **PYQ Q116 For Exam:** **Ex-situ conservation** protects species **outside** natural habitats. **Plants:** botanical gardens, **seed banks**, field gene banks, **cryopreservation** (-196°C), tissue culture. **Animals:** zoos, captive breeding. India: **Project Tiger (1973)**, **Crocodile Breeding (1975)** — most successful ex-situ, **Project Elephant (1992)** — 12 states. Allows controlled breeding, genetic techniques and reintroduction.

For Exam

Ex-situ conservation removes species from threatened habitats to zoos, botanical gardens, gene banks, seed banks or cryopreservation (-196°C). It allows controlled breeding and reintroduction. India's Project Tiger (1973) protects tiger habitats across reserves; Crocodile Breeding (1975) successfully restocked waterways; Project Elephant (1992) covers 12 states. In-situ protected area networks remain the best approach for preserving total ecosystem diversity.

Previously Asked Questions

- **Q116** (10 marks, May 2024) — What is meant by ex-situ conservation? Describe various strategies of ex-situ cons...
- **Q121** (10 marks, Apr 2025, May 2024) ★🔥 — Write an essay on biodiversity conservation strategies.

Unit Recap (from SLM)

- Conservation: protect genetic diversity, prevent extinction, safeguard ecosystems
- In-situ: National Parks (strict), Sanctuaries (limited use), Biosphere Reserves (3 zones)
- Biosphere zones: core (no humans), buffer (research), transition (sustainable use)
- Ex-situ: zoos, gene banks, seed banks, cryopreservation (-196°C)
- Project Tiger 1973; Crocodile Breeding 1975; Project Elephant 1992
- India signed CITES and World Heritage Convention; Global 200 ecoregions

Block 3: Social Issues and Sustainable Development

Unit 1: Environment and Human Health

3.1.1 Environment and Human Health

Theory

Human health and well-being are intimately linked to the state of the environment. Good quality natural environments provide clean air, water, fertile land, and energy inputs; environmental degradation significantly increases the risk of cancer, heart disease, asthma, and many other illnesses. Global climate change will have wide-ranging health impacts — negative impacts are anticipated to outweigh positive ones through changes in temperature extremes, floods, droughts, infectious disease occurrence, food production, air pollutants, and population displacement.

Important Points

- Environmental hazards = chemical, physical, or biological factors with negative short- or long-term health impacts
- Nearly 25% of global deaths and total disease burden attributed to environmental factors (WHO)
- Poor environmental quality hits hardest those already at health risk — children and elderly especially vulnerable
- Climate change impacts: heat/cold extremes, floods/droughts, infectious diseases, nutritional adequacy, air pollutants, climate refugees
- IAOH (Indian Association of Occupational Health) and NIOH (National Institute of Occupational Health, Ahmedabad) promote occupational health

For Exam

Human health is intimately linked to environmental quality. Natural environments provide basic needs (clean air, water, food, energy), while degradation increases disease risk. Environmental hazards are chemical, physical, or biological factors causing illness. Climate change will negatively impact health through temperature extremes, altered disease patterns, food insecurity, and displacement. Nearly 25% of global deaths are environment-related. Occupational health bodies in India include IAOH and NIOH.

Diagram (refer SLM): Fig. 3.1.1 Impact of climate change on human health; Fig. 3.1.2 Climate change and respiratory diseases; Fig. 3.1.3 Environmental impacts on human health (WHO, 2017).

3.1.2 Environmental Degradation and Health Hazards

Theory

Environmental degradation is the deterioration of the environment through consumption of air, water, and soil resources and destruction of environments and wildlife. Industrialization brought health benefits globally but also occupational and environmental hazards — biological, chemical, physical, biomechanical, or psychosocial. Occupational health is the branch of medicine dealing with worker health and safety, focused on preventing hazards at ground level. These health problems are especially severe in developing nations where hazard controls are not widely adopted.

Important Points

- Environmental degradation = disintegration of earth through resource consumption and wildlife eradication
- Occupational health = highest level of worker's physical, mental, and social well-being; essentially preventive medicine
- Hazard types: biological, chemical, physical, biomechanical, psychosocial
- Traditional hazards: poor sanitation/shelter + industrial contamination of air, water, food, land
- Poor air quality linked to premature death, cancer, respiratory and cardiovascular damage; second-hand smoke harms non-smokers
- **PYQ Q115 For Exam: Occupational health hazards** are workplace conditions causing illness/injury. Types: **biological** (bacteria, viruses); **chemical** (toxic fumes, pesticides, asbestos); **physical** (noise, radiation, heat); **biomechanical** (repetitive strain); **psychosocial** (stress). Examples: silicosis, asbestosis, pesticide poisoning, hearing loss, cancer. Developing countries face highest burden. Prevention: PPE, ventilation, safe exposure limits, NIOH/IAOH programmes.

For Exam

Environmental degradation deteriorates air, water, and soil through human activities, increasing health risks. Environmental health hazards include traditional poor sanitation and industrial contamination. Occupational health focuses on preventing workplace hazards — monitoring toxic substances, setting safe exposure levels, and protecting workers. Hazard types are biological, chemical, physical, biomechanical, and psychosocial. Developing countries face the severest burden due to inadequate hazard control.

Previously Asked Questions

- **Q115** (10 marks, Aug 2023) 🔥 — Write an essay on occupational health hazards with examples.

3.1.3–3.1.4 Environmental Diseases

Theory

Environmental diseases are influenced by population size, housing, vector control, health care access, and hygiene. Climate change may increase diarrheal diseases and expand risk areas for dengue. Diseases of concern include allergies, asthma, neurotoxic effects, cancer, heart disease, and obesity linked to environment and diet. Outdoor air pollution especially influences asthma and allergies in children. Flooding increases diarrheal illness, cholera, dysentery, and typhoid in developing nations.

Important Points

- Revival of infectious diseases driven by population growth, urbanization, deforestation, international travel
- IPCC warns climate change increases diarrheal disease burden and dengue risk areas
- Integrated approach needed — people breathe both indoor and outdoor air
- Industrialized nations reduce flood disease through flood control, sanitary infrastructure, monitoring
- Environmental health = branch of public health focusing on inter-relationships between people and environment
- **PYQ Q106 For Exam: Environmental diseases** arise from demographic, social and environmental factors. Examples: **Asthma** (air pollution, allergens); **Allergies** (~40% global allergic rhinitis); **Cancer, heart disease** (pollution-linked); **Diarrheal diseases** (poor sanitation; IPCC links climate change to expanded range); **Cholera, typhoid** (flood/water contamination). Prevention: pollution control, clean water, integrated public health.

For Exam

Environmental diseases arise from complex interactions of demographic, social, and environmental factors. Climate change expands infectious disease risk (diarrhea, dengue). Growing concerns include asthma, allergies, cancer, and heart disease linked to pollution and lifestyle. Outdoor air pollution worsens

asthma, especially in children. Flood-related diseases (cholera, typhoid) are reduced in industrialized nations through infrastructure but remain serious in developing countries.

Previously Asked Questions

- **Q106** (5 marks, Apr 2025) — What are environmental diseases? Give examples.

3.1.5 Occupational Diseases

Theory

Occupational disease is any illness associated with a particular occupation, resulting from biological, chemical, physical, and psychological factors in the work environment. Major occupational diseases of concern in India include silicosis, asbestosis, musculoskeletal injuries, coal workers' pneumoconiosis, byssinosis, pesticide poisoning, and noise-induced hearing loss. Common environmental diseases also include asbestosis, silicosis, fluorosis, asthma, and allergies.

Important Points

- **Asbestosis:** Prolonged inhalation of asbestos fibres (mining, construction) — Chronic fibrotic lung disease; pulmonary fibrosis; can occur from secondhand exposure
- **Silicosis:** Inhaling crystalline silica dust (glass manufacturing, tunnel work, stone cutting) — Progressive scarring of lung tissue; cough, sputum, breathlessness; acute silicosis from massive single exposure
- **Fluorosis:** Excess fluoride via drinking water/food/industrial pollutants over long period — Crippling disease; dental staining to skeletal fluorosis (joint stiffness, bone changes); low fluoride prevents caries
- **Asthma:** Environmental allergens, air pollution, dust mites, mould, occupational chemicals, tobacco smoke — Airways narrow and swell; wheezing, coughing; urbanization increases prevalence
- **Allergies:** Airborne allergens (dust mites, pollens, fungi, animal dander); urbanization and vehicle emissions — Allergic rhinitis in ~40% of world population; environmental pollutants worsen allergies
- IAOH = association of health professionals promoting occupational health

- NIOH = WHO collaborative centre for occupational health, Ahmedabad
- **PYQ Q111 For Exam: Occupational diseases** from workplace exposure. Examples: **Silicosis** (silica dust — mining/construction); **Asbestosis** (asbestos fibres — lung fibrosis); **Pesticide poisoning** (agriculture); **Fluorosis** (excess fluoride — dental/skeletal); **Asthma** (occupational chemicals); **Allergies** (industrial dust/fumes). Prevention: PPE, ventilation, safe limits, monitoring. **NIOH** (Ahmedabad) promotes occupational health in India.

For Exam

Occupational diseases result from workplace exposure to harmful agents. In India, major concerns include silicosis, asbestosis, and pesticide poisoning. Asbestosis is chronic lung fibrosis from asbestos fibre inhalation. Silicosis results from silica dust causing lung scarring in mining/construction workers. Fluorosis is a crippling disease from excess fluoride affecting teeth and bones. Asthma involves narrowed airways triggered by pollution and allergens. Allergies are worsened by urbanization, vehicle emissions, and airborne allergens affecting 40% globally with allergic rhinitis.

Previously Asked Questions

- **Q8** (1 mark, Aug 2023) 🔥 — What is Occupational disease?
- **Q59** (2 marks, May 2024) 🔥 — What is meant by occupational disease? Give example.
- **Q106** (5 marks, Apr 2025) — What are environmental diseases? Give examples.
- **Q111** (5 marks, Apr 2025, Aug 2023, May 2024) ★🔥 — What are occupational diseases? Give examples.
- **Q115** (10 marks, Aug 2023) 🔥 — Write an essay on occupational health hazards with examples.

Unit Recap (from SLM)

- Health hazards = chemical, physical, or biological environmental factors harming health
- Environmental health hazards include poor sanitation and industrial contamination of air, water, food, land

- Occupational health monitors toxic substances, sets safe exposure levels, and protects workers
- Asbestosis, silicosis, fluorosis, asthma, and allergies are key environment-linked diseases
- Climate change negatively impacts human health through multiple pathways
- IAOH and NIOH are India's key occupational health institutions
- Nearly 25% of global disease burden is environment-related
- Environmental health = public health branch studying people-environment relationships

Unit 2: Epidemiological Issues in Health

3.2.1 Epidemiology and Health Issues

Theory

Epidemiology studies the frequency and pattern of health events in populations — how many cases occur relative to population size. It researches disease causation, transmission, outbreak investigation, surveillance, and treatment comparisons. The Epidemiologic Triad consists of an external agent, a host, and an environment — optionally including a vector (e.g., *Anopheles* mosquito transmitting malaria plasmodium; *Ixodes* ticks transmitting Lyme disease).

Important Points

- Epidemiology = study of disease frequency, distribution, and determinants in populations
- Major areas: disease causation, transmission, outbreak investigation, surveillance, occupational epidemiology
- Epidemiologic Triad: **Agent + Host + Environment** ($\pm$ Vector)
- Malaria: plasmodium stored in mosquito salivary glands, injected into next human host
- Epidemiologists use biology, statistics, social sciences, and engineering for exposure assessment

For Exam

Epidemiology studies the frequency and pattern of health events in defined populations. It addresses disease causation, transmission, surveillance, and outbreak investigation. The Epidemiologic Triad comprises agent, host, and environment — with vector as optional fourth element. Example: Anopheles mosquito vector transmits malaria plasmodium without being affected itself. Epidemiology relies on biology, statistics, and social sciences to understand and control disease spread.

Diagram (refer SLM): Fig. 3.2.1 Epidemiological triad of disease.

3.2.2–3.2.3 Water-Related and Water-Borne Diseases

Theory

Water-related diseases include those from direct/indirect water contact, ingestion, water aerosols, vector habitats, and water shortage during floods/droughts. Water-borne diseases involve direct transmission via microbiological or chemical pollutants. Seven categories exist: waterborne microbiological, waterborne chemical, water hygiene, water contact, water vector habitat, excretal disposal, and water aerosol diseases. About 502,000 diarrheal deaths annually linked to contaminated water; 2 billion people use fecally contaminated drinking water sources.

Important Points

- **Cholera:** *Vibrio cholerae* (bacteria); Contaminated water/food; Severe diarrhoea, dehydration; rapid outbreak after floods
- **Diarrhoea:** Multiple pathogens; Fecally contaminated water; Most common water-borne disease; mainly affects children
- **Typhoid:** *Salmonella typhi* (bacteria); Contaminated drinking water; Fever, abdominal pain; common in areas with poor sanitation
- **Amebiasis:** *Entamoeba histolytica* (protozoa); Contaminated water; Dysentery, liver abscess
- **Hepatitis:** Hepatitis A/E viruses; Contaminated water; Liver inflammation; jaundice
- **Gastroenteritis:** Bacteria/viruses; Contaminated water; Vomiting, stomach cramps, diarrhoea

- **Giardiasis:** Giardia lamblia (protozoa); Contaminated water; Diarrhoea, abdominal cramps
- **Campylobacteriosis:** Campylobacter (bacteria); Contaminated water; Diarrhoea, fever
- **Scabies:** Sarcoptes scabiei (mite); Water contact; Skin infestation
- **Worm infections:** Helminths; Contaminated water/soil; Intestinal parasites; soil colonization from contaminated irrigation

Seven Categories of Water-Related Disease

- 1. Waterborne microbiological disease
- 2. Waterborne chemical disease
- 3. Water hygiene disease
- 4. Water contact disease
- 5. Water vector habitat disease
- 6. Excretal disposal disease
- 7. Water aerosol disease
- WHO: 844 million lack basic drinking water; 159 million depend on surface water; 2 billion use fecally contaminated sources
- Climate-linked outbreaks: heavy rains overflow sewage plants; drought concentrates pathogens in limited water
- Prevention: good engineering, water management, personal hygiene, antibacterial/anti-parasitic treatment
- Water-borne diseases mainly affect children due to poor hygiene and weak immunity

For Exam

Water-related diseases encompass all illnesses from water exposure — ingestion, contact, aerosols, vectors, and shortage. Water-borne diseases are directly transmitted by pathogens in water. Major examples: cholera (*Vibrio cholerae*), diarrhoea (most common), typhoid (*Salmonella typhi*), hepatitis, giardiasis, and worm infections. Seven categories of water-related disease are recognized. WHO reports 502,000 annual diarrheal deaths; 2 billion use contaminated water. Climate change increases outbreaks — floods

contaminate water, droughts concentrate pathogens. Prevention requires engineering, hygiene, and effective water management.

3.2.4–3.2.5 Vector-Borne and Air-Borne Diseases

Theory

Vectors are living organisms transmitting pathogens between humans or from animals to humans during blood meals. Airborne diseases spread via pathogenic microbes discharged through coughing, sneezing, or close contact, remaining suspended on dust particles and respiratory droplets. Over one billion people are infected annually from vector-borne diseases; airborne diseases like TB and influenza spread rapidly in crowded, poorly ventilated settings.

Important Points

- **Vector-borne:** Malaria; Anopheles mosquito; Plasmodium via blood meal; Most common in tropical/subtropical regions; IPCC links climate expansion
- **Vector-borne:** Dengue; Aedes mosquito; Virus via blood meal; Substantial economic burden; thrives in poor housing
- **Vector-borne:** Lyme disease; Ixodes tick; Borrelia via blood meal; Example in Epidemiologic Triad
- **Vector-borne:** Yellow fever; Mosquito; Virus via blood meal; Tropical regions
- **Vector-borne:** Schistosomiasis; Freshwater snails; Parasite in water habitat; Water vector habitat disease
- **Vector-borne:** Leishmaniasis; Sand flies; Parasite via blood meal; Kala-azar in India
- **Vector-borne:** Lymphatic filariasis; Mosquito; Worm via blood meal; Elephantiasis
- **Air-borne:** Tuberculosis; Mycobacterium tuberculosis; Cough/sneeze droplets; Bacterial; poor ventilation increases spread
- **Air-borne:** Influenza; Influenza virus; Respiratory droplets; Seasonal epidemics

- **Air-borne:** Measles; Paramyxovirus; Cough/sneeze aerosols; Highly contagious
- **Air-borne:** Chickenpox; Varicella zoster virus; Airborne droplets; Viral exanthem
- **Air-borne:** COVID-19; SARS-CoV-2; Respiratory aerosols; Pandemic potential
- **Air-borne:** Whooping cough; Bordetella pertussis; Cough droplets; Bacterial; affects children
- **Air-borne:** Common cold; Rhinovirus; Droplet/contact; Most frequent respiratory infection
- **Air-borne:** Anthrax; Bacillus anthracis spores; Airborne spores; From soil/garbage sources
- **Air-borne:** Meningitis; Bacteria/virus/fungus; Airborne particles; Brain/spinal cord inflammation
- Vectors: mosquitoes, ticks, flies, sand flies, fleas, freshwater snails
- Over 1 billion infected and 1 million+ die annually from vector-borne diseases
- Airborne prevention: PPE, ventilation, face masks; pathogen survival depends on temperature, humidity, sunlight
- Aerosols = extremely small particles suspended in atmosphere
- Malnourished and immunocompromised most vulnerable to vector-borne diseases
- **PYQ Q98 For Exam: Vector-borne diseases** transmitted by blood-feeding organisms: **Malaria** (Anopheles mosquito, Plasmodium); **Dengue** (Aedes); **Leishmaniasis** (sand flies); **Lyme disease** (ticks); **Schistosomiasis** (freshwater snails). Over **1 billion** infected annually. Most common in tropical/subtropical regions; IPCC links climate change to expanded range. Prevention: vector control, nets, sanitation. Malnourished most vulnerable.

For Exam

Vector-borne diseases are transmitted by blood-feeding organisms — mosquitoes (malaria, dengue), ticks (Lyme disease), sand flies (leishmaniasis), and snails (schistosomiasis). Over one billion people infected annually. Airborne diseases spread via cough/sneeze droplets and aerosols — tuberculosis,

influenza, measles, COVID-19, and chickenpox (Varicella zoster virus). Prevention requires vector control, ventilation, protective equipment, and face masks. Both types thrive in poor living conditions with overcrowding and inadequate sanitation.

Previously Asked Questions

- **Q24** (1 mark, May 2024) — Which is the vector that transmits Malaria?
- **Q38** (1 mark, Apr 2025, May 2024) ★ — Name two Vector borne diseases.
- **Q98** (5 marks, May 2024) — Give an account on vector borne diseases.

Diagram (refer SLM): Fig. 3.2.2 Biodiversity and infectious diseases; Fig. 3.2.3 Airborne vs droplet transmission.

Unit Recap (from SLM)

- Epidemiology studies disease frequency, distribution, and causes in populations
- Epidemiologic Triad = Agent + Host + Environment ($\pm$ Vector); malaria via *Anopheles* mosquito
- Water-borne diseases: cholera, diarrhoea, typhoid, hepatitis, giardiasis, worm infections
- Seven categories of water-related disease including microbiological, chemical, and vector habitat
- Vector-borne: malaria, dengue, Lyme disease, leishmaniasis; 1 billion+ infected annually
- Airborne: TB, influenza, measles, COVID-19, chickenpox; spread via cough/sneeze droplets
- WHO: 502,000 annual diarrheal deaths from contaminated water; 2 billion use unsafe water
- Climate change and floods increase water-borne and vector-borne disease outbreaks

Unit 3: Need for Public Awareness on Aspects Related to Environment

3.3.1 Need for Public Awareness

Theory

Environmental awareness is critical because it minimises pollution and global warming, promotes renewable resources, and educates people about consumption risks. Public environmental awareness is among the most significant markers of a nation's civilization — encompassing knowledge, attitudes, and actions toward a sustainable society. It involves educating people so they can decide for themselves, using print and electronic media, with NGOs active from grassroots to policy levels.

Important Points

- Awareness = understanding environmental changes, cause-effect relationships, and responsibility to preserve
- All citizens equally responsible — not only government action
- Public awareness ≠ directing behaviour; it provides information for informed decisions
- Media (print + electronic) strongly influences public opinion
- Strong public support makes politicians respond positively to environmental activities

For Exam

Public environmental awareness is essential for sustainable society — it encompasses knowledge, attitudes, and responsible action. All citizens share responsibility for environmental protection, not just government. Awareness educates people about environmental risks (global warming, sustainable development) enabling informed decisions. NGOs, media, and community action from grassroots to policy level are vital. It is a key marker of national civilization.

3.3.2–3.3.3 Environmental Movements in India

Theory

Environmental movements are social/political movements for conservation or environmental improvement — also called green or conservation movements. They favour sustainable resource management and policy changes, ranging from highly organized to radically informal, local to global. In India, Chandi Prasad Bhatt's Dasholi Gram Swarajya Sangh (DGSM) preceded the Chipko movement (1974). Major movements up to 2000 include Chipko, Save Silent Valley, Jungle Bachao Andolan, Appiko, and Narmada Bachao Andolan.

Important Points

- **Chipko Movement:** 1973 (Khumtung, Himalayas); Sunderlal Bahuguna, Chandi Prasad Bhatt, Gaura Devi, women activists; Forest conservation; "Chipko" = embrace trees to prevent felling; Non-violent Satyagraha; 15-year tree-felling ban (1980, Indira Gandhi); Right Livelihood Award (1987); slogan "Ecology is permanent economy"
- **Narmada Bachao Andolan (NBA):** 1985; Medha Patkar, Baba Amte; Protest against large dams on Narmada (Sardar Sarovar); fight for tribal/farmer rehabilitation; Right Livelihood Award (1991); opposed displacement of ~1 million people; World Bank exit (1990–93); questioned development vs. ecology
- **Save Silent Valley Movement:** 1970s; Kerala environmentalists; Prevent hydroelectric dam in Silent Valley rainforest, Kerala; Successfully prevented dam; protected pristine tropical rainforest
- **Jungle Bachao Andolan:** 1982 (Bihar); Tribal communities; Protest against government order to replace natural sal forest with teak plantations; Tribal resistance to commercial forestry replacing native forests
- **Appiko Movement:** 1983 (Karnataka); Pandurang Hegde; Protect Western Ghats forests in Karnataka (similar to Chipko); Hugged trees to prevent felling; raised awareness about forest conservation in South India
- DGSM founded by Chandi Prasad Bhatt; inaugurated by Sucheta Kriplani; land donated by Shyma Devi
- Women played vital role in Chipko — broadened agenda beyond forests to alcoholism and social issues
- Sunderlal Bahuguna: 5000 km trans-Himalaya foot march (1981–83) spreading Chipko message

- NBA: Narmada is India's largest west-flowing river; 3000+ planned dams; Sardar Sarovar displaced 250,000+ people
- Green movement = conservation movement = environmental movement (alternative terms)
- Environmental awareness evokes human responsibility to respect, protect, and preserve nature
- **PYQ Q96 For Exam: Narmada Bachao Andolan (1985)** opposed **Sardar Sarovar Dam** on Narmada. Led by **Medha Patkar** and **Baba Amte** — fought displacement of tribals, farmers (~1 million affected). Non-violent resistance, rallies, legal action. Won **World Bank withdrawal (1990–93)**. Raised tribal rights, resettlement and sustainable development. One of India's most significant anti-dam environmental movements.

For Exam

Environmental movements are social movements for conservation and policy change. The Chipko Movement (1973, Himalayas) used non-violent tree-hugging to protect forests, led by Sunderlal Bahuguna and women activists, winning a 15-year felling ban in 1980. Narmada Bachao Andolan (1985) led by Medha Patkar and Baba Amte opposed Sardar Sarovar Dam displacement of tribals and farmers. Other movements: Save Silent Valley (protected Kerala rainforest), Jungle Bachao Andolan (Bihar tribal resistance), and Appiko (Karnataka, similar to Chipko). Both major movements won Right Livelihood Awards.

Previously Asked Questions

- **Q23** (1 mark, May 2024) — Name the environmental movement which began in 1973 in parts of Uttar Pradesh.
- **Q39** (1 mark, Apr 2025, May 2024) ★ — Who is the leader of Narmada Bachao Andolan?
- **Q72** (2 marks, Apr 2025, May 2024) ★ — What do you mean by green movement? Name any one green movement in India.
- **Q96** (5 marks, May 2024) — Write a short note on Narmada Bachao Andolan.

Diagram (refer SLM): Fig. 3.3.1 Chipko Movement poster; Fig. 3.3.2 Narmada Bachao Andolan poster.

Unit Recap (from SLM)

- Environmental awareness evokes human responsibility to respect and preserve nature
- Environmental movement = individuals/groups acting for environmental policy change
- Chipko (1973) protected Himalayan forests through non-violent tree embrace
- Narmada Bachao Andolan (1985) opposed Narmada dam displacement and ecological damage
- Public awareness uses media and NGOs to educate and mobilize citizens
- Green/conservation movements favour sustainable natural resource management
- Women played central role in Chipko Movement
- Chipko slogan: "Ecology is the permanent economy" (Sunderlal Bahuguna)

Unit 4: Current Environment Conservation Activities

3.4.1 Environmental Conservation

Theory

Conservation safeguards, maintains, and restores natural areas and ecological communities, using natural resources sensibly to prevent extinction.

Environmental conservation protects against species extinction and ecosystem collapse caused by pollution and human activity. Conservation (proper resource management for present and future) differs from preservation (protecting nature from all use) though terms are sometimes used interchangeably.

Important Points

- Conservation = safeguarding + managing natural resources for present and future use
- Works against reckless commercial/corporate environmental destruction

- Trees transform industrial CO₂ into oxygen — essential for human respiration
- Voluntary activities target pollution control, global conservation, and natural environment maintenance
- Conservation essential because all life forms are interdependent
- **PYQ Q122 For Exam: Environmental conservation activities: (1) Swachh Bharat Abhiyan** — cleanliness, toilets, waste management; **(2) Haritha Keralam** — waste, water, organic farming; **(3) Eco-clubs** (National Green Corps) — student environmental action; **(4) Nature Clubs** — vermicomposting, campus gardens; **(5) Chipko/Narmada movements; (6) Afforestation; (7) National Parks/Sanctuaries; (8) Rainwater harvesting; (9) Public awareness.** Conservation = managed sustainable use for present and future generations.

For Exam

Environmental conservation is protecting ecosystems from species extinction and degradation through sensible resource management for present and future generations. It safeguards natural areas, prevents reckless exploitation, and maintains ecological balance. Trees convert CO₂ to oxygen, making forest conservation vital. Conservation differs from preservation — it allows managed sustainable use. All species are interdependent, making conservation essential for survival.

Previously Asked Questions

- **Q122** (10 marks, Apr 2025) — Write an essay on various environmental conservation activities.

3.4.2 Environment Conservation Activities

Theory

India has launched major conservation initiatives including Swachh Bharat Abhiyan (cleanliness), Haritha Keralam Mission (green Kerala), Eco-clubs, and Nature Clubs in educational institutions. These programmes promote waste management, water conservation, organic farming, sanitation, and biodiversity awareness through people's participation.

For Exam

India's key environmental conservation activities include Swachh Bharat Abhiyan (clean India mission), Haritha Keralam Mission (integrated waste, water and organic farming in Kerala), Eco-clubs under National Green Corps, and Nature Clubs in schools — all promoting public participation in waste management, water conservation, sanitation, and biodiversity awareness.

Previously Asked Questions

- **Q122** (10 marks, Apr 2025) — Write an essay on various environmental conservation activities.

3.4.2.1 Swachh Bharat Abhiyan

Theory

Swachh Bharat Abhiyan is the most significant cleanliness campaign by the Government of India, launched on 2 October 2014 (Mahatma Gandhi's birth anniversary) by the Ministry of Housing and Urban Affairs. Prime Minister Narendra Modi led a cleanliness pledge at India Gate with ~30 lakh government employees, flagging off a walkathon at Rajpath. The logo features Mahatma Gandhi's glasses with the National Tricolor on the bridge, symbolising the nation uniting for a clean India.

Important Points

- **Launch:** 2 October 2014; inspired by Mahatma Gandhi's vision of clean India
 - **Ministry:** Ministry of Housing and Urban Affairs, Government of India
 - **Mantra:** "Won't mess up, won't let" (Na gandagi karenge, na karne denge)
 - **Model:** Each person invites 9 others — chain participation for mass mobilisation
 - **Status:** Became Jan Andolan (people's movement) with celebrities, sportspersons, jawans, industrialists participating
- Five objectives:**
- **1.** Promote cleanliness and hygiene holistically
 - **2.** Reduce incidence of open defecation
 - **3.** Improve quality of life in rural areas
 - **4.** Encourage sustainable sanitation practices

- **5.** Create awareness about health and hygiene
- **ODF (Open Defecation Free):** Key target — many panchayats declared ODF
- **Activities:** Street sweeping, garbage cleaning, sanitation drives, cleanliness campaigns via plays and music
- **Slogan:** "Cleanliness is next to Godliness"

For Exam

Swachh Bharat Abhiyan (2 October 2014) is India's largest cleanliness campaign inspired by Mahatma Gandhi, launched by Ministry of Housing and Urban Affairs. Objectives: holistic cleanliness, reduce open defecation, improve rural quality of life, sustainable sanitation, and health awareness. It became a Jan Andolan with mass participation using the "Won't mess up, won't let" mantra. Logo shows Gandhi's glasses with tricolor. Key target is ODF (Open Defecation Free) status for panchayats.

Previously Asked Questions

- **Q52** (2 marks, Aug 2023) 🔥 — Write a very short note on Swachh Bharat Abhiyan.
- **Q60** (2 marks, May 2024) 🔥 — What are the objectives of Swachh Bharat Abhiyan?

3.4.2.2 Haritha Keralam Mission

Theory

Haritha Keralam Mission (10 November 2016) is a people-centric endeavour under Nava Kerala Mission, envisioned by the Government of Kerala to bestow an eco-friendly land with greenery and serenity to posterity. Despite receiving twice the national average rainfall, Kerala loses bulk rainwater to the sea within 48–72 hours, making water conservation critical. The mission uses a biocentric approach placing humans within the greater natural environment context.

Important Points

- **Launch:** 10 November 2016; under Nava Kerala Mission umbrella
 - **Slogan:** To safeguard the natural resources
- Three sub-missions:**

- **Hygienic Waste Management:** Effective waste disposal at source — Decentralized disposal (Thumbboormoozhy model); biogas plants; block-level plastic/e-waste collection; hospital waste disposal; centralized disposal in Trivandrum, Kochi, Kozhikode
- **JalaSamirdhi (Water Conservation):** Rejuvenate water bodies — Phase 1: wells, ponds, springs, canals, streams; Phase 2: rivers, reservoirs, lakes, backwaters; dead river revival; rainwater harvesting; groundwater recharging; 3Rs (Reduce, Reserve, Recycle)
- **Sujalam Suphalam (Agriculture):** Organic farming development — Organic vegetables in all households; paddy on barren lands; rooftop cultivation; sapling planting; zero pesticide cultivation
- **Threats addressed:** Water depletion, unscientific waste management, air pollution, chemical effluents in water, deforestation, grass cover loss
- **Goals:** Eco-friendly environment, sustainable natural resources, zero pesticide food, biodiversity protection, social hygiene
- **Implementation:** Through Local Self-Government bodies in people-centric mode
- **Participation:** Voluntary organisations, youth associations, students, philanthropic groups
- **PYQ Q109 For Exam: Haritha Keralam Mission** (10 November 2016) — Kerala's green initiative under **Nava Kerala Mission**. Three sub-missions: **(1) Hygienic Waste Management** — decentralized disposal, biogas, e-waste; **(2) JalaSamirdhi** — water body rejuvenation, rainwater harvesting, 3Rs; **(3) Sujalam Suphalam** — organic farming, rooftop cultivation, zero pesticide. Despite high rainfall, Kerala loses water to sea in 48–72 hours. Implemented through **Local Self-Government** with community participation.

For Exam

Haritha Keralam Mission (10 November 2016) is Kerala's people-centric green initiative under Nava Kerala Mission. Three sub-missions: Hygienic Waste Management (decentralized disposal, biogas, e-waste), JalaSamirdhi (water body rejuvenation, rainwater harvesting, 3Rs approach), and Sujalam Suphalam (organic farming, rooftop cultivation, zero pesticide). Despite high rainfall, Kerala loses water to sea within 48–72 hours. Implemented through Local Self-Government bodies with community participation.

Previously Asked Questions

- **Q109** (5 marks, Apr 2025) — Write a note on Haritha Keralam Mission.

3.4.2.3 Eco-club and Nature Club

Theory

Eco-clubs (National Green Corps, started 2001) and Nature Clubs engage students in hands-on environmental conservation on campus and in communities, creating awareness and promoting sustainable practices.

Important Points

- **Eco-club:** 2001 (National Green Corps) — Tree planting, water conservation, waste segregation at source, anti-plastic campaigns, stop indiscriminate waste burning, nature trails in sanctuaries, quiz/essay/painting/rally/nukkad natak
- **Nature Club:** — — Sensitize students/staff, reduce local pollution, protect biodiversity, vermicomposting, herbal/butterfly gardens, plantation day, collaborate with regional nature clubs
- **PYQ Q97 For Exam: Eco-club** (2001, National Green Corps): tree planting, water conservation, **waste segregation** at source, anti-plastic campaigns, stop waste burning, nature trails in sanctuaries, quiz/essay/painting/rally/**nukkad natak**. **Nature Club:** sensitizes students/staff, reduces local pollution, protects biodiversity through vermicomposting, herbal/butterfly gardens, plantation drives. Both emphasise student participation for environmental learning and action.

For Exam

Eco-clubs (2001, National Green Corps) motivate students to plant trees, conserve water, segregate waste, reduce plastic use, and organise environmental awareness programmes. Nature Clubs create campus conservation through vermicomposting, gardens, and community drives. Both emphasise student participation for environmental learning and action.

Previously Asked Questions

- **Q97** (5 marks, May 2024) — What is Eco-club? Comment on its activities.

Diagram (refer SLM): Fig. 3.4.1 Eco-club of National Green Corps.

Unit Recap (from SLM)

- Environmental conservation protects natural world from collapse due to human activities
- Swachh Bharat Abhiyan (2014) promotes universal sanitation and hygienic practices
- Haritha Keralam focuses on waste management, water conservation, and organic farming
- Eco-clubs promote student participation in environmental learning and action
- Nature Clubs sensitize students and society about conservation in campus and community
- Conservation uses resources wisely; preservation protects without use
- 3Rs approach: Reduce, Reserve, Recycle (water in Haritha Keralam)
- ODF = Open Defecation Free — key Swachh Bharat target

Unit 5: Population Growth, Population Explosion and Associated Issues

3.5.1 Population Growth

Theory

Population is growing rapidly, outpacing the planet's support capacity. Overpopulation causes negative environmental and economic outcomes — over-farming, deforestation, water pollution, eutrophication, and global warming. Population impacts environment through natural resource use and waste production. In India, rapid growth threatens environment through agricultural expansion, uncontrolled urbanization, industrialization, and habitat destruction, declining per capita agricultural land, forest, and water resources.

Important Points

- Malthusian theory: population grows geometrically, resources arithmetically
- Overpopulation = existing population exceeds Earth's carrying capacity
- Population growth factors: birth rate, death rate, immigration, emigration
- Population is development source but environmental degradant when exceeding threshold limits

- Poverty is both cause and effect of environmental degradation

For Exam

Population growth outpaces resource availability, causing environmental degradation. Overpopulation occurs when human numbers exceed Earth's carrying capacity. Four factors govern growth: birth rate, death rate, immigration, and emigration. In India, rapid growth drives deforestation, water scarcity, pollution, and habitat loss, reducing per capita land and resources. Population is vital for development but degrades environment beyond threshold limits. Poverty and environmental damage are mutually reinforcing.

3.5.2–3.5.3 Population Issues and Explosion

Theory

Population explosion is the rapid, alarming increase in human numbers — pyramiding of biological population through increased survival and exponential growth. India's growth rate is ~1.8% per annum. World population was 6.3 billion in 2000, predicted to grow fourfold in next 100 years. Malthusian theory states population multiplies geometrically while resources multiply arithmetically. Effects include resource depletion, environmental degradation, unemployment, social problems, and hindered economic development.

Important Points

Ten Fatal Effects of Overpopulation

- **1 — Resource depletion:** Earth produces limited water/food; environmental damage from reckless hunting, deforestation, pollution in last 50 years
- **2 — Environmental degradation:** Overuse of coal, oil, natural gas; vehicle/industry growth degrades air; CO₂ emissions → global warming, polar melt, sea level rise
- **3 — Unemployment:** Fewer jobs for large population; rise in crime, theft to feed families
- **4 — Conflicts and wars:** Strain on resources in developing countries; water conflicts; diseases spread faster; starvation fuels child mortality
- **5 — High cost of living:** Demand-supply gap raises prices of food, shelter, healthcare

- **6 — Malnutrition and famine:** Resource scarcity causes starvation; famine correlates with poverty in less-developed regions
- **7 — Pandemics and epidemics:** Overcrowding, unhygienic conditions, malnutrition, inaccessible healthcare spread infectious diseases
- **8 — Water shortage:** Only ~1% of world's water is fresh and accessible; 70% freshwater demand by 2025
- **9 — Faster climate change:** Large nations (China, India) increase industrial emissions; larger populations speed climate change
- **10 — Intensive farming:** Evolved farming damages local ecosystems; machinery contributes to climate change

Ten Effects of Population Explosion (India-specific)

- **1 — Investment requirement:** 1.8% growth needs larger investment for same per capita income growth
- **2 — Capital formation:** 35% under 14 years = dependents reduce savings; rate of capital formation falls
- **3 — Per capita income:** Large population + rapid growth = low per capita capital availability
- **4 — Food problem:** Rapid growth = root cause of India's food problem
- **5 — Unemployment:** Large labour force but insufficient capital; disguised rural + open urban unemployment
- **6 — Low standard of living:** Rapid growth accounts for low living standards; bare necessities inadequate
- **7 — Poverty:** Rising population increases poverty; less savings, low capital formation, low productivity
- **8 — Social problems:** Rural-urban migration, slums, unhygienic conditions, frustration, crime, prostitution
- **9 — Land pressure:** Per capita land diminishes; sub-division and fragmentation of holdings
- **10 — Environmental pressure:** Higher birth rate → more pollution, toxic wastes, biosphere damage; hinders development
- Population growth factors: birth rate, death rate, immigration, emigration
- Overpopulation = existing population exceeds Earth's carrying capacity

- Population explosion must be controlled effectively
- **PYQ Q107 For Exam: Population growth degrades environment** through: **resource depletion; deforestation; pollution; global warming; biodiversity loss; water scarcity** (70% demand by 2025); **intensive farming** damage; **urbanisation** (slums, waste). More people → more demand → over-exploitation → degradation. India: 1.8% growth, 35% under 14 years. Controlling population is essential for sustainable development.
- **PYQ Q117 For Exam: Fatal effects of overpopulation:** (1) resource depletion; (2) environmental degradation/global warming; (3) unemployment; (4) conflicts; (5) high cost of living; (6) malnutrition/famine; (7) pandemics; (8) water shortage; (9) accelerated climate change; (10) intensive farming damage. **India-specific:** low per capita income, food shortage, disguised unemployment, poverty, slums, land fragmentation. Population explosion must be controlled effectively.

For Exam

Population explosion is rapid human number increase beyond economic coping capacity. India's growth rate is 1.8% annually with 35% population under 14 years (high dependency). Fatal effects: resource depletion, environmental degradation (global warming), unemployment, conflicts, high cost of living, famine, pandemics, water shortage (70% demand by 2025), accelerated climate change, and intensive farming damage. India-specific effects include reduced capital formation, low per capita income, food problems, disguised unemployment, poverty, social problems (slums, crime), and land fragmentation. Must be controlled effectively.

Previously Asked Questions

- **Q9** (1 mark, Aug 2023) — Name two major issues associated with population growth.
- **Q73** (2 marks, Apr 2025, Aug 2023, May 2024) ★🔥 — Give an account of population explosion.
- **Q107** (5 marks, Apr 2025, Aug 2023, May 2024) ★🔥 — What is the relation between population growth and degradation of environment?
- **Q117** (10 marks, May 2024, Apr 2025) ★🔥 — Write an essay on various fatal effects of over population.

Diagram (refer SLM): Fig. 3.5.1 Population growth statistics of India (2010–2020).

Unit Recap (from SLM)

- Overpopulation = population exceeds Earth's carrying capacity
- Population growth = imbalance between births and deaths
- Effects: food/water shortages, climate vulnerability, intergenerational poverty
- Rapid growth declines per capita agricultural land, forest, and water resources
- Population explosion = economy cannot cope with rapid growth rate
- India: 1.8% annual growth; 35% under 14 years (high dependency)
- Effects: pollution, toxic waste, biosphere damage, economic hindrance
- Intensive farming and deforestation linked to population pressure

Unit 6: Sustainable Development

3.6.1–3.6.3 Sustainable Development: Concept and Principles

Theory

The concept of sustainable development was described by the **1987 Brundtland Commission Report** (*Our Common Future*) as:

"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

Sustainable development is a way for people to use resources without the resources running out. There are four dimensions — society, environment, culture, and economy — which are intertwined, not separate. Sustainability balances environmental, societal, and economic considerations in pursuit of improved quality of life. Three main pillars: **Economic, Environmental, and Social** — informally referred to as **people, planet, and profits**.

Important Points

- **Brundtland Commission (1987):** World Commission on Environment and Development (WCED)

- **Full definition:** Present needs met without compromising future generations' ability to meet their own needs
- **Three pillars:** Economic growth, Environmental protection, Social equity (people, planet, profits)
- **Four dimensions:** Society, Environment, Culture, Economy
- **Main features:** (i) respect all life forms, (ii) improve human life quality, (iii) minimise resource depletion, (iv) enable communities to care for environment
- **Six principles:** Conservation of ecosystem; sustainable society; biodiversity conservation; population control; human resource development; public participation
- Criticised as vague by some; others feared it meant reversion to pre-modern lifestyles

For Exam

Sustainable development (Brundtland Commission, 1987) is defined as: "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs." Three pillars are economic, environmental, and social (people, planet, profits). Four dimensions — society, environment, culture, economy — are intertwined. Features include respecting all life, improving quality of life, minimising resource depletion, and community environmental stewardship. Six principles cover ecosystem conservation, sustainable society, biodiversity, population control, human resources, and public participation.

Previously Asked Questions

- **Q10** (1 mark, Aug 2023) 🔥 — The concept of sustainable development was proposed by?
- **Q49** (2 marks, Aug 2023, May 2024) ★🔥 — What is Sustainable development? / Define sustainable development.

Diagram (refer SLM): Fig. 3.6.1 Elements of sustainable development.

3.6.4 Examples of Sustainable Development

Theory

Practical sustainable development examples demonstrate how communities can reduce environmental impact while maintaining quality of life. These include renewable energy, sustainable agriculture, water efficiency, and urban green spaces that improve health and well-being.

Important Points

- **Wind energy:** Low cost, minimal land footprint; coexists with farming and recreation
- **Solar energy:** Cuts 94% of coal plant pollution; increasingly cost-competitive globally
- **Crop rotation:** Successive planting of different crops to increase soil fertility without chemicals
- **Water-efficient fixtures:** Dual flush toilets, low-flow faucets reduce water use by 50%+
- **Green spaces:** Parks and forests cool cities, produce oxygen, reduce air pollution; WHO: inactivity from poor walkability causes 3.3% global deaths

For Exam

Sustainable development examples include wind and solar energy (renewable, low-pollution power), crop rotation (chemical-free soil fertility), water-efficient fixtures (dual flush toilets, low-flow taps reducing use 50%+), and urban green spaces (parks cooling cities, producing oxygen, improving mental health). WHO notes 3.3% of global deaths from physical inactivity linked to poor access to recreational green areas. These demonstrate practical paths to ecological development.

Previously Asked Questions

- **Q26** (1 mark, May 2024) — Name the method of cultivating crops on the same ground to increase soil fertility.

3.6.5 Sustainable Development Goals (SDGs)

Theory

The 17 SDGs (Global Goals) were adopted by the UN in 2015 as a blueprint for sustainable future by 2030, replacing Millennium Development Goals. They are interconnected objectives addressing poverty, hunger, health, education,

equality, clean energy, climate action, and partnerships. The UN General Assembly established SDGs as part of the Post-2015 Development Agenda.

Important Points

- **1:** No Poverty; End poverty in all forms; Social protection systems; equal resource access; disaster resilience
- **2:** Zero Hunger; Food security, nutrition, sustainable agriculture; End malnutrition; resilient agricultural practices; genetic diversity of seeds
- **3:** Good Health & Well-being; Healthy lives for all ages; Reduce maternal/infant mortality; combat communicable/non-communicable diseases; universal health coverage
- **4:** Quality Education; Inclusive, equitable, lifelong learning; Free primary/secondary education; vocational training; eliminate discrimination
- **5:** Gender Equality; Empower all women and girls; End discrimination; equal opportunities; end gender-based violence
- **6:** Clean Water & Sanitation; Safe water and sanitation for all; Universal access to safe drinking water; adequate sanitation and hygiene
- **7:** Affordable & Clean Energy; Sustainable energy access; Universal access to affordable, reliable, renewable energy
- **8:** Decent Work & Economic Growth; Inclusive, sustainable economic growth; Full employment; eradicate forced labour; protect labour rights
- **9:** Industry, Innovation & Infrastructure; Resilient infrastructure, innovation; Sustainable industrialisation; scientific research; upgrade infrastructure
- **10:** Reduced Inequality; Reduce inequality within and among countries; Income growth of bottom 40%; social, economic, political inclusion
- **11:** Sustainable Cities & Communities; Safe, inclusive, sustainable cities; Affordable housing; sustainable transport; protect cultural/natural heritage
- **12:** Responsible Consumption & Production; Sustainable consumption patterns; Sustainable management of natural resources; reduce waste generation
- **13:** Climate Action; Combat climate change and its impacts; Integrate climate measures into policies; improve education and capacity

- **14:** Life Below Water; Conserve oceans and marine resources; Reduce marine pollution; protect marine ecosystems; regulate fishing
- **15:** Life On Land; Protect terrestrial ecosystems and biodiversity; Combat desertification; halt biodiversity loss; restore degraded land
- **16:** Peace, Justice & Strong Institutions; Promote peaceful, inclusive societies; Reduce violence; end abuse/exploitation; ensure access to justice
- **17:** Partnerships for the Goals; Strengthen global partnership; Mobilise financial resources; technology transfer; capacity building
- Adopted 2015; target achievement by 2030; replace Millennium Development Goals (completed 2015)
- SDGs are interconnected — progress in one goal supports others
- SDG 3: 13 targets including WHO Framework Convention on Tobacco Control
- SDG 4: education for sustainable development and global citizenship
- **PYQ Q113 For Exam:** **17 SDGs** adopted UN **2015**, target **2030**, replacing MDGs. Goals: No Poverty, Zero Hunger, Good Health, Quality Education, Gender Equality, Clean Water (SDG 6), Affordable Clean Energy (SDG 7), Climate Action (SDG 13), Life Below Water (SDG 14), Life On Land (SDG 15), Peace, Partnerships. **Interconnected** — progress in one supports others. Global blueprint for **economic, social and environmental** sustainability.

For Exam

The 17 Sustainable Development Goals (SDGs) were adopted by the UN in 2015 to achieve a sustainable future by 2030, replacing Millennium Development Goals. Key goals include No Poverty, Zero Hunger, Good Health, Quality Education, Gender Equality, Clean Water, Affordable Clean Energy, Climate Action, Life Below Water, and Life On Land. They are interconnected — addressing poverty, environment, equality, and peace together. SDGs provide a global blueprint for sustainable development across economic, social, and environmental dimensions.

Previously Asked Questions

- **Q40** (1 mark, Apr 2025, Aug 2023) ★🔥 — What are Sustainable Development Goals?

- **Q113** (10 marks, Aug 2023, Apr 2025) ★🔥 — Write an essay on Sustainable Development Goals.

Diagram (refer SLM): Fig. 3.6.2 Sustainable Development Goals.

Unit Recap (from SLM)

- SD (Brundtland, 1987) = meet present needs without compromising future generations
- Three pillars: economic, environmental, social (people, planet, profits)
- Six principles: ecosystem conservation, sustainable society, biodiversity, population control, human resources, public participation
- Examples: wind/solar energy, crop rotation, water fixtures, green spaces
- 17 SDGs adopted by UN in 2015; target 2030
- SD balances society, environment, culture, and economy
- Green spaces improve health; inactivity causes 3.3% global deaths (WHO)
- SD requires policy changes for ecological global development

Block 4: Environmental Ethics and Contemporary Environmental Issues

Unit 1: Concept of Environmental Ethics 4.1.1 Environmental Ethics

Theory

Environmental ethics extends traditional ethical boundaries from humans to the non-human world, studying the moral relationship between humans and the environment. It establishes that humans are part of society alongside plants and animals, requiring ethical values that include rights of non-human living beings. It influences law, sociology, theology, economics, ecology, and geography.

Important Points

- Environmental ethics = moral relationship of humans to environment and non-human contents
- Extends ethics beyond humans to include animals, plants, and ecosystems

- Humans are part of nature, not separate from or superior to it
- Key to establishing human-earth relationship in environmental studies
- Ensures humans do their part to keep environment safe and protected
- **Scope of environmental ethics:** Extends to law, sociology, theology, economics, ecology, geography; moral duty to non-human world
- **PYQ Q74 For Exam: Scope of environmental ethics** covers the moral relationship between humans and the environment — extending ethical boundaries to **plants, animals and ecosystems**. It establishes human duties toward nature, guides protection behaviour, and influences **law, sociology, theology, economics, ecology and geography**. Foundational to environmental studies; ensures sustainable coexistence with the non-human world.

For Exam

Environmental ethics is the philosophical discipline studying the moral relationship between humans and the environment, extending ethical consideration to non-human world. It holds that humans are part of nature alongside plants and animals, not above them. It establishes moral values including rights of non-human beings and guides behaviour toward environmental protection. It is foundational to environmental studies and influences multiple disciplines.

Previously Asked Questions

- **Q41** (1 mark, Apr 2025, Aug 2023) ★🔥 — Define environmental ethics.
- **Q51** (2 marks, Aug 2023) — Explain Environmental ethics.
- **Q74** (2 marks, Apr 2025, Aug 2023) ★🔥 — What is the scope of environmental ethics?

4.1.2–4.1.5 Environmental Ethical Perspectives

Theory

Four major ethical perspectives govern how humans relate to the environment. Anthropocentrism places humans as primary moral agents; biocentrism extends moral standing to all living beings; ecocentrism values entire ecosystems including abiotic factors; ecofeminism links the exploitation of

women and nature under patriarchal ideology. Together they form the philosophical foundation of environmental ethics.

Important Points

Environmental Ethical Perspectives

Perspective	Key Idea	Focus
Anthropocentrism	Humans are primary moral agents; superior to nature	Human-centred; nature as life-support system for humans
Biocentrism	All life deserves equal moral consideration (Paul Taylor, 1986)	Living organisms; interdependence of species
Ecocentrism	Ecosystems (living + nonliving) have inherent value	Nature-centred; holistic biodiversity
Ecofeminism	Nature and women linked through patriarchal exploitation (d'Eaubonne, 1974)	Gender–nature link; reproductive/nurturing role

- **Anthropocentrism:** humans as "roof and crown" of creation — led to environmental destruction
- **Biocentrism:** two moral concerns — avoiding harm to sentient beings + preserving nature's purity
- **Ecocentrism:** most holistic; considers abiotic factors (sea levels, ocean acidity, weather patterns)
- **Ecofeminism:** reconceptualize men-women-nature relations in non-hierarchical ways

For Exam

Anthropocentrism is human-centred ethics regarding humans as morally superior, potentially justifying environmental exploitation. Biocentrism (Paul Taylor, 1986) gives equal moral standing to all living beings through four principles emphasizing interdependence and denying human superiority. Ecocentrism values entire ecosystems including abiotic factors, taking a holistic nature-centred approach. Ecofeminism (1974, d'Eaubonne) links exploitation of women and nature through patriarchal ideology, advocating non-hierarchical relationships. Ecocentrism is most holistic; biocentrism focuses on living organisms; anthropocentrism is most human-centred; ecofeminism adds gender dimension.

Previously Asked Questions

- **Q11** (1 mark, Aug 2023) 🔥 — The term Anthropocentrism refers to?
- **Q61** (2 marks, May 2024) 🔥 — Define anthropocentrism.

Diagram (refer SLM): Fig. 4.1.1 Anthropocentrism model; Fig. 4.1.2 Difference between three views; Fig. 4.1.3 Concept comparison.

4.1.6–4.1.7 Environmental Equity, Justice, and Social Security

Theory

Environmental equity ensures equitable distribution of environmental burden, disaster hazards, and pollution across all social, economic, and political sectors — no community privileged over another. It comprises fair treatment (no disproportionate impact) and meaningful involvement (community input considered in decisions). Environmental justice ensures all people enjoy equal protection from environmental hazards regardless of race, class, or origin. Food security means all people have physical and economic access to adequate, nutritious, safe, culturally appropriate food produced sustainably.

Important Points

- **Environmental equity** = equal rights to environmental benefits regardless of disparity; basic human right
- **Environmental justice** = fair treatment and involvement of all communities in environmental law-making
- **Food security** = physical + economic access to adequate nutritious safe food; sustainable production; informed food choices
- **Food system (SLM):** includes production, processing, distribution, marketing, acquisition, and consumption of food
- **Sustainable food systems** require: environmental health, economic vitality, human health & social equity
- **Social security** = state protection against poverty/vulnerability (retirement, disability, death); promotes social transformation; India's shift from family-based to institutional support with urbanization

Sustainable Food Systems (SLM — PYQ Q82)

- **Environmental Health:** Food production and procurement must not compromise land, air, or water — now or for future generations

- **Economic Vitality:** People producing food must earn a decent living wage so they can continue sustainable food production
- **Human Health & Social Equity:** Healthy foods must be available economically and physically to all; community health is prioritised; people access food in a dignified manner
- **Food security link:** Food access depends on food supply; food security requires a healthy and sustainable food system for all people at all times
- **Core principle:** Nutritious, safe, culturally appropriate food produced in environmentally sustainable and socially just manner; producers earn decent living; people make informed food choices
- **PYQ Q82 For Exam:** A healthy, sustainable food system focuses on three pillars (SLM): **Environmental Health** — food production must not harm land, air or water for present or future generations; **Economic Vitality** — food producers must earn a decent living wage to sustain production; **Human Health & Social Equity** — nutritious, safe food must be available economically and physically to all communities with dignified access. Food security means all people always have physical and economic access to adequate, nutritious, safe, culturally appropriate food produced sustainably. The food system spans production, processing, distribution, marketing, acquisition and consumption — sustainable food systems are essential for long-term food security.
- **PYQ Q108 For Exam: Environmental equity and justice** ensure fair distribution of environmental benefits and burdens. **Equity:** all communities deserve clean air, water and safe environment. **Justice:** polluters accountable; vulnerable groups (tribals, poor) protected from disproportionate harm. Links to **food security, climate justice, intergenerational equity**. Environmental decisions must consider social fairness with ecological sustainability.

For Exam

Environmental equity distributes environmental burdens and pollution fairly across all communities — a basic human right with fair treatment and meaningful involvement. Environmental justice ensures equal protection from hazards and participation in environmental decision-making regardless of race, class, or nationality. Food security guarantees access to adequate, nutritious, safe food through sustainable systems balancing environmental health, economic vitality, and social equity. Social security is state protection against

poverty and vulnerability, increasingly important as urbanization weakens traditional family support in India.

Previously Asked Questions

- **Q82** (5 marks, Aug 2023) 🔥 — Discuss sustainable food system.
- **Q108** (5 marks, Apr 2025) — Give an account of environmental equity and justice.

Unit Recap (from SLM)

- Environmental ethics studies moral human-environment relationships including non-human world
- Anthropocentrism = human-centred; humans have primary moral worth
- Biocentrism = all living beings have equal moral standing (Taylor's 4 principles)
- Ecocentrism = ecosystems (living + nonliving) have inherent value; holistic
- Ecofeminism (1974) links women's and nature's exploitation under patriarchy
- Environmental equity = fair distribution of environmental burden across communities
- Environmental justice = equal protection and participation in environmental decisions
- Food security and social security are linked human rights in environmental ethics

Unit 2: Overview of Solid Waste Segregation and Management

4.2.1–4.2.2 Solid Waste Management and Segregation

Theory

Solid waste is unwanted garbage from industrial, residential, and commercial activities. Solid Waste Management (SWM) collects, treats, and disposes this waste through six functional components. **Waste segregation at source** is critical — the **3-bin system** separates waste into three categories for efficient recycling, composting, and disposal. The **3Rs (Reduce, Reuse, Recycle)** are the

core waste-management principles for minimising waste and conserving resources (SLM also adds **Refuse** in zero-waste approach).

Important Points

3Rs in Waste Management (SLM — PYQ Q83)

- **Reduce:** Minimise waste generation at source — consume less, avoid unnecessary products, reduce consumption and discards; fewer products are made when people buy less
- **Reuse:** Use items again before discarding — reusing discards; products made to last; extends product life and reduces waste volume
- **Recycle:** Recover usable materials from waste to manufacture new goods — keeps waste out of landfills and incinerators; provides recycled raw materials instead of virgin resources
- **Refuse (4th R — Zero Waste):** Refuse unnecessary packaging and single-use items; say no to waste-generating products
- **3R goal:** Conserve natural resources; reduce pollution from extraction, manufacturing and disposal; support zero waste and sustainable living
- **SLM principle:** “Reduce, Reuse, Recycle and Refuse are what we need for a safe and healthy living”
- **PYQ Q83 For Exam:** The 3Rs in waste management are **Reduce, Reuse, and Recycle** — fundamental strategies for sustainable solid waste management (SLM adds **Refuse** for zero waste). **Reduce** means minimising waste at source by consuming less and avoiding unnecessary products — fewer products are made when people buy less. **Reuse** means using items repeatedly before discarding — reusing discards and making products last longer. **Recycle** means recovering materials from waste to make new products, keeping waste out of landfills and incinerators and supplying recycled raw materials to manufacturers. Together, 3Rs conserve natural resources, reduce pollution from extraction and disposal, and support zero waste alongside source segregation, composting and safe disposal.

3-Bin Segregation System

- **Green bin:** Wet / Organic waste; Food scraps, vegetable peels, garden waste, biodegradable matter; Composting, biogas, vermicomposting
- **Blue bin:** Dry / Recyclable waste; Paper, cardboard, plastic bottles, metal cans, glass; Recycling, reuse

- **Red bin:** Hazardous / Reject waste; Sanitary waste, diapers, batteries, e-waste, medical waste, broken glass; Incineration, sanitary landfill, special treatment
- Segregation at source prevents mixed waste contamination and enables efficient processing
- Eco-clubs promote: source separation of waste, disposing to nearest storage point
- Swachh Bharat and Haritha Keralam emphasise waste disposal at source of generation

Six Functional Components of SWM

- **1 — Waste generation:** Identifying and gathering unusable materials
- **2 — Onsite handling, storage, processing:** Containers (household bins, communal drums, shallow pits) before collection
- **3 — Waste collection:** Placing bins, collecting waste, loading collection vehicles
- **4 — Waste transfer and transport:** Moving waste to regional disposal site (hand-cart, animal-cart, truck)
- **5 — Waste processing and recovery:** Recovering reusable/recyclable materials
- **6 — Disposal:** Final systematic disposal — landfilling, composting, incineration, recycling
- SWM = effective measures for managing plant, animal, and human waste
- Collection intervals must prevent storage facility overload
- **PYQ Q112 For Exam: Solid waste management strategies: (1) Source segregation — 3-bin (green=wet, blue=dry, red=hazardous); (2) 3Rs — Reduce, Reuse, Recycle; (3) Composting; (4) Recycling; (5) Landfill (reduce over time); (6) Biogas plants; (7) E-waste treatment; (8) Swachh Bharat/Haritha Keralam. Six components: generation → handling → collection → transport → processing → disposal.**

For Exam

Solid waste management uses six functional elements from generation to disposal. The 3-bin segregation system separates waste at source: green bin

(wet/organic — composting), blue bin (dry/recyclable — recycling), and red bin (hazardous/reject — special treatment). Segregation enables efficient composting, recycling, and safe disposal. Six SWM components: generation, onsite handling, collection, transport, processing/recovery, and disposal.

Previously Asked Questions

- **Q75** (2 marks, Apr 2025, Aug 2023) ★🔥 — Define solid waste management.
- **Q83** (5 marks, Aug 2023) 🔥 — Explain 3Rs in waste management.
- **Q112** (10 marks, Aug 2023) 🔥 — Discuss briefly the different solid waste management strategies.

Diagram (refer SLM): Fig. 4.2.1 Flowchart of solid waste management processes.

4.2.3–4.2.4 On-Site and Off-Site Disposal

Theory

On-site disposal handles waste at the generation site. Off-site disposal transports waste to external facilities. Both approaches must consider local conditions, community safety, and environmental impact.

Important Points

- **On-site: Communal pit:** Direct disposal into fenced pit — 6 m³ per 50 people; cover weekly with soil; max 100 m from dwellings
- **On-site: Family pit:** Shallow pit (up to 1 m) per family — Best for large plots with organic waste; cover regularly
- **On-site: Communal bins:** Oil drums (halved, perforated base) — Prevents wind scatter and animal access
- **Off-site: Landfilling:** Waste in ground pit/trench, covered daily — 0.5 m soil cover; fenced sites; consultation with local authorities
- **Off-site: Incineration:** Burning combustible waste off-site — Reduces volume; best for pathogen destruction; cover ash with soil
- **Off-site: Composting:** Organic waste decomposition — Adds humus to soil; central management needs expertise

- **Off-site: Recycling:** Reuse of plastic, tins, glass — Strong tradition in developing countries reduces waste volumes

For Exam

On-site disposal manages waste at source: communal pits (6 m³/50 people), family pits (shallow, organic waste), and communal bins (perforated oil drums). Off-site options include landfilling (daily soil cover, 0.5 m), incineration (destroys pathogens, reduces volume), composting (organic waste to fertilizer), and recycling (plastic, tins, glass). Incineration is best for complete pathogen destruction. Landfill gases include CO₂ and methane.

Previously Asked Questions

- **Q12** (1 mark, Aug 2023) 🔥 — Propose two important methods for the disposal of solid waste.

4.2.5 Zero Waste

Theory

Zero Waste is a philosophy eliminating materials with no viable end-of-use management — designing products and processes to avoid and eliminate waste volume and toxicity. It conserves resources and reduces pollution from extraction, manufacturing, and disposal through reduce, reuse, recycle, and refuse strategies.

Important Points

- Zero Waste = ethical, economical, efficient, visionary goal for sustainable natural cycles
- All discarded materials designed to become resources for others
- No burning or burying; conserve and recover all resources
- Strategies: reusing discards, reducing consumption, producer accountability, comprehensive recycling/composting, citizen participation, ban on incineration/illegal dumping, reducing landfilling over time
- Reduce + Reuse = fewer products made; products made to last; recycling keeps waste from landfills
- **PYQ Q99 For Exam: Zero Waste eliminates waste by designing systems so all materials become resources — no burning or burying. Strategies:**

Reduce consumption; **Reuse**; **Recycle/compost**; **Refuse** (4th R); producer accountability; citizen participation; reduce landfilling. Conserves resources, reduces pollution — ethical, economical, efficient approach to sustainable living.

For Exam

Zero Waste eliminates waste generation by designing products and processes so all materials become resources for reuse. It avoids burning and burying waste, conserving natural resources and reducing pollution. Key strategies include reducing consumption, reusing materials, comprehensive recycling and composting, producer accountability, citizen participation, and progressively reducing landfilling. It follows Reduce, Reuse, Recycle, and Refuse principles for safe, healthy living.

Previously Asked Questions

- **Q99** (5 marks, May 2024) 🔥 — Explain the concept of Zero Waste.

4.2.6–4.2.7 Plastics and Microplastics

Theory

Plastics are polymeric materials (long chains of monomers) used widely due to low cost, durability, and chemical resistance. Microplastics are particles <5 mm ("mermaid's tears") from primary sources (deliberately manufactured — cosmetics, toothpaste) or secondary sources (breakdown of larger plastics, synthetic fibres, tyre wear). They pass through water filtration, enter oceans, and harm aquatic and human health.

Important Points

- **Primary microplastics:** Deliberately manufactured <5 mm — Facial cleansers, toothpaste, microbeads (PE); ~6% EU skin-cleaning products contain microplastics
- **Secondary microplastics:** Breakdown of larger plastics — Degraded bags/bottles; synthetic clothing fibres from washing; tyre wear fragments
- **Effects on wildlife:** Entanglement (79% fatal) and ingestion (4% fatal) — Digestive blockage, reduced nutrition, starvation; gill lodging reduces respiration

- **Human health:** Ingestion of microplastics — Physical blockage, chemical additive leakage, POP adsorption; effects not fully understood
- IUPAC: plastics = polymeric materials with additives for performance/cost
- Found globally: Arctic, Antarctic, oceans, rivers, lakes, soil, atmosphere (indoor + outdoor)
- Entanglement far more fatal (79%) than ingestion (4%) for marine organisms

For Exam

Plastics are synthetic polymers widely used but persist in environment. Microplastics (<5 mm) are primary (manufactured for cosmetics/hygiene — 6% of EU skin products) or secondary (degraded larger plastics, clothing fibres, tyre fragments). They pass water filters, enter oceans and food chains, causing entanglement (79% fatal) and ingestion harm. Effects include digestive blockage, reduced nutrition, respiration impairment, and potential chemical toxicity. Global presence includes oceans, soil, and atmosphere.

Previously Asked Questions

- **Q62** (2 marks, May 2024) — What are microplastics?

Diagram (refer SLM): Fig. 4.2.2 Urban SWM flowchart; Fig. 4.2.4 Fate of microplastics; Fig. 4.2.5 Lifecycle of microplastics.

Unit Recap (from SLM)

- Solid waste = unwanted garbage from human, animal, industrial activities
- Six SWM components: generation, handling, collection, transport, processing, disposal
- 3-bin system: green (wet/organic), blue (dry/recyclable), red (hazardous/reject)
- On-site: communal pit, family pit, communal bins; Off-site: landfill, incineration, composting, recycling
- Zero Waste = eliminate waste through reduce, reuse, recycle; no burning/burying
- Microplastics <5 mm: primary (cosmetics) and secondary (plastic breakdown)

- Incineration best destroys pathogenic bacteria; landfill produces CO₂ and methane
- Sanitary landfill = engineered facility for municipal solid waste disposal

Unit 3: Concept of Global Warming and Climate

4.3.1–4.3.2 Global Warming and Implications

Theory

Global warming is the gradual increase in Earth's surface temperature due to rising greenhouse gases (CO₂, CO) trapping reflected sunlight. Essential in normal proportions for life, excess GHGs cause damaging heat accumulation — the greenhouse effect. IPCC (1998): world already 0.5°C warmer than pre-industrial period. Implications include desertification, droughts, reduced agriculture, polar ice melt, sea level rise, and climate refugees.

Important Points

- Global warming = central environmental issue; earth surface getting progressively warmer
- Sunlight absorbed by GHGs (CO₂) normally warms earth; excess GHGs trap more heat
- Also called greenhouse effect (similar to greenhouse heat trapping)
- Implications: fertile zones → deserts; droughts; reduced agriculture; polar melt → sea level rise
- 5% sea level rise displaces millions in low-tide areas; small island nations may disappear
- **Climate refugees** = people displaced by drought, floods, starvation — most visible symbol of global warming

Causes of Climate Change (SLM — PYQ Q84)

- **1. Rising greenhouse gases:** Increasing CO₂, CO, CH₄, N₂O and other GHGs from human activities
- **2. Fossil fuel burning:** Coal, oil, natural gas combustion since Industrial Revolution (+25% CO₂ since 1750)

- **3. Deforestation:** Removes carbon sinks; releases stored CO₂ into atmosphere
- **4. Industrial & transport emissions:** Factories, power plants, vehicles emit CO₂ and pollutants
- **5. Agriculture & livestock:** Methane (CH₄) from livestock and rice paddies; nitrous oxide (N₂O) from fertilisers
- **6. Greenhouse effect mechanism:** Excess GHGs absorb reflected sunlight → heat trapped → Earth's surface temperature rises gradually (global warming)

Consequences of Climate Change (SLM — PYQ Q84)

- **1. Desertification:** Fertile zones lose moisture → turn into dry lands and deserts
- **2. Droughts:** Many countries suffer prolonged water shortages
- **3. Reduced agriculture:** Crop yields fall; heat waves, floods, pests; lower protein/nutrients in crops
- **4. Ongoing temperature rise:** IPCC (1998) — world already 0.5°C warmer than pre-industrial period
- **5. Polar ice melt & sea-level rise:** Melting polar regions; even 5% sea-level rise displaces millions; small island nations may disappear
- **6. Climate refugees:** People displaced by drought, floods, starvation — most visible symbol of global warming
- **7. Sector impacts:** Livestock heat stress; fisheries species migration; forest wildfires/drought; water quality degradation and saltwater intrusion
- **PYQ Q84 For Exam:** Climate change is caused primarily by increasing greenhouse gas emissions from human activities — fossil fuel combustion, deforestation, industrial and transport emissions, and agriculture (CO₂, methane, N₂O). Excess GHGs strengthen the greenhouse effect, trapping reflected solar heat and raising Earth's temperature. Consequences include desertification and droughts, reduced agricultural production, polar ice melt and sea-level rise (5% rise could displace millions), creation of climate refugees, heat stress on livestock, crop yield losses, forest fires, and degradation of water resources. IPCC notes the world is already 0.5°C warmer than pre-industrial levels.

For Exam

Global warming is the gradual rise in Earth's temperature due to increasing greenhouse gases trapping reflected solar heat — the greenhouse effect. Normal GHG levels sustain life; excess levels cause damaging warming. IPCC notes 0.5°C rise since pre-industrial era. Consequences include desertification, droughts, reduced crop yields, polar ice melt, sea level rise (5% rise displaces millions), and creation of climate refugees displaced by extreme weather and food insecurity.

Previously Asked Questions

- **Q13** (1 mark, Aug 2023) 🔥 — Define global warming.
- **Q78** (2 marks, Apr 2025, Aug 2023) ★🔥 — What are the major implications of global warming?
- **Q84** (5 marks, Aug 2023) 🔥 — What are the causes and consequences of climate change?
- **Q114** (10 marks, Aug 2023) 🔥 — Explain global warming and climate change and propose suitable mitigation measures...

4.3.3–4.3.4 Greenhouse Gases, Greenhouse Effect, and International Agreements

Theory

Major GHG drivers include water vapour (most abundant, positive feedback loop), CO₂ (primary human-emitted, from fossil fuel combustion since Industrial Revolution ~1750), methane (25× more potent than CO₂ over 100 years), nitrous oxide (300× CO₂ warming impact), and fluorocarbons (HFCs, PFCs, SF₆, NF₃ — most potent, no natural sources). John Tyndall (1861) proposed CO₂ concentration raises surface temperature. International agreements — **Kyoto Protocol (1997/2005)** and **Paris Agreement (2015)** — aim to reduce global GHG emissions.

Important Points

Complete GHG List

- **Water vapour:** H₂O; Evaporation (feedback, not direct emission); Days; Most abundant; positive feedback loop

- **Carbon dioxide:** CO₂; Fossil fuels, deforestation, vehicles, industry; 50–200 years; Baseline (1×); +25% since Industrial Revolution
- **Methane:** CH₄; Livestock, natural gas leaks, wetlands, rice paddies; ~12 years; 25× CO₂ over 100 years
- **Nitrous oxide:** N₂O; Agriculture, fuel combustion, wastewater; 114 years; 300× CO₂; 40% from human activities
- **HFCs:** Various; Refrigerants, air conditioning; Years to centuries; Up to 12,000× CO₂
- **PFCs:** Various; Aluminium/semiconductor manufacturing; Thousands of years; Up to 10,000× CO₂
- **SF₆:** SF₆; Electrical equipment; 3,200 years; 23,900× CO₂
- **NF₃:** NF₃; Electronics manufacturing; 740 years; 17,200× CO₂
- **CFCs:** CFCs; Aerosols, refrigerants (now regulated); 50–100 years; Ozone-depleting + GHG

International Climate Agreements

- **UNFCCC:** 1992 (Rio Earth Summit) — United Nations Framework Convention on Climate Change; foundation for climate negotiations
- **Kyoto Protocol:** Adopted 1997; effective 2005 — Legally binding emission reduction targets for developed nations; origin of carbon trading; Clean Development Mechanism (CDM) for developing countries; Marrakesh Accords formalised carbon credit market; divided countries into industrialised (emission trading) and developing (CDM)
- **Paris Agreement:** 2015 (COP21) — All nations set voluntary emission reduction targets (NDCs — Nationally Determined Contributions); aim to limit warming to well below 2°C, pursue 1.5°C; global stocktake every 5 years; no legally binding targets (unlike Kyoto)
- **Montreal Protocol:** 1987 — Phase out ozone-depleting substances (CFCs, halons); indirect climate benefit
- GHG concentrations measured in ppm, ppb, ppt
- Greenhouse effect: CO₂, CFCs, N₂O, CH₄ form heat-trapping cloud over earth
- UNFCCC = United Nations Framework Convention on Climate Change

For Exam

Major greenhouse gases: water vapour (most abundant, feedback loop), CO₂ (primary human emission, +25% since 1750), methane (25× CO₂ potency), nitrous oxide (300× CO₂), and fluorocarbons (highest GWP, human-only). Kyoto Protocol (1997/2005) set binding emission targets for developed nations and created carbon trading/CDM. Paris Agreement (2015) requires all nations to submit NDCs aiming to limit warming below 2°C. UNFCCC (1992) is the foundation convention. Montreal Protocol (1987) phased out ozone-depleting substances.

Previously Asked Questions

- **Q76** (2 marks, Apr 2025, Aug 2023) ★🔥 — List out the major greenhouse gases.
- **Q84** (5 marks, Aug 2023) 🔥 — What are the causes and consequences of climate change?

Diagram (refer SLM): Fig. 4.3.1 GHG concentrations; Fig. 4.3.2 Greenhouse effect and global warming.

4.3.5–4.3.7 Climate Change Impacts on Sectors

Theory

Climate change affects agriculture, livestock, fisheries, forests, and water resources globally. Agriculture faces yield changes, pest proliferation, and nutritional decline. Livestock suffers heat stress and drought-reduced forage. Fisheries see species migration and disease outbreaks. Forests experience altered growing seasons, drought, wildfire risk, and species range shifts. Water resources face changed evaporation, flooding, quality degradation, and saltwater intrusion.

Important Points

Impacts on Crops

- **Higher CO₂ effect:** May increase plant growth in lab, but reduced protein/nitrogen in alfalfa, soybean
- **Temperature extremes:** Prevent crop growth; heat waves reduced US corn yields; Michigan cherries lost \$220M (2012)

- **Floods and droughts:** Harm crops; waterlogged roots cannot breathe; prolonged flooding fatal to plants
- **Pest proliferation:** Warmer temps help weeds, pests, fungi; higher pesticide rates needed
- **Nutritional decline:** Rising CO₂ reduces protein and essential minerals in wheat, soybeans, rice
- **Irrigation needs:** Changing rainfall may increase irrigation demand in future

Impacts on Livestock, Fisheries, Forests, and Water

- **Livestock:** Heat stress reduces fertility and milk production; drought reduces forage quality; parasites increase with rainfall; pesticide residues risk in food chain
- **Fisheries:** Species migrate to colder waters; oyster parasites spread in warmer estuaries; reproduction/migration timing disrupted by temperature
- **Forests:** Longer growing seasons; species shift north/upslope; drought increases wildfire; CO₂ boosts growth in nutrient-rich areas only
- **Forest disturbances:** Insect outbreaks, invasive species, wildfires, storms alter forest composition; drought + fire interaction worsens damage
- **Forest sector economy:** Timber supply shifts poleward; prices may fall; biofuel demand increases wood biomass use
- **Water: cycle:** Warmer temps increase evaporation; changed rainfall patterns; more intense storms
- **Water: supply:** Low water and droughts affect agriculture, forestry, energy, drinking water; groundwater recharge reduced
- **Water: quality:** Heavy rain overwhelms sewer systems; runoff carries sediment, nutrients, pollutants; sea level rise causes saltwater intrusion into coastal aquifers
- Climate change threatens global food security; may cause massive agricultural disruption
- Plant breeders developing drought-tolerant crop varieties
- Heat waves during pollination/pod-set most damaging to crop yields
- **PYQ Q119 For Exam: Climate change impacts: Crops — reduced yields, pests, lower protein in wheat/rice, extreme weather. Livestock — heat**

stress, reduced forage, disease. **Fisheries** — species migration, parasite spread. Threatens **global food security**. Forests: range shifts, drought, wildfire. Water: changed rainfall, flooding, saltwater intrusion.

For Exam

Climate change impacts multiple sectors. Crops face reduced yields, pest increase, nutritional decline (lower protein in wheat/rice), and extreme weather damage. Livestock suffers heat stress, reduced forage, and disease. Fisheries see species migration and parasite spread. Forests experience species range shifts, drought, wildfire, and altered timber supply. Water resources face changed rainfall, flooding, quality degradation, saltwater intrusion, and groundwater depletion. Overall impacts threaten global food security.

Previously Asked Questions

- **Q119** (10 marks, May 2024) 🔥 — Write an essay on major impacts of climate change on crops, livestock and fisheries.

Diagram (refer SLM): Fig. 4.3.3 Impact on agriculture; Fig. 4.3.4 Potential impacts on environments; Fig. 4.3.5 Impact on water resources.

4.3.8–4.3.11 Management, Acid Rain, and Ozone Depletion

Theory

Climate management requires reducing CO₂ emissions, national emission strategies, international cooperation, public awareness, and global environmental ethics. Acid rain (coined 1852, Robert Angus Smith) is precipitation with sulfuric/nitric acid from SO₂ and NO₂ emissions, harming vegetation and ecosystems — severe in Mumbai, Delhi, Kanpur, Bangalore, Ahmedabad, Kolkata. Ozone layer (stratosphere, 15–30 km) absorbs 97–99% UV radiation; depletion by chlorine/bromine from ODS (CFCs, halons); Montreal Protocol (1987) phased out ODS.

Important Points

Eight Management Options to Tackle Climate Change

- **1.** Radical re-thinking of western industrialised lifestyle (reduce fossil fuel dependence)
- **2.** Reduce CO₂ emissions — close fossil fuel power stations, replace with nuclear/renewable

- **3.** Industrialised nations assist developing countries in limiting emissions
- **4.** All countries implement national strategies to limit toxic emissions
- **5.** Recognise climatic changes in economic, social, and environmental policies
- **6.** Increase public awareness through education and training on global warming
- **7.** Continuous international research cooperation on climate change
- **8.** Develop global environmental ethics — share resources equitably, not just consume
- **Acid rain:** $\text{SO}_2 + \text{NO}_2 \rightarrow$ sulfuric/nitric acid precipitation; coined 1852 by Robert Angus Smith; harms trees, lakes, soil minerals; reduces aquatic life resistance; Indian cities: Mumbai, Delhi, Kanpur, Bangalore, Ahmedabad, Kolkata
- **Acid rain control:** Reduce CO_2 , SO_2 , NO_2 emissions; regulate gas spread in atmosphere; control emissions from power plants and refineries
- **Ozone layer:** Stratosphere (15–30 km altitude); absorbs 97–99% UV radiation; ozone = O_3 molecule
- **ODS (Ozone Depleting Substances):** CFCs, carbon tetrachloride, HCFCs, methyl chloroform (chlorine); halons, methyl bromide (bromine); one chlorine destroys 100,000 ozone molecules
- **Ozone depletion effects:** Humans: skin cancer, sunburn, cataract, weak immunity; Animals: skin/eye cancer; Plants: reduced growth/photosynthesis; Marine: plankton destruction
- **Montreal Protocol:** 1987; stop ODS production/import; minimize atmospheric concentration; Antarctic ozone “hole” each spring since early 1980s
- **Mt. Pinatubo (1991):** Volcanic aerosols temporarily accelerate ozone depletion

Global Warming & Mitigation (SLM — PYQ Q114)

- **Global warming:** Gradual rise in Earth’s temperature; GHGs trap reflected solar heat; IPCC: +0.5°C since pre-industrial era

- **Mitigation = Eight Management Options** (see numbered list above): reduce emissions, national strategies, international cooperation, public awareness, global environmental ethics
- **PYQ Q114 For Exam: Global warming** is the gradual increase in Earth's surface temperature as greenhouse gases (CO₂, CH₄, N₂O) trap reflected solar heat — the greenhouse effect (IPCC: already 0.5°C warmer than pre-industrial). **Mitigation measures:** (1) rethink fossil-fuel-dependent lifestyle (2) reduce CO₂ — shift to renewable/nuclear (3) developed nations assist developing countries (4) national emission strategies (5) integrate climate into economic/social/environmental policy (6) public education on global warming (7) international research cooperation (8) develop global environmental ethics for equitable resource sharing.
- **PYQ Q100 For Exam: Eight management options to tackle climate change:** (1) rethink fossil-fuel lifestyle; (2) reduce CO₂ — renewable/nuclear; (3) developed nations assist developing countries; (4) national emission strategies; (5) integrate climate into economic/social/environmental policy; (6) public education on global warming; (7) international research cooperation; (8) global environmental ethics — equitable resource sharing.

For Exam

Climate change management requires eight approaches: reduce CO₂ emissions, national emission strategies, international cooperation, public education, and global environmental ethics. Acid rain (Robert Angus Smith, 1852) forms when SO₂ and NO₂ create sulfuric/nitric acid precipitation, damaging forests, lakes, and ecosystems — affecting Indian cities like Mumbai and Delhi. Ozone layer in stratosphere (15–30 km) absorbs 97–99% UV radiation; depleted by ODS (CFCs, halons) — one chlorine destroys 100,000 ozone molecules. Effects include skin cancer, cataracts, and marine plankton destruction. Montreal Protocol (1987) phased out ODS globally.

Previously Asked Questions

- **Q42** (1 mark, Apr 2025) — Which gases are the main contributors of acid rain?
- **Q50** (2 marks, Aug 2023) — Explain any four effects of ozone layer depletion.
- **Q100** (5 marks, May 2024) 🔥 — Briefly describe different management options to tackle climate change.

- **Q114** (10 marks, Aug 2023) 🔥 — Explain global warming and climate change and propose suitable mitigation measures...

Unit Recap (from SLM)

- Global warming = increasing average air temperatures from GHG trapping heat
- Major GHGs: water vapour, CO₂, methane, N₂O, fluorocarbons
- Impacts: temperature rise, rainfall changes, droughts, sea level rise, climate refugees
- Climate change affects agriculture, livestock, fisheries, forests, and water resources
- Acid rain from SO₂/NO₂ emissions damages ecosystems (Robert Angus Smith, 1852)
- Ozone depletion by ODS (CFCs); Montreal Protocol (1987) for protection
- Ozone layer in stratosphere absorbs 97–99% harmful UV radiation
- Management: reduce emissions, international cooperation, public awareness

Unit 4: Concept and Definition of Carbon Usage

4.4.1 Carbon Sequestration

Theory

Carbon sequestration is long-term storage of carbon in plants, soils, geologic formations, and oceans to reduce atmospheric CO₂ and global climate change. It occurs naturally (photosynthesis) and artificially (capturing industrial CO₂ for underground storage). Types include biological (oceans, soils, forests, grasslands), geological, industrial (pre-combustion, post-combustion, oxyfuel), and technological (DAC, graphene, engineered molecules).

Important Points

- **Biological: Oceans:** Absorb ~25% of human CO₂ emissions — Colder, nutrient-rich polar regions are best carbon sinks; positive/negative flux

- **Biological: Soils:** SOC via photosynthesis; carbonates in arid soil — Agricultural practices restore depleted SOC; caliche in desert soils
- **Biological: Forests:** Capture ~25% of global emissions — Carbon released on fires/deforestation; trees = key carbon sinks
- **Biological: Grasslands:** More reliable than forests under climate stress — Carbon stays in roots/soil during burns; less vulnerable to wildfire
- **Geological:** Store CO₂ in underground rock formations — From industrial sources (steel, cement, power plants); injected into porous rocks
- **Industrial: Pre-combustion:** Remove carbon from fuel before burning — Coal reacted with oxygen → synthesis gas (CO + H₂)
- **Industrial: Post-combustion:** Scrub CO₂ from flue gases after burning — Waste gases captured and cleaned at smokestacks
- **Industrial: Oxyfuel:** Burn fuel in pure oxygen; capture all gases — All combustion products stored
- **Technological: DAC:** Direct Air Capture from atmosphere — Energy-intensive and costly at mass scale
- **Technological: Engineered molecules:** Molecules shaped to capture CO₂ from air — Filter approach targeting CO₂ specifically

Advantages of Carbon Sequestration

- Promotes tree planting and forest preservation (forests capture 25% of emissions)
- Reduces global warming (45% of CO₂ stays in atmosphere; sequestration removes it)
- Reduces ocean acidification (~30% of CO₂ absorbed by oceans raises acidity)
- Industrial carbon capture from coal plants and smokestacks

Disadvantages of Carbon Sequestration

- Reservoir leaks via rock fissures and tectonic processes may release stored CO₂
- Adds 1–5 cents/kWh to energy cost; nearly doubles modern coal technology cost
- DAC and technological methods too costly for mass-scale implementation

- **PYQ Q85 For Exam: Carbon sequestration** stores CO₂ to reduce atmospheric levels. **Ways: (1) Biological** — oceans (~25%), forests (~25%), soil/grassland; **(2) Geological** — inject CO₂ underground; **(3) Industrial pre-combustion** — capture before burning; **(4) Post-combustion/oxyfuel** — scrub flue gas. Advantages: reduces warming, reforestation; disadvantages: leak risk, increased cost.

For Exam

Carbon sequestration stores carbon in plants, soils, oceans, and geological formations to reduce atmospheric CO₂. Biological methods: oceans (25% absorption), soil organic carbon, forests (25% capture), grasslands (more climate-resilient). Geological: inject industrial CO₂ underground. Industrial capture: pre-combustion, post-combustion, oxyfuel. Advantages: reforestation, warming reduction, less ocean acidification. Disadvantages: leak risk, increased energy costs (1–5 cents/kWh).

Previously Asked Questions

- **Q85** (5 marks, Aug 2023) — What are the different ways of Carbon sequestration?

Diagram (refer SLM): Fig. 4.4.1 Biological sequestration; Fig. 4.4.2 Geological sequestration; Fig. 4.4.3 Industrial sequestration.

4.4.2 Carbon Footprint and Reduction Methods

Theory

Carbon footprint is the total greenhouse gases — primarily carbon dioxide — released into the atmosphere by a particular human activity, measured in tonnes of CO₂ per year. According to WHO, it measures the impact of activities on CO₂ produced through burning fossil fuels. It includes direct emissions (transport, heating) and indirect emissions (electricity, goods consumed, food production).

Important Points

Carbon Footprint Calculator Factors

- Home electricity use (depends on fuel type of power plant and consumption amount)
- Furnace/boiler emissions (efficiency, house size, insulation, fuel type)

- Vehicle use (fuel efficiency, distance driven, idling in traffic)
- Recycling habits (reducing waste sent to landfills reduces GHG from raw material processing)

Methods of Reducing Carbon Footprint

- **Drive efficient vehicles:** Lower fuel consumption = less CO₂ per km
- **Maintain vehicles properly:** Well-tuned engines burn fuel more efficiently
- **Use public transportation:** Shared transport reduces per-person emissions
- **Use energy-efficient appliances:** Less electricity consumed = less power plant emissions
- **Insulate home:** Reduces heating and air conditioning energy needs
- **Switch to LED lighting:** Transition from incandescent bulbs saves energy
- **Eat less meat:** Meat production has higher carbon footprint than fruits/vegetables
- **Buy local food:** Less transportation = less food-mile emissions
- **Recycle and reuse:** Reduces manufacturing emissions from raw material extraction
- **Purchase low-carbon products:** Manufacturers publishing carbon footprints enable informed choice
- **Buy carbon offsets:** Fund tree planting or renewable energy projects
- **Use farmer's market on foot:** Zero transport emissions for food purchase
- Carbon footprint includes methane, N₂O, CFCs — expressed as CO₂ equivalent (tonnes)
- Offsetting: purchase carbon credits funding tree planting or renewable energy projects

For Exam

Carbon footprint measures total GHGs (tonnes CO₂/year) from human activities including transport, heating, electricity, and consumption. Reduction methods include driving efficient vehicles, using public transport, insulating homes, energy-efficient appliances, LED lighting, eating less meat, buying local food, recycling, and purchasing low-carbon products. Carbon offsets allow

individuals to fund tree planting or renewable energy. WHO defines it as CO₂ from fossil fuel burning in tonnes.

Previously Asked Questions

- **Q77** (2 marks, Apr 2025, Aug 2023) ★ — What is meant by Carbon footprint?

Carbon Credits and Trading

Carbon footprint is total greenhouse gases (primarily CO₂) released by an activity, individual, organization, or nation — measured in tonnes CO₂/year. Carbon credits are tradable permits allowing emission of one tonne CO₂ equivalent. Carbon trading (originated in Kyoto Protocol, 2005) buys/sells these credits to reduce overall emissions. Cap-and-trade sets emission caps declining annually; companies with surplus sell to those exceeding limits.

Important Points

- **Carbon footprint:** GHGs released by human activity (tonnes CO₂/year) — Direct emissions (transport, heating) + indirect (electricity, goods consumed); reduce via efficient vehicles, public transport, insulation, less meat, recycling
- **Carbon credit:** Tradable permit to emit 1 tonne CO₂ equivalent — VER (voluntary market) vs CER (certified, regulatory); formalized in Kyoto Protocol/Marrakesh Accords
- **Carbon trading:** Buying/selling emission credits — Kyoto Protocol (2005); ERPA contract; CDM for developing countries (CER credits)
- **Cap and trade:** Government sets emission cap; declining annually — Surplus sellers help over-emitters comply; aggregate pollution below baseline
- **Carbon offset:** Purchase credits funding tree planting/renewable energy — Individuals and companies offset unavoidable emissions

Carbon Credit Trading Example (Cap-and-Trade)

- **1. Government allocation** — Receives 1000 credits (cap). Receives 1000 credits (cap)
- **2. Actual emissions** — Emits only 700 tonnes CO₂. Emits 1200 tonnes CO₂
- **3. Credit status** — **Surplus. 300 credits: Deficit: 200 credits** (exceeds cap)

- **4. Options for B** — —. Pay government fine OR buy credits
- **5. Trade agreement** — Sells 200 surplus credits to B. Buys 200 credits from A
- **6. Outcome** — Earns money + positive image. Avoids fine; meets compliance
- **7. Net effect** — Total emissions capped at 2000 tonnes. Market incentivises emission reduction
- 2016 global CO₂ emissions ~36 billion metric tonnes
- Kyoto Protocol divided countries: industrialised (emission trading market) vs developing (CDM/CER credits)
- ERPA (Emission Reduction Purchase Agreement) = legal contract for carbon credit trading
- China started national emissions-trading programme (July 2021)

For Exam

Carbon footprint measures GHGs released (tonnes CO₂/year). Carbon credits are tradable permits for one tonne CO₂, created under Kyoto Protocol. In cap-and-trade: if Company A emits less than its cap (surplus credits), it sells to Company B which exceeds its cap — B avoids fines, A earns revenue, and total emissions stay within the national cap. Credit types: VER (voluntary) and CER (certified via CDM). Carbon trading incentivises emission reduction through market mechanisms.

4.4.6–4.4.10 Environmental Economics, Green Economy, and Circular Economy

Theory

Environmental economics applies economic tools to allocate environmental resources efficiently, addressing market failure, externalities, and sustainable development. Originated in 1960s industrialisation concerns; led to UNEP (1972). Green economy harmonizes human-nature interaction with full-cost accounting. Circular economy extends product life through sharing, leasing, reusing, repairing, and recycling — designing out waste with renewable energy and non-toxic materials.

Important Points

Five Concepts of Environmental Economics

- **Sustainable development:** Meets present needs without compromising future generations — Four components: economic growth, environmental protection, social equity, institutional capacity
- **Market failure:** Perfect market cannot efficiently allocate environmental resources — Clean oceans — no market to price degree of cleanliness
- **Externalities:** Unintended consequences of economic activity affecting third parties — Negative: industrial pollution harms community; Positive: community park benefits visitors (free riders)
- **Valuation:** Assigning economic value to environmental resources — Use method (willingness to pay); non-use method (contingent valuation)
- **Cost-benefit analysis (CBA):** Weighing policy benefits against costs with discount factor — Best policy = greatest surplus of benefits over costs
- **Environmental economics:** Allocates environmental resources efficiently; prescriptive or incentive-based approaches; 3Es: Energy, Economy, Environment
- **Green economy:** Harmonious human-nature interaction; natural capital has economic value; full-cost accounting; polluters held liable (Blueprint for Green Economy, UK 1989); low carbon, resource efficient, socially inclusive
- **Circular economy:** Share, lease, reuse, repair, refurbish, recycle; design out waste; consumables = biological nutrients; durables = technical nutrients for reuse; C2C = Cradle to Cradle
- **Circular principles:** (1) Design out waste — products optimised for disassembly; (2) Separate consumables (biological) from durables (technical); (3) Renewable energy fuels the cycle
- Environmental economics challenges: transnational nature of pollution; impact on multiple society sectors
- Green economists advocate full cost accounting — entities harming natural assets held liable
- Circular economy replaces “end-of-life” concept with restoration; eliminates toxic chemicals
- **PYQ Q123 For Exam: Environmental economics** applies economic tools to environmental problems: **market failure, externalities** (pollution = negative

externality), **valuation**, **cost-benefit analysis**. **Green economy** — full-cost accounting, polluters liable. **Circular economy** — share, reuse, repair, recycle. **3Es**: Energy, Economy, Environment. Sustainable resource allocation for current and future generations.

For Exam

Environmental economics (1960s origin, UNEP 1972) applies economic analysis to environmental resource allocation, addressing market failure (no market for clean oceans), externalities (pollution as negative externality), valuation, and cost-benefit analysis. Green economy harmonizes human-nature needs with full-cost accounting where polluters are liable. Circular economy minimizes waste through sharing, reusing, repairing, and recycling — three principles: design out waste, separate biological/technical nutrients, use renewable energy. 3Es: Energy, Economy, Environment.

Previously Asked Questions

- **Q48** (2 marks, Aug 2023) — Define the term green economy.
- **Q123** (10 marks, Apr 2025, Aug 2023) ★ — Write an essay on environmental economics.

Diagram (refer SLM): Fig. 4.4.5 Environment-social-economic interlinking; Fig. 4.4.6 Green economy; Fig. 4.4.7 Linear vs circular economy; Fig. 4.4.8 Circular economy elements.

4.4.11–4.4.12 Environmental Impact Assessment (EIA)

Theory

EIA evaluates likely environmental impacts of proposed projects before decision-making, considering socio-economic, cultural, and health impacts. UNEP defines EIA as a tool identifying environmental, social, and economic impacts prior to decisions. It predicts effects early, proposes mitigation, and enables environmentally sound project design. The EIA process is cyclical with interaction between various steps.

Important Points

EIA Process Steps (Complete Table)

- **1: Screening**; Project plan screened for scale of investment, location, type of development; determine if statutory clearance needed; Decision: EIA

required or exempt

- **2: Scoping;** Identify potential impacts, zone of impacts, mitigation possibilities, monitoring needs; Terms of Reference (TOR) for EIA study
- **3: Baseline data collection;** Document environmental status of study area — air, water, soil, biodiversity, noise, socio-economic conditions; Baseline environmental report
- **4: Impact prediction;** Predict positive/negative, reversible/irreversible, temporary/permanent impacts on all environmental components; Impact prediction matrix
- **5: Mitigation measures;** Propose prevention, minimisation, or bypassing of adverse impacts; compensation for environmental damage; Mitigation plan
- **6: EIA report preparation;** Compile all findings into formal Environmental Impact Assessment report; Draft EIA report
- **7: Public hearing;** Inform and consult public and environmental groups living near project site; collect objections/suggestions; Public hearing proceedings
- **8: Decision making;** Impact Assessment Authority reviews EIA + EMP (Environment Management Plan) with experts and project proponent; Grant or reject environmental clearance
- **9: Monitoring & EMP implementation;** Monitor compliance during all project phases — construction, operations, decommissioning; Compliance reports
- **10: Assessment of alternatives;** Compare project location and process technology alternatives; select least-impact option; Alternative analysis
- **11: Risk assessment;** Inventory analysis; hazard probability and index for catastrophic risks; Risk assessment report

EIA Stakeholders

- Project proponent (applicant)
- Environmental consultant (prepares EIA on behalf of proponent)
- Pollution Control Board — CPCB (central) / SPCB (state)
- Public (right to express opinion at hearing)
- Impact Assessment Authority

- MoEFCC Regional Centre

EIA Benefits: cost-effective impact elimination; informed decision-making; community participation; resource optimization; protection of environment; saves time and cost

EIA Shortcomings: exempt projects; inexperienced expert committees; inadequate public hearing; fraudulent data; confidential EMPs for strategic industries (nuclear)

- **PYQ Q86 For Exam: Importance of EIA:** evaluates environmental, social and economic impacts **before** project approval — informed decisions, cost-effective impact prevention, **community participation**, resource optimization, sustainable development. Identifies adverse effects early for mitigation. Stakeholders: proponent, consultant, CPCB/SPCB, public, MoEFCC. Prevents irreversible environmental damage from unchecked development.
- **PYQ Q63 For Exam: Environmental Impact Assessment (EIA)** evaluates likely **environmental, social and economic impacts** of proposed projects **before** decision-making. **Aim:** enable informed decisions, identify adverse effects early, incorporate **mitigation measures**, ensure **community participation**, optimize resources and promote **sustainable development**. UNEP defines EIA as a tool for identifying impacts prior to decisions.

For Exam

Environmental Impact Assessment (EIA) evaluates environmental, social, and economic impacts before project approval. Key stages: (1) Screening — determine if EIA needed; (2) Scoping — identify impact zone; (3) Baseline data — document existing environment; (4) Impact prediction; (5) Mitigation measures; (6) EIA report; (7) Public hearing; (8) Decision making by Impact Assessment Authority; (9) Monitoring and EMP implementation; (10) Alternative assessment; (11) Risk assessment. Stakeholders include proponent, consultant, CPCB/SPCB, public, and MoEFCC. Benefits include cost-effective protection and community participation.

Previously Asked Questions

- **Q43** (1 mark, Apr 2025, Aug 2023, May 2024) ★🔥 — How many stages are there in EIA? What are they?
- **Q63** (2 marks, May 2024) 🔥 — What is Environmental Impact Assessment? What is its aim?

- **Q86** (5 marks, Aug 2023) 🔥 — Write a short note on Importance of EIA.

Diagram (refer SLM): Fig. 4.4.9 Scoping stages; Fig. 4.4.10 EIA process stages.

Unit Recap (from SLM)

- Carbon sequestration stores carbon in plants, soils, oceans, and geological formations
- Four sequestration approaches: biological, geological, industrial, technological
- Carbon footprint = GHGs released by activity; carbon credit = permit to emit 1 tonne CO₂
- Carbon trading under Kyoto Protocol; cap-and-trade sets declining emission caps
- Environmental economics addresses market failure, externalities, and resource valuation
- Green economy harmonizes human-nature interaction; circular economy eliminates waste
- EIA evaluates project impacts before decision — screening through monitoring
- EIA benefits: cost-effective protection, community participation, informed decisions

Unit 5: A Brief Overview of Prominent Natural Disasters in India

4.5.1–4.5.3 Landslides

Theory

A landslide is mass movement of rock, debris, or earth down a slope under gravity (mass wasting). Caused by slope oversteepening, water saturation, earthquakes, volcanic activity, weathering, and human activity (mining, deforestation). Types by movement: slides, flows, spreads, topples, falls. Mitigation includes restricting populations in hazard zones, early warning systems, slope modification, retaining walls, and drainage diversion.

Important Points

Causes and Susceptibility Factors

- **Heavy rainfall:** Saturates slope material; reduces friction and cohesion; adds weight
- **Erosion:** Undercuts slope base (river, sea, excavation); slope becomes too steep
- **Earthquakes:** Short-term stress triggers slope failure; common in hilly regions
- **Volcanic activity:** Seismic shaking + debris accumulation on slopes
- **Deforestation:** Removes vegetation binding soil; increases surface runoff
- **Mining/traffic vibrations:** Disturbs slope stability; alters drainage patterns
- **Weathering:** Freeze-thaw reduces cohesion between rock grains
- **Slope angle:** Steeper slopes more susceptible; coastal erosion increases angle
- **Rock type:** Sand: high friction, low cohesion; Clay: high cohesion, low friction
- **Water content:** Weakens sandy materials (reduces interparticle friction); dissolves clay cements

Types of Landslides

Type	Movement	Material / Feature
Slides	Along discrete shearing surface	Translational or rotational/slump
Flows	Viscous fluid-like movement	Debris, earth, mud; water fluidizes
Spreads	Lateral movement on weak substrate	Liquefaction; riverbank retrogression
Topples	Outward rotation of rock mass	Rotation from steep slope face
Falls	Free fall of rock/debris	Detachment from steep cliff or slope

Mitigation and Prevention

- Hazard zoning: restrict populations in landslide-prone areas
- Early warning: monitor strain in rocks/soils, slope displacement, groundwater levels

- Engineering: retaining walls, piles, grouting rock joints, debris pathway diversion
- Drainage: reroute surface and underwater drainage away from unstable slopes
- Slope geometry modification; chemical agents to reinforce slope material

For Exam

A landslide is gravity-driven mass movement of rock, debris, or earth down a slope. Causes include heavy rainfall (water saturation), erosion, earthquakes, deforestation, and mining. Five types: slides (translational/rotational), flows (viscous, water-fluidized), spreads (lateral on liquefied substrate), topples (outward rotation), and falls. Water weakens sandy slopes (reduces friction) and clay slopes (dissolves cements). Mitigation uses hazard zoning, early warning systems, retaining walls, grouting, and drainage management.

Diagram (refer SLM): Fig. 4.5.1 Landslide in South Asia; Fig. 4.5.2 Types of landslides.

4.5.4 Floods

Theory

A flood is water overflow submerging normally dry land — most common weather-related natural disaster. Caused by massive rainfall, river overflow, dam collapse, snowmelt, deforestation, climate change, and greenhouse gas emissions. Five types: river flood, coastal flood, storm surge, inland flood, and flash flood.

Important Points

Causes of Floods (SLM — PYQ Q101)

#	Cause	Mechanism
1	Heavy rainfall	Drainage systems overwhelmed; excess water cannot be stored
2	River overflow	Rivers exceed bank capacity; dams mismanaged
3	Dam/levee collapse	Structural failure releases stored water suddenly
4	Snow/glacier melt	Rapid melting increases river water volume

#	Cause	Mechanism
5	Deforestation	Loss of trees increases surface runoff and erosion
6	Climate change	Altered rainfall patterns; more extreme weather events
7	Urbanisation	Paved surfaces and poor drainage worsen inland flooding
8	Coastal/storm surge	Abnormal sea-level rise during cyclones and high tides

Five Types of Flooding

Type	Description	Cause / Example
River flood	Water exceeds river banks	Tropical storms, snowmelt (Kerala 2018)
Coastal flood	Land submerged by seawater	Tidal surges, high winds, tsunamis
Storm surge	Abnormal coastal water rise above tide	Wind + waves + low pressure
Inland flood	Not coastal; rainfall/snowmelt flooding	Worse in urban areas with poor drainage
Flash flood	Begins within 3–6 hours of heavy rain	Dam/levee failure, ice jam release

- Chennai floods (2016), Kerala floods (2018, 2019) are prominent Indian examples
- Flood size measured by peak water level ('flood peak')
- Environmental consequences: weed dispersal, soil erosion, pollutant release, habitat destruction
- Flash floods most dangerous due to sudden onset with little warning
- **PYQ Q101 For Exam:** The main **causes of floods** include **heavy rainfall**, **river overflow**, **dam or levee collapse**, **snow/glacier melt**, **deforestation** (increases surface runoff), **climate change** (extreme weather), **urbanisation** (paved surfaces and poor drainage), and **storm surges** along coasts. Deforestation and climate change increase flood risk. Indian examples: Chennai 2016, Kerala 2018–19.

For Exam

Floods overflow water onto normally dry land — the most common natural disaster. Eight causes include heavy rainfall, river overflow, dam collapse, snowmelt, deforestation, climate change, and urbanisation. Five types: river flood (water exceeds banks), coastal flood (sea submergence), storm surge (abnormal coastal rise), inland flood (rainfall/snowmelt, worse in cities), and flash flood (within 3–6 hours). Indian examples: Chennai 2016, Kerala 2018–19. Deforestation and climate change increase flood risk.

Previously Asked Questions

- **Q44** (1 mark, Apr 2025) — What is a storm surge?
- **Q101** (5 marks, May 2024) — Explain the different causes of flood.

Diagram (refer SLM): Fig. 4.5.3 River flooding (Kerala 2018); Fig. 4.5.4 Coastal flood; Fig. 4.5.5 Storm surge; Fig. 4.5.6 Inland flooding.

4.5.5–4.5.7 Cyclones and Earthquakes

Theory

Cyclones are atmospheric disturbances around low-pressure areas with destructive circular winds. Tropical cyclones (10° – 25° N/S) are called hurricanes/typhoons in northern hemisphere (anti-clockwise); cyclones in southern hemisphere (clockwise). Extratropical cyclones form in mid-latitudes at frontal zones. Earthquakes are sudden ground shaking from seismic waves when rock masses fracture and slip along fault planes — recorded by seismographs; located by triangulation using three seismographs.

Important Points

Cyclones

- **Tropical cyclone:** Forms over warm tropical oceans (10° – 25° N/S); winds >119 km/h (74 mph); energy from warm sea surface; organised surface wind circulation — Bulbul struck West Bengal (2019); rotates clockwise in Southern Hemisphere, anti-clockwise in Northern Hemisphere
- **Extratropical cyclone:** Mid/high latitude frontal zones; wave cyclones; less violent than tropical; temperature variations drive formation — Common globally at mid-latitudes
- **Eye of cyclone:** Calm, cloudless centre; no rain; very light winds; low pressure area — Most distinctive feature of cyclone structure

- **Storm components:** Wind + waves + low atmospheric pressure create storm surge — Can exceed 20 feet when combined with high tide

Earthquakes

- **Hypocentre (Focus):** Location below earth's surface where earthquake starts — Point of initial rock fracture
- **Epicentre:** Location directly above hypocentre on earth's surface — Area of maximum surface damage; located by triangulation
- **Fault / Fault plane:** Fracture surface where rock masses slip past each other — Normal fault: hanging wall moves down relative to footwall
- **P waves (Primary):** Longitudinal body waves; fastest; travel through solids and liquids — First waves to hit seismographs; direction of motion = direction of propagation
- **S waves (Secondary/Shear):** Transverse body waves; slower than P waves; travel through solids only — Motion perpendicular to propagation; used to map earth's interior
- **Surface waves:** Travel on earth's surface; cause maximum destruction — Most damaging to buildings and infrastructure
- **Seismograph:** Instrument recording earthquake waves — Base fixed to ground; heavy weight hangs free on spring; records as seismogram
- **Triangulation:** Method using 3 seismographs to locate epicentre — Requires three stations at different locations
- Cyclones classified: (i) tropical cyclones, (ii) extratropical cyclones
- Earthquakes most frequent in Pacific Ocean
- Energy stored at stuck fault edges released as seismic waves when friction overcome
- Bulbul = severe cyclonic storm, West Bengal 2019; hurricanes = Atlantic tropical cyclones

For Exam

Cyclones are revolving tropical storms around low-pressure centres. Tropical cyclones form over warm oceans (10°–25°) with winds exceeding 119 km/h; the calm “eye” has no rain. Bulbul struck West Bengal in 2019. Extratropical cyclones occur at mid-latitude frontal zones. Earthquakes: hypocentre

(underground origin), epicentre (surface point above), fault (slip plane). P waves (fastest, longitudinal, solids+liquids), S waves (transverse, solids only), surface waves (maximum destruction). Seismographs record earthquakes; triangulation with 3 stations locates epicentre. Earthquakes most frequent in Pacific Ocean.

Previously Asked Questions

- **Q15** (1 mark, Aug 2023) — Define Earthquakes.

Diagram (refer SLM): Fig. 4.5.7 Normal fault; Fig. 4.5.8 Seismograph operation.

Unit Recap (from SLM)

- Landslide = mass of rock/debris/earth moving down slope under gravity
- Landslide types: slides, flows, spreads, topples, falls
- Flood = water overflow on dry land; most common weather disaster
- Five flood types: river, coastal, storm surge, inland, flash flood
- Cyclones = revolving storms around low pressure; eye is calm centre
- Tropical cyclones: hurricanes (Atlantic), typhoons; Bulbul hit West Bengal (2019)
- Earthquake: hypocentre (below ground), epicentre (surface); recorded by seismograph
- P waves, S waves, and surface waves — surface waves cause most destruction

Unit 6: Important Acts and Rules for the Conservation of Environment

4.6.1–4.6.4 Wildlife Protection Act, 1972

Theory

The Wildlife Protection Act, 1972 protects wild animals, birds, and plant species for environmental and ecological security, restricting hunting and establishing protected area networks. Constitutional basis: Article 48A (protect environment, wildlife, forests — 42nd Amendment 1976) and Article 51A(g) (fundamental duty to protect environment). History: Wild Birds Protection Act

1887 → 1912 → 1935 amendment; serious protection from 1960s. India became CITES party; established National Board for Wildlife and National Tiger Conservation Authority.

Important Points

Salient Features of Wildlife Protection Act, 1972

- Protects listed species of animals, birds, and plants; ensures ecological security
- **Six schedules** giving varying protection degrees:
 - Schedule I and Schedule II (Part II) = **absolute protection**, maximum penalties
 - Schedules III–VI = decreasing protection levels; some species may be hunted under licence
- Prohibits hunting of endangered species; regulates trade of scheduled animals
- Establishes wildlife sanctuaries, national parks, conservation reserves, community reserves, tiger reserves
- Provides licences for sale, transfer, and possession of some wildlife species
- India became party to **CITES** (Convention on International Trade in Endangered Species / Washington Convention)
- First comprehensive list of endangered wildlife of India prepared under this Act

Key Bodies Established

- **National Board for Wildlife:** Under WPA 1972 — Statutory advisory body; chaired by Prime Minister; approves national park/sanctuary projects; promotes conservation
- **National Tiger Conservation Authority (NTCA):** Under WPA 1972 — Statutory body under MoEFCC; strengthens Project Tiger (1973); protects tigers from extinction
- **Central Zoo Authority:** 1992 — Oversees zoos in India; ensures proper animal care and breeding programmes

Five Protected Area Types

- **1. Sanctuaries** — limited human activity; refuge for injured/abused wildlife

- **2. National Parks** — no human activity; fixed boundaries; Bandipur, Karnataka
- **3. Conservation Reserves** — state-declared adjacent to sanctuaries/parks
- **4. Community Reserves** — private/community land for wildlife conservation
- **5. Tiger Reserves** — for tiger protection on NTCA recommendation

For Exam

Wildlife Protection Act, 1972 protects wild animals, birds, and plants, restricting hunting and establishing protected areas. Constitutional support: Article 48A and Article 51A(g). Six schedules provide graded protection — Schedule I gets absolute protection with maximum penalties. India joined CITES (Washington Convention). National Board for Wildlife (PM-chaired) advises on conservation; NTCA oversees Project Tiger (1973); Central Zoo Authority (1992) oversees zoos. Five protected area types: sanctuaries, national parks, conservation reserves, community reserves, and tiger reserves.

Previously Asked Questions

- **Q14** (1 mark, Aug 2023) — Write down the Salient Features of Wildlife Protection Act.

4.6.5–4.6.7 Key Environmental Acts — Provisions Comparison

Theory

India's four major environmental acts form an interconnected legal framework: Wildlife Protection Act (1972) for biodiversity, Water Act (1974) for water pollution, Air Act (1981) for air pollution, and Environment Protection Act (1986) as umbrella legislation coordinating all environmental laws. CPCB (central) and SPCBs (state) enforce pollution control under Air and Water Acts.

Important Points

Key Environmental Acts — Comparison

Act	Year	Purpose	Key Features
Wildlife Protection Act	1972 (amended 2006)	Protect wild animals, birds, plants	6 schedules; sanctuaries, NPs, NTCA; Article 48A, 51A(g)

Act	Year	Purpose	Key Features
Water Pollution Act	1974	Prevent and control water pollution	CPCB + SPCB; Water Cess Act 1977
Air Pollution Act	1981	Prevent and control air pollution	CPCB + SPCB; noise added 1987
Environment Protection Act	1986	Umbrella environmental framework	Coordinates Water/Air Acts; up to 5–7 years imprisonment

Sanctuary vs National Park (Wildlife Act)

Feature	Wildlife Sanctuary	National Park
Human activity	Limited (timber, minor forest products)	No human activity allowed
Boundaries	Not generally fixed	Fixed and defined
Grazing/livestock	Permitted with restrictions	Not permitted
Declaration	State government notification	State government notification
Upgrade/downgrade	Can upgrade to National Park	Cannot downgrade to sanctuary
Example	Refuge for injured/abused wildlife	Bandipur National Park, Karnataka

- Wildlife Act history: Wild Birds Protection Act 1887 → 1912 → 1935 amendment → serious protection from 1960s
- Environment Act = first concrete step for air pollution (Air Act) + umbrella for all environmental coordination
- Water Cess Act 1977 augments CPCB and SPCB resources
- **PYQ Q87 For Exam: Air (Prevention and Control of Pollution) Act, 1981** — prevents, controls and abates air pollution. Defines **air pollutant** (including **noise**, 1987 amendment). Establishes **CPCB** (central) and **SPCB** (state) with powers to inspect, sample, regulate emissions, set standards and penalize violations. Works with Environment Protection Act 1986 for comprehensive pollution control in India.

For Exam

India's four key environmental acts: (1) Wildlife Protection Act (1972) — protects species via 6 schedules, establishes sanctuaries/national parks, NTCA, National Board for Wildlife; (2) Water Act (1974) — prevents water pollution via

CPCB/SPCB, prohibits excess discharge; (3) Air Act (1981) — prevents air pollution, defines pollutants including noise (1987), establishes CPCB/SPCB; (4) Environment Protection Act (1986) — umbrella legislation coordinating all acts, sets emission standards, hazardous waste rules, penalties up to 5–7 years imprisonment. Sanctuaries allow limited human activity; National Parks prohibit all activity with fixed boundaries.

Previously Asked Questions

- **Q28** (1 mark, May 2024) — Mention the year in which the Air Prevention and Control of Pollution Act passed b...
- **Q87** (5 marks, Aug 2023) — Comment on Air Act.

Unit Recap (from SLM)

- Wildlife Protection Act 1972 protects animals, birds, plants; amended 2006
- Constitutional: Article 48A (state duty), Article 51A(g) (citizen duty)
- Six schedules; Schedule I = absolute protection; CITES = Washington Convention
- National Board for Wildlife (PM-chaired); NTCA for Project Tiger
- Air Act 1981: CPCB + SPCB; defines air pollutants including noise
- Environment Protection Act 1986: umbrella legislation; emission standards; penalties
- Water Act 1974: water pollution control; CPCB + SPCB; Water Cess Act 1977

Sanctuaries allow limited activity; National Parks prohibit all human activity

Quick Reference — Exam Essentials

Topic	Must-Know
Four segments	Atmosphere, Hydrosphere, Lithosphere, Biosphere
Atmosphere	N ₂ 78%, O ₂ 21%; troposphere = weather zone
Water on Earth	71% hydrosphere; hydrological cycle
Forest types	Tropical moist/dry, montane, alpine (16 types)

Topic	Must-Know
Ecosystem	Producers → Consumers → Decomposers
Food chain/web	Linear vs interconnected feeding relationships
Ecological pyramid	Numbers, biomass, energy (energy always upright)
Biodiversity	Genetic, species, ecosystem levels
India hotspots	Western Ghats, Himalaya, Indo-Burma, Sundaland
Protected areas	NP (strict), Sanctuary (habitat), Biosphere Reserve (UNESCO)
Threats	Habitat loss, poaching, man-wildlife conflict
SD definition	Brundtland 1987 — needs of present without compromising future
GHGs	CO ₂ , CH ₄ , N ₂ O, CFCs — greenhouse effect
Ozone	Stratosphere; depleted by CFCs
Carbon credit	Tradeable permit to emit CO ₂ ; cap and trade
Acts	Wildlife 1972, Water Act, Air Act, Environment Protection Act
Movements	Chipko (trees), Narmada Bachao (dam)
Waste	Reduce, Reuse, Recycle; 3-bin segregation

About the Author

Abdul Vahab A A

Website: <https://abdulvahabaa.in>

Thank you for using these notes. I hope this Environmental Studies file supports your semester learning, revision, and exam confidence.

All the best for your exams. Keep learning consistently and care for the environment around you.

Dua mein Yaad Rakhna.

*Study notes based on SGOU SLM — Environmental Studies (B21ES01AC).
Covers Blocks 1–4, 24 Units, theory only.*

Prepared by Abdul Vahab A A | abdulvahabaa.in